

1. Results on test functions and test suites

1.1 Quantitative analysis and convergence analysis

1.1.1 Evaluation on standard test functions

23 standard test functions including 7 unimodal functions (F1-F7) and 16 multimodal functions (F8-F23) are applied to test the performance of all compared optimizers. The number of population members N and maximum number of iterations T are set to 80 and 1000 in the following experiment. And the number of decision variable m is inherent to the test functions.

In quantitative analysis, AVE, STD, BEST, and RANK of ten optimizers for each function are listed in Table 1. The best indexes for each test function are highlighted in bold font. For 7 unimodal functions, VOA obtains the best performance in F1, F2, F3, F5, and F6 functions. Their AVE, STD and Best indexes are all 0.00E+00, and also, RANK indexes are 1. Although VOA is slightly worse than COA and FLO on F4 and F7 functions, the AVE, STD and Best indexes of VOA remain the same order of magnitude overall like that in COA and FLO. For 16 multimodal functions, VOA achieves the best AVE, STD and BEST in all functions except F15. While only STD of VOA is worse than that in EEFO on F15. Overall, VOA has an average ranking of 1.261 on 23 standard test functions and perform the best on 86.96% (20/23) of the standard test functions, outperforming other optimizers. Based on these results, it is evident that VOA shows great exploration and exploitation ability.

Table 1
Quantitative analysis for 23 standard test functions.

Fun.	Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
F1	AVE	0.00E+00	7.33E-50	0.00E+00	0.00E+00	0.00E+00	6.75E-288	3.65E-197	1.03E-75	9.49E-81	1.29E-12
	STD	0.00E+00	1.51E-49	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	5.35E-75	1.94E-80	2.20E-12
	BEST	0.00E+00	7.66E-55	0.00E+00	0.00E+00	0.00E+00	0.00E+00	2.74E-217	2.31E-81	7.30E-83	7.93E-15
	RANK	1	9	1	1	1	5	6	8	7	10
F2	AVE	0.00E+00	3.24E-25	8.41E-288	0.00E+00	0.00E+00	4.04E-133	2.30E-103	1.11E-47	9.93E-47	9.01E-06
	STD	0.00E+00	6.54E-25	0.00E+00	0.00E+00	0.00E+00	2.21E-132	1.20E-102	2.52E-47	9.37E-47	4.46E-05
	BEST	0.00E+00	4.82E-28	1.19E-299	0.00E+00	0.00E+00	4.27E-200	2.49E-116	9.58E-50	3.95E-48	4.36E-09
	RANK	1	9	4	1	1	5	6	7	8	10
F3	AVE	0.00E+00	1.63E-09	0.00E+00	0.00E+00	0.00E+00	1.30E-145	2.72E-180	5.54E-02	2.62E-24	4.46E+02
	STD	0.00E+00	2.88E-09	0.00E+00	0.00E+00	0.00E+00	6.74E-145	0.00E+00	1.91E-01	1.07E-23	1.25E+03
	BEST	0.00E+00	6.67E-11	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.00E-196	1.61E-06	7.30E-30	2.46E+01
	RANK	1	8	1	1	1	6	5	9	7	10
F4	AVE	3.07E-206	2.21E-01	1.58E-273	0.00E+00	0.00E+00	2.76E-130	2.17E-99	3.62E-06	1.89E-20	1.74E+00
	STD	0.00E+00	2.14E-01	0.00E+00	0.00E+00	0.00E+00	1.51E-129	1.12E-98	1.45E-05	1.94E-20	6.95E-01
	BEST	1.89E-230	2.09E-02	1.43E-283	0.00E+00	0.00E+00	9.07E-204	9.17E-111	9.06E-10	1.98E-22	8.16E-01
	RANK	4	9	3	1	1	5	6	8	7	10
F5	AVE	0.00E+00	4.06E+00	0.00E+00	0.00E+00	0.00E+00	2.36E+01	4.78E-04	2.51E+01	2.62E+01	4.04E+02
	STD	0.00E+00	4.26E+00	0.00E+00	0.00E+00	0.00E+00	1.05E+00	7.12E-04	1.62E+00	6.17E-01	9.04E+02
	BEST	0.00E+00	1.06E-05	0.00E+00	0.00E+00	0.00E+00	2.28E+01	6.71E-07	2.30E+01	2.51E+01	5.71E+00
	RANK	1	6	1	1	1	7	5	8	9	10
F6	AVE	0.00E+00	4.31E-33	0.00E+00	0.00E+00	0.00E+00	5.63E-23	4.24E-06	7.13E-02	1.91E-01	1.18E-12
	STD	0.00E+00	9.83E-33	0.00E+00	0.00E+00	0.00E+00	1.12E-22	4.90E-06	1.28E-01	1.69E-01	2.90E-12
	BEST	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	6.58E-25	3.28E-08	5.03E-06	5.00E-06	1.02E-14
	RANK	1	5	1	1	1	6	8	9	10	7
F7	AVE	2.51E-05	2.05E-02	3.53E-05	1.10E-05	1.03E-05	1.45E-03	3.61E-05	1.14E-03	2.52E-04	1.04E-01
	STD	2.48E-05	6.59E-03	3.50E-05	1.06E-05	1.08E-05	1.53E-03	3.57E-05	8.97E-04	1.26E-04	4.91E-01
	BEST	7.23E-06	1.12E-02	1.50E-07	4.88E-07	1.77E-06	5.55E-05	6.02E-07	3.97E-05	3.20E-05	7.23E-03
	RANK	3	9	4	2	1	8	5	7	6	10
F8	AVE	-1.26E+04	-9.18E+03	-1.26E+04	-9.96E+03	-1.26E+04	-9.49E+03	-1.26E+04	-7.83E+03	-6.20E+03	-9.38E+03
	STD	0.00E+00	5.33E+02	0.00E+00	1.60E+03	5.11E-04	1.01E+03	2.24E-02	6.50E+02	7.55E+02	5.65E+02
	BEST	-1.26E+04	-1.03E+04	-1.26E+04	-1.26E+04	-1.26E+04	-1.21E+04	-1.26E+04	-1.03E+04	-7.77E+03	-1.05E+04
	RANK	1	8	1	5	3	6	4	9	10	7
F9	AVE	0.00E+00	7.33E+01	0.00E+00	0.00E+00	0.00E+00	3.29E+01	0.00E+00	2.96E+01	1.05E-01	3.71E+01
	STD	0.00E+00	1.58E+01	0.00E+00	0.00E+00	0.00E+00	5.14E+01	0.00E+00	3.11E+01	5.75E-01	1.05E+01
	BEST	0.00E+00	4.38E+01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.99E+01
	RANK	1	10	1	1	1	8	1	7	6	9
F10	AVE	8.88E-16	2.00E+01	8.88E-16	8.88E-16	8.88E-16	8.88E-16	8.88E-16	6.62E-01	1.13E-14	3.02E-07
	STD	0.00E+00	5.91E-03	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	3.63E+00	3.22E-15	2.57E-07
	BEST	8.88E-16	1.99E+01	8.88E-16	8.88E-16	8.88E-16	8.88E-16	8.88E-16	4.44E-15	7.99E-15	3.93E-08
	RANK	1	10	1	1	1	1	1	9	7	8
F11	AVE	0.00E+00	1.03E-02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	2.95E-03	1.29E-03	1.27E-02
	STD	0.00E+00	1.19E-02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	7.15E-03	4.10E-03	1.31E-02
	BEST	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.80E-14
	RANK	1	9	1	1	1	1	1	7	8	10
F12	AVE	1.57E-32	4.02E-01	1.57E-32	1.57E-32	1.57E-32	2.72E-10	4.53E-07	5.19E-03	1.38E-02	2.07E-02
	STD	0.00E+00	4.98E-01	0.00E+00	0.00E+00	0.00E+00	1.47E-09	4.50E-07	4.94E-03	1.28E-02	6.33E-02

	BEST	1.57E-32	1.57E-32	1.57E-32	1.57E-32	1.57E-32	2.66E-26	1.73E-09	9.37E-07	6.19E-07	1.50E-16
	RANK	1	10	1	1	1	5	6	7	8	9
F13	AVE	1.35E-32	1.18E-01	1.35E-32	1.35E-32	1.35E-32	5.83E-02	4.74E-06	1.38E-01	1.85E-01	1.46E-03
	STD	0.00E+00	3.47E-01	0.00E+00	0.00E+00	0.00E+00	7.14E-02	7.36E-06	1.24E-01	1.44E-01	3.80E-03
	BEST	1.35E-32	1.35E-32	1.35E-32	1.35E-32	1.35E-32	1.78E-23	1.13E-08	5.50E-06	1.02E-05	2.26E-15
	RANK	1	8	1	1	1	7	5	9	10	6
F14	AVE	9.98E-01	9.98E-01	9.98E-01	9.98E-01	9.98E-01	9.98E-01	9.98E-01	9.98E-01	1.86E+00	9.98E-01
	STD	0.00E+00	0.00E+00	0.00E+00	0.00E+00	8.44E-12	0.00E+00	3.25E-11	3.60E-15	1.00E+00	0.00E+00
	BEST	9.98E-01	9.98E-01	9.98E-01	9.98E-01	9.98E-01	9.98E-01	9.98E-01	9.98E-01	9.98E-01	9.98E-01
	RANK	1	1	1	1	8	1	9	7	10	1
F15	AVE	3.07E-04	1.73E-03	3.07E-04	4.47E-04	3.49E-04	6.61E-04	3.17E-04	4.60E-04	3.74E-03	1.65E-03
	STD	3.07E-19	4.76E-03	0.00E+00	1.69E-04	6.39E-05	3.29E-04	8.04E-06	3.47E-04	7.57E-03	5.09E-03
	BEST	3.07E-04	3.07E-04	3.07E-04	3.09E-04	3.08E-04	3.07E-04	3.08E-04	3.07E-04	3.07E-04	3.07E-04
	RANK	2	9	1	5	4	7	3	6	10	8
F16	AVE	-1.03E+00	-1.03E+00	-1.03E+00	-1.03E+00	-1.03E+00	-1.03E+00	-1.03E+00	-1.03E+00	-1.03E+00	-1.03E+00
	STD	0.00E+00	0.00E+00	2.07E-07	1.07E-03	1.44E-05	0.00E+00	6.26E-15	3.70E-13	1.10E-09	0.00E+00
	BEST	-1.03E+00	-1.03E+00	-1.03E+00	-1.03E+00	-1.03E+00	-1.03E+00	-1.03E+00	-1.03E+00	-1.03E+00	-1.03E+00
	RANK	1	1	8	10	9	1	5	6	7	1
F17	AVE	3.98E-01	3.98E-01	3.98E-01	3.99E-01	3.98E-01	3.98E-01	3.98E-01	3.98E-01	3.98E-01	3.98E-01
	STD	0.00E+00	0.00E+00	5.11E-06	3.93E-03	1.69E-04	0.00E+00	9.46E-09	5.29E-09	2.59E-07	0.00E+00
	BEST	3.98E-01	3.98E-01	3.98E-01	3.98E-01	3.98E-01	3.98E-01	3.98E-01	3.98E-01	3.98E-01	3.98E-01
	RANK	1	1	8	10	9	1	6	5	7	1
F18	AVE	3.00E+00	3.00E+00	3.00E+00	3.88E+00	3.02E+00	3.00E+00	3.00E+00	3.00E+00	3.00E+00	3.00E+00
	STD	0.00E+00	3.93E-15	0.00E+00	3.49E+00	7.97E-02	2.57E-15	4.08E-11	6.04E-12	1.20E-06	0.00E+00
	BEST	3.00E+00	3.00E+00	3.00E+00	3.00E+00	3.00E+00	3.00E+00	3.00E+00	3.00E+00	3.00E+00	3.00E+00
	RANK	1	5	1	10	9	4	7	6	8	1
F19	AVE	-3.86E+00	-3.86E+00	-3.86E+00	-3.80E+00	-3.85E+00	-3.86E+00	-3.86E+00	-3.86E+00	-3.86E+00	-3.86E+00
	STD	0.00E+00	2.71E-15	2.71E-15	4.74E-02	2.35E-02	1.44E-03	3.50E-04	2.99E-03	2.25E-03	2.71E-15
	BEST	-3.86E+00	-3.86E+00	-3.86E+00	-3.86E+00	-3.86E+00	-3.86E+00	-3.86E+00	-3.86E+00	-3.86E+00	-3.86E+00
	RANK	1	2	2	10	9	6	5	8	7	2
F20	AVE	-3.32E+00	-3.29E+00	-3.31E+00	-2.96E+00	-2.93E+00	-3.27E+00	-3.24E+00	-3.25E+00	-3.28E+00	-3.28E+00
	STD	0.00E+00	5.54E-02	3.02E-02	2.25E-01	2.42E-01	6.11E-02	6.77E-02	7.60E-02	6.30E-02	5.83E-02
	BEST	-3.32E+00	-3.32E+00	-3.32E+00	-3.26E+00	-3.27E+00	-3.32E+00	-3.32E+00	-3.32E+00	-3.32E+00	-3.32E+00
	RANK	1	3	2	9	10	6	8	7	5	4
F21	AVE	-1.02E+01	-6.45E+00	-1.02E+01	-1.02E+01	-1.02E+01	-6.21E+00	-5.90E+00	-9.13E+00	-9.48E+00	-7.21E+00
	STD	0.00E+00	3.01E+00	0.00E+00	1.35E-04	4.34E-06	2.58E+00	1.93E+00	2.07E+00	1.75E+00	2.88E+00
	BEST	-1.02E+01	-1.02E+01	-1.02E+01	-1.02E+01	-1.02E+01	-1.02E+01	-1.02E+01	-1.02E+01	-1.02E+01	-1.02E+01
	RANK	1	8	1	4	3	9	10	6	5	7
F22	AVE	-1.04E+01	-5.77E+00	-1.04E+01	-1.04E+01	-1.04E+01	-6.57E+00	-5.80E+00	-9.16E+00	-1.04E+01	-9.48E+00
	STD	0.00E+00	3.01E+00	0.00E+00	1.77E-04	3.50E-05	2.59E+00	1.84E+00	2.29E+00	1.16E-04	2.12E+00
	BEST	-1.04E+01	-1.04E+01	-1.04E+01	-1.04E+01	-1.04E+01	-1.04E+01	-1.04E+01	-1.04E+01	-1.04E+01	-1.04E+01
	RANK	1	10	1	5	3	8	9	7	4	6
F23	AVE	-1.05E+01	-5.58E+00	-1.05E+01	-1.05E+01	-1.05E+01	-6.04E+00	-5.67E+00	-9.18E+00	-1.03E+01	-9.92E+00
	STD	0.00E+00	3.22E+00	9.03E-15	2.76E-05	2.86E-05	2.41E+00	1.65E+00	2.54E+00	1.48E+00	1.91E+00
	BEST	-1.05E+01	-1.05E+01	-1.05E+01	-1.05E+01	-1.05E+01	-1.05E+01	-1.05E+01	-1.05E+01	-1.05E+01	-1.05E+01
	RANK	1	10	2	3	4	8	9	7	5	6
Mean rank		1.261	6.957	2.087	3.304	3.609	5.261	5.652	7.348	7.435	6.652
Total rank		1	8	2	3	4	5	6	9	10	7

In convergence analysis, convergence curves of ten optimizers are shown in Fig. 1. For 7 unimodal functions (Fig. 1.F1-F7), VOA has great convergence speed except F4 and F7. Although the convergence speed of VOA is slightly lower than COA and FLO on F4 and F7, it still performs better than other compared optimizers. For 16 multimodal functions (Fig. 1.F8-F23), VOA could avoid falling into the local optimum and find the best solution quickly. These results align with the quantitative analysis in Table 1, which illustrate that VOA has great superiority.

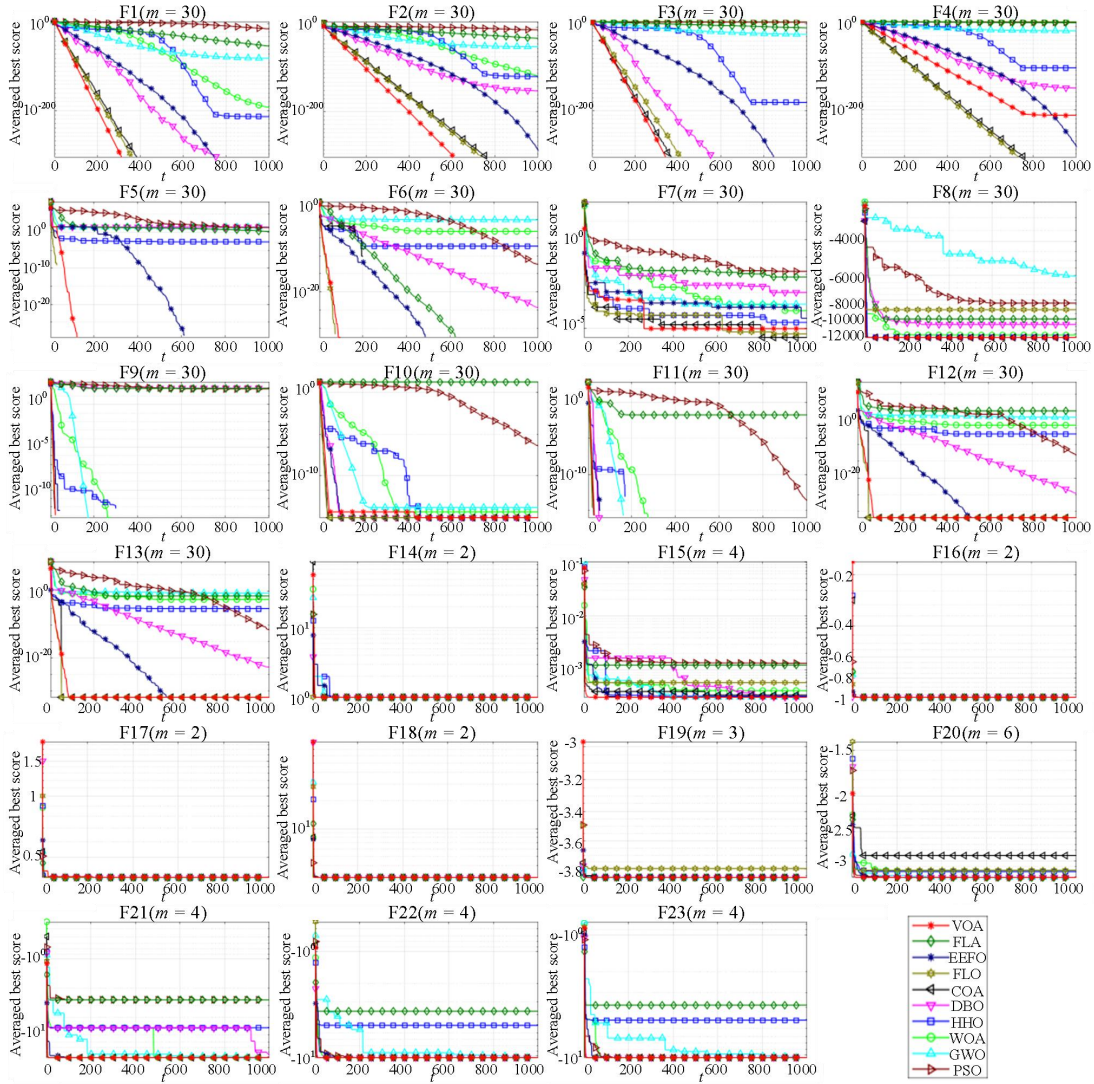


Fig. 1 Convergence analysis for 23 standard test functions.

1.1.2 Evaluation on CEC2017 test suite

The CEC2017 test suite has 29 standard benchmark functions with four different types, i.e., 2 unimodal functions (C17F1 and C17F3), 7 multimodal functions (C17F4-C17F10), 10 hybrid functions (C17F11-C17F20) and 10 composition functions (C17F21-C17F30). In this test suite, m can be set to 10, 30, 50 or 100. This evaluation uses a low dimension of $m = 10$ for quantitative and convergence analyses, and further supplements the high-dimensional optimization results with convergence analysis at dimensions of $m = 30$ and 50. In this experiment, N is 100 and T is 5000.

In quantitative analysis, when $m = 10$, AVE, STD, BEST, and RANK of ten optimizers for each function are listed in Table 2. In comparison with other optimizers, VOA has been ranked as the first best optimizer with good AVE, STD and BEST indexes on 19 functions (C17F1, C17F3, C17F5, C17F6, C17F9-C17F11, C17F16 and C17F20-C17F30). For the other 10 functions (C17F4, C17F7, C17F8, C17F12-C17F15, C17F17-C17F19), although the performance of VOA is worse than that in EEFO or PSO, VOA still obtains the RANK 2. In addition, the STD of VOA in these 10 functions are also low, indicating the stability of its optimization effect. VOA has an average ranking of 1.345 on CEC2017 test suite, perform the best on 68.97% (20/29) of the test suite and be second on 34.48% (10/29), outperforming other optimizers. Based on these results, VOA shows great superiority.

Table 2
Quantitative analysis for 29 CEC2017 test suite with $m = 10$.

Fun.	Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
C17F1	AVE	1.00E+02	9.03E+02	1.00E+02	6.64E+09	6.05E+09	5.54E+03	5.70E+04	6.45E+03	1.91E+06	2.34E+06
	STD	0.00E+00	9.28E+02	0.00E+00	2.72E+09	2.96E+09	4.63E+03	4.65E+04	1.19E+04	7.26E+06	1.28E+07
	BEST	1.00E+02	1.00E+02	1.00E+02	1.51E+09	6.54E+08	1.00E+02	2.39E+03	1.46E+02	3.42E+02	1.00E+02
	RANK	1	3	1	6	10	4	6	5	8	9
C17F3	AVE	3.00E+02	3.00E+02	3.00E+02	1.06E+04	7.84E+03	3.00E+02	3.00E+02	3.04E+02	6.69E+02	3.00E+02

	STD	0.00E+00	0.00E+00	0.00E+00	2.59E+03	2.35E+03	0.00E+00	1.39E-01	1.15E+01	8.52E+02	0.00E+00
	BEST	3.00E+02	3.00E+02	3.00E+02	4.22E+03	1.75E+03	3.00E+02	3.00E+02	3.00E+02	3.00E+02	3.00E+02
	RANK	1	1	1	7	9	1	6	7	8	1
C17F4	AVE	4.00E+02	4.06E+02	4.00E+02	8.62E+02	7.59E+02	4.08E+02	4.04E+02	4.13E+02	4.10E+02	4.13E+02
	STD	9.21E-02	1.70E+01	2.99E-05	2.43E+02	2.22E+02	1.32E+01	1.77E+00	2.21E+01	1.06E+01	2.14E+01
	BEST	4.00E+02	4.00E+02	4.00E+02	5.17E+02	5.10E+02	4.00E+02	4.00E+02	4.00E+02	4.03E+02	4.03E+02
	RANK	2	4	1	8	10	5	3	8	6	7
C17F5	AVE	5.05E+02	5.18E+02	5.08E+02	5.75E+02	5.77E+02	5.24E+02	5.34E+02	5.41E+02	5.10E+02	5.07E+02
	STD	2.31E+00	7.68E+00	3.67E+00	1.32E+01	1.34E+01	1.12E+01	9.00E+00	1.50E+01	4.16E+00	3.14E+00
	BEST	5.01E+02	5.05E+02	5.02E+02	5.44E+02	5.47E+02	5.08E+02	5.14E+02	5.19E+02	5.03E+02	5.02E+02
	RANK	1	5	3	8	10	6	7	8	4	2
C17F6	AVE	6.00E+02	6.00E+02	6.00E+02	6.40E+02	6.40E+02	6.01E+02	6.12E+02	6.25E+02	6.00E+02	6.00E+02
	STD	0.00E+00	7.02E-01	3.67E-04	8.23E+00	7.27E+00	1.71E+00	9.50E+00	1.21E+01	1.60E-01	1.38E-01
	BEST	6.00E+02	6.00E+02	6.00E+02	6.21E+02	6.25E+02	6.00E+02	6.00E+02	6.05E+02	6.00E+02	6.00E+02
	RANK	1	5	2	8	10	6	7	8	4	3
C17F7	AVE	7.18E+02	7.20E+02	7.19E+02	7.91E+02	7.82E+02	7.25E+02	7.59E+02	7.69E+02	7.22E+02	7.15E+02
	STD	3.44E+00	6.12E+00	4.60E+00	1.59E+01	1.81E+01	7.75E+00	1.11E+01	2.07E+01	5.28E+00	2.72E+00
	BEST	7.13E+02	7.09E+02	7.12E+02	7.60E+02	7.53E+02	7.13E+02	7.34E+02	7.44E+02	7.12E+02	7.11E+02
	RANK	2	4	3	10	9	6	7	8	5	1
C17F8	AVE	8.07E+02	8.17E+02	8.08E+02	8.45E+02	8.43E+02	8.21E+02	8.26E+02	8.33E+02	8.09E+02	8.05E+02
	STD	2.51E+00	9.25E+00	3.40E+00	1.15E+01	1.13E+01	8.50E+00	7.41E+00	1.31E+01	3.85E+00	2.14E+00
	BEST	8.03E+02	8.04E+02	8.03E+02	8.25E+02	8.22E+02	8.06E+02	8.12E+02	8.13E+02	8.02E+02	8.02E+02
	RANK	2	5	3	10	9	6	7	8	4	1
C17F9	AVE	9.00E+02	9.00E+02	9.00E+02	1.32E+03	1.37E+03	9.01E+02	1.16E+03	1.14E+03	9.02E+02	9.00E+02
	STD	0.00E+00	3.39E-01	0.00E+00	1.69E+02	1.44E+02	5.12E+00	2.24E+02	2.58E+02	8.22E+00	0.00E+00
	BEST	9.00E+02	9.00E+02	9.00E+02	1.00E+03	1.04E+03	9.00E+02	9.00E+02	9.22E+02	9.00E+02	9.00E+02
	RANK	1	4	1	9	10	5	8	7	6	1
C17F10	AVE	1.13E+03	1.77E+03	1.14E+03	2.54E+03	2.43E+03	1.63E+03	1.81E+03	1.89E+03	1.36E+03	1.34E+03
	STD	1.66E+02	2.95E+02	9.78E+01	2.16E+02	1.49E+02	2.77E+02	3.18E+02	3.07E+02	2.26E+02	1.70E+02
	BEST	1.01E+03	1.23E+03	1.01E+03	1.93E+03	2.07E+03	1.15E+03	1.26E+03	1.15E+03	1.01E+03	1.01E+03
	RANK	1	7	2	10	5	6	8	9	4	3
C17F11	AVE	1.10E+03	1.13E+03	1.10E+03	2.13E+03	1.44E+03	1.14E+03	1.13E+03	1.18E+03	1.12E+03	1.11E+03
	STD	1.30E+00	4.16E+01	2.38E+00	9.50E+02	2.23E+02	5.58E+01	1.63E+01	7.74E+01	1.09E+01	1.31E+01
	BEST	1.10E+03	1.10E+03	1.10E+03	1.18E+03	1.17E+03	1.10E+03	1.11E+03	1.11E+03	1.10E+03	1.10E+03
	RANK	1	6	2	10	9	7	5	8	4	3
C17F12	AVE	1.36E+03	1.78E+04	1.28E+03	1.13E+08	5.10E+07	1.36E+06	1.44E+05	2.17E+06	2.88E+05	1.88E+04
	STD	1.17E+02	1.60E+04	6.69E+01	7.91E+07	4.81E+07	3.05E+06	3.15E+05	2.67E+06	5.44E+05	1.71E+04
	BEST	1.20E+03	1.81E+03	1.20E+03	9.35E+05	1.54E+06	2.49E+03	3.19E+03	5.32E+03	5.02E+03	1.49E+03
	RANK	2	3	1	10	9	7	5	8	6	4
C17F13	AVE	1.38E+03	1.42E+03	1.31E+03	1.28E+05	2.49E+04	4.28E+03	1.46E+04	1.67E+04	7.01E+03	5.60E+03
	STD	6.57E+01	1.07E+02	4.01E+00	4.00E+05	2.29E+04	5.22E+03	8.63E+03	1.56E+04	3.44E+03	4.94E+03
	BEST	1.30E+03	1.30E+03	1.30E+03	2.06E+03	2.54E+03	1.34E+03	1.90E+03	1.54E+03	1.67E+03	1.36E+03
	RANK	2	3	1	10	9	4	7	8	6	5
C17F14	AVE	1.42E+03	1.47E+03	1.40E+03	3.20E+03	1.50E+03	1.46E+03	1.49E+03	1.50E+03	1.59E+03	1.43E+03
	STD	9.62E+00	3.13E+01	1.16E+00	2.23E+03	2.11E+01	3.92E+01	2.00E+01	2.99E+01	6.66E+02	1.72E+01
	BEST	1.40E+03	1.41E+03	1.40E+03	1.46E+03	1.45E+03	1.41E+03	1.44E+03	1.43E+03	1.44E+03	1.40E+03
	RANK	2	5	1	10	7	4	6	8	9	3
C17F15	AVE	1.51E+03	1.54E+03	1.50E+03	1.03E+04	4.50E+03	1.63E+03	1.58E+03	1.71E+03	1.75E+03	1.52E+03
	STD	8.15E+00	4.49E+01	9.19E-01	3.47E+03	3.10E+03	1.38E+02	6.40E+01	1.02E+02	5.45E+02	2.49E+01
	BEST	1.50E+03	1.50E+03	1.50E+03	2.72E+03	1.65E+03	1.50E+03	1.51E+03	1.56E+03	1.52E+03	1.50E+03
	RANK	2	4	1	10	9	6	5	7	8	3
C17F16	AVE	1.60E+03	1.70E+03	1.61E+03	2.02E+03	1.95E+03	1.71E+03	1.83E+03	1.79E+03	1.64E+03	1.67E+03
	STD	2.96E-01	8.57E+01	3.01E+01	1.20E+02	1.01E+02	1.09E+02	1.24E+02	1.29E+02	6.48E+01	9.51E+01
	BEST	1.60E+03	1.60E+03	1.60E+03	1.70E+03	1.68E+03	1.60E+03	1.60E+03	1.61E+03	1.60E+03	1.60E+03
	RANK	1	5	2	10	9	6	8	7	3	4
C17F17	AVE	1.71E+03	1.75E+03	1.70E+03	1.79E+03	1.77E+03	1.74E+03	1.76E+03	1.78E+03	1.74E+03	1.72E+03
	STD	7.07E+00	4.14E+01	5.52E+00	2.47E+01	1.79E+01	1.55E+01	2.38E+01	2.83E+01	2.06E+01	1.02E+01
	BEST	1.70E+03	1.70E+03	1.70E+03	1.75E+03	1.74E+03	1.72E+03	1.73E+03	1.74E+03	1.71E+03	1.70E+03
	RANK	2	6	1	10	8	4	7	9	5	3
C17F18	AVE	1.84E+03	2.18E+03	1.80E+03	4.54E+06	6.70E+05	1.24E+04	1.56E+04	1.45E+04	2.71E+04	1.55E+04
	STD	1.39E+01	1.18E+03	7.27E-01	1.00E+07	1.56E+06	1.34E+04	1.24E+04	1.03E+04	1.19E+04	1.42E+04
	BEST	1.81E+03	1.82E+03	1.80E+03	3.53E+03	5.61E+03	1.87E+03	2.17E+03	1.91E+03	2.36E+03	1.83E+03
	RANK	2	3	1	10	9	4	7	5	8	6
C17F19	AVE	1.91E+03	1.94E+03	1.90E+03	3.19E+04	3.21E+03	1.98E+03	3.57E+03	9.88E+03	3.05E+03	1.91E+03
	STD	3.06E+00	4.91E+01	3.73E-01	3.91E+04	1.07E+03	1.43E+02	2.61E+03	9.90E+03	3.45E+03	7.15E+00
	BEST	1.90E+03	1.90E+03	1.90E+03	3.54E+03	1.97E+03	1.90E+03	1.92E+03	2.07E+03	1.91E+03	1.90E+03
	RANK	2	4	1	10	7	5	8	9	6	3
C17F20	AVE	2.00E+03	2.06E+03	2.00E+03	2.19E+03	2.19E+03	2.05E+03	2.09E+03	2.09E+03	2.04E+03	2.03E+03
	STD	1.44E-01	7.51E+01	5.99E-01	6.02E+01	5.40E+01	4.74E+01	5.57E+01	4.38E+01	3.23E+01	4.82E+01
	BEST	2.00E+03	2.00E+03	2.00E+03	2.08E+03	2.10E+03	2.00E+03	2.02E+03	2.03E+03	2.02E+03	2.00E+03
	RANK	1	6	2	10	9	5	8	7	4	3
C17F21	AVE	2.20E+03	2.20E+03	2.20E+03	2.32E+03	2.32E+03	2.20E+03	2.29E+03	2.28E+03	2.30E+03	2.27E+03

	STD	0.00E+00	1.18E+00	1.95E+01	5.44E+01	5.47E+01	1.53E+00	6.94E+01	6.68E+01	3.37E+01	5.23E+01
	BEST	2.20E+03	2.20E+03	2.20E+03	2.22E+03	2.23E+03	2.20E+03	2.20E+03	2.20E+03	2.20E+03	2.20E+03
	RANK	1	2	4	9	10	3	7	6	8	5
C17F22	AVE	2.24E+03	2.29E+03	2.28E+03	2.69E+03	2.65E+03	2.30E+03	2.31E+03	2.33E+03	2.30E+03	2.31E+03
	STD	4.28E+01	2.31E+01	4.10E+01	2.22E+02	2.13E+02	1.15E+01	1.39E+01	1.25E+02	1.74E+01	2.41E+01
	BEST	2.20E+03	2.23E+03	2.20E+03	2.33E+03	2.31E+03	2.25E+03	2.25E+03	2.26E+03	2.22E+03	2.21E+03
	RANK	1	3	2	10	9	4	6	8	5	7
C17F23	AVE	2.61E+03	2.63E+03	2.61E+03	2.71E+03	2.68E+03	2.63E+03	2.64E+03	2.64E+03	2.61E+03	2.62E+03
	STD	2.34E+00	1.26E+01	3.78E+00	2.30E+01	2.31E+01	1.16E+01	1.82E+01	2.05E+01	5.75E+00	5.99E+00
	BEST	2.60E+03	2.61E+03	2.60E+03	2.65E+03	2.63E+03	2.61E+03	2.62E+03	2.61E+03	2.60E+03	2.61E+03
	RANK	1	6	2	10	9	5	7	8	3	4
C17F24	AVE	2.51E+03	2.75E+03	2.62E+03	2.84E+03	2.79E+03	2.60E+03	2.73E+03	2.72E+03	2.73E+03	2.72E+03
	STD	4.40E+01	5.56E+01	1.16E+02	7.93E+01	8.34E+01	1.28E+02	1.20E+02	1.03E+02	4.50E+01	9.07E+01
	BEST	2.50E+03	2.50E+03	2.50E+03	2.61E+03	2.61E+03	2.50E+03	2.50E+03	2.50E+03	2.50E+03	2.40E+03
	RANK	1	8	3	10	9	2	7	5	6	4
C17F25	AVE	2.90E+03	2.92E+03	2.91E+03	3.19E+03	3.19E+03	2.93E+03	2.93E+03	2.93E+03	2.93E+03	2.92E+03
	STD	1.15E+01	2.47E+01	2.18E+01	1.18E+02	1.42E+02	2.67E+01	2.91E+01	6.59E+01	1.63E+01	2.47E+01
	BEST	2.90E+03	2.90E+03	2.90E+03	3.00E+03	3.02E+03	2.90E+03	2.90E+03	2.60E+03	2.90E+03	2.90E+03
	RANK	1	3	2	9	10	6	7	8	5	3
C17F26	AVE	2.85E+03	2.98E+03	2.90E+03	3.77E+03	3.81E+03	2.97E+03	3.18E+03	3.16E+03	3.09E+03	2.94E+03
	STD	1.14E+02	1.77E+02	0.00E+00	3.89E+02	3.43E+02	7.03E+01	3.80E+02	3.88E+02	3.64E+02	6.28E+01
	BEST	2.60E+03	2.90E+03	2.90E+03	3.23E+03	3.17E+03	2.80E+03	2.81E+03	2.60E+03	2.90E+03	2.80E+03
	RANK	1	5	2	9	10	4	8	7	6	3
C17F27	AVE	3.09E+03	3.09E+03	3.09E+03	3.23E+03	3.15E+03	3.10E+03	3.13E+03	3.11E+03	3.09E+03	3.11E+03
	STD	2.22E+00	4.28E+00	2.48E+00	6.48E+01	2.35E+01	5.52E+00	2.78E+01	2.36E+01	2.67E+00	1.73E+01
	BEST	3.09E+03	3.09E+03	3.09E+03	3.13E+03	3.12E+03	3.09E+03	3.09E+03	3.09E+03	3.09E+03	3.09E+03
	RANK	1	4	2	10	9	5	8	7	3	6
C17F28	AVE	3.09E+03	3.24E+03	3.10E+03	3.67E+03	3.64E+03	3.29E+03	3.34E+03	3.29E+03	3.31E+03	3.34E+03
	STD	5.48E+01	1.27E+02	6.49E+01	1.60E+02	1.38E+02	1.14E+02	1.38E+02	1.72E+02	1.11E+02	1.70E+02
	BEST	2.80E+03	3.10E+03	2.80E+03	3.40E+03	3.29E+03	3.10E+03	3.10E+03	3.10E+03	3.10E+03	3.10E+03
	RANK	1	3	2	10	9	4	7	5	6	8
C17F29	AVE	3.15E+03	3.21E+03	3.15E+03	3.40E+03	3.32E+03	3.21E+03	3.27E+03	3.32E+03	3.15E+03	3.16E+03
	STD	9.65E+00	5.76E+01	1.11E+01	9.44E+01	5.51E+01	5.66E+01	7.30E+01	1.02E+02	1.39E+01	3.14E+01
	BEST	3.13E+03	3.14E+03	3.13E+03	3.24E+03	3.24E+03	3.13E+03	3.17E+03	3.20E+03	3.13E+03	3.13E+03
	RANK	1	6	2	10	8	5	7	9	3	4
C17F30	AVE	3.73E+03	8.45E+04	4.01E+03	1.27E+07	2.01E+06	3.68E+05	1.46E+05	2.10E+05	3.89E+05	3.51E+05
	STD	1.72E+02	2.47E+05	7.96E+02	8.70E+06	1.28E+06	5.78E+05	3.64E+05	3.32E+05	5.69E+05	5.09E+05
	BEST	3.46E+03	3.45E+03	3.52E+03	9.17E+04	3.59E+04	3.69E+03	3.17E+03	6.43E+03	4.73E+03	3.55E+03
	RANK	1	3	2	10	9	7	4	5	8	6
Mean Rank		1.345	4.276	1.828	9.414	8.931	4.897	6.655	7.310	5.552	3.966
Total Rank		1	4	2	10	9	5	7	8	6	3

In convergence analysis, convergence curves of ten optimizers under $m = 10, 30$ and 50 are shown in Fig. 2, Fig. 3 and Fig. 4, respectively. When $m = 10$, VOA can approach the best solution quickly and show best performance in solving most functions. When $m = 30$, VOA performs a little worse than EEFO or FLA on C17F10 and C17F12, while it still obtains better performance than other optimizers. When $m = 50$, the results of VOA on C17F3, C17F5, C17F7-10, C17F12, C17F19 and C17F20 are not as good as EEFO, FLA, or PSO. However, it still performs better than other functions. When m is small, VOA is very fast to find the global best solution. When m is larger, VOA can also quickly find a local best solution and constantly explore the global best solution. In a word, VOA has a good ability in most functions.

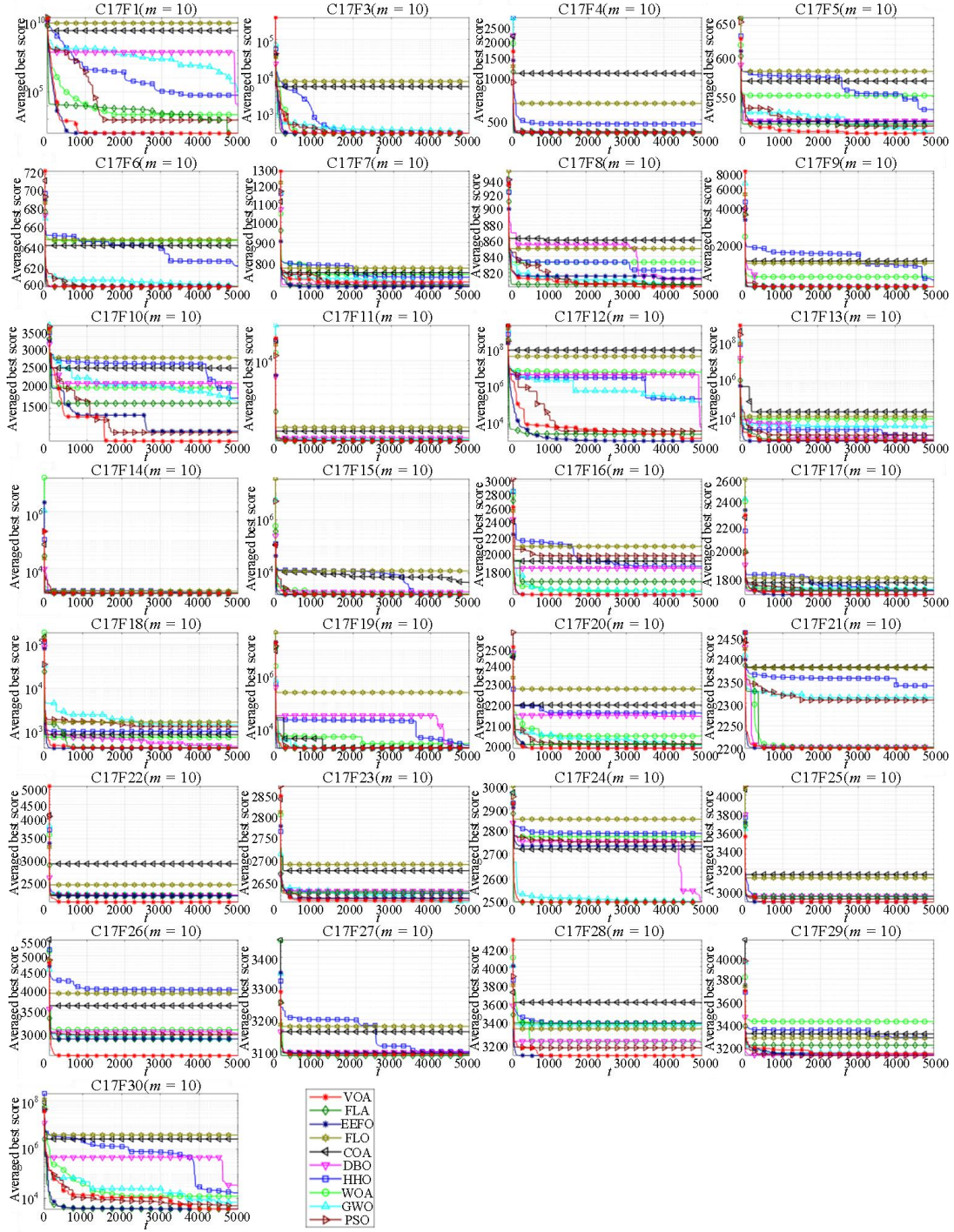


Fig. 2 Convergence analysis for CEC2017 with $m = 10$.

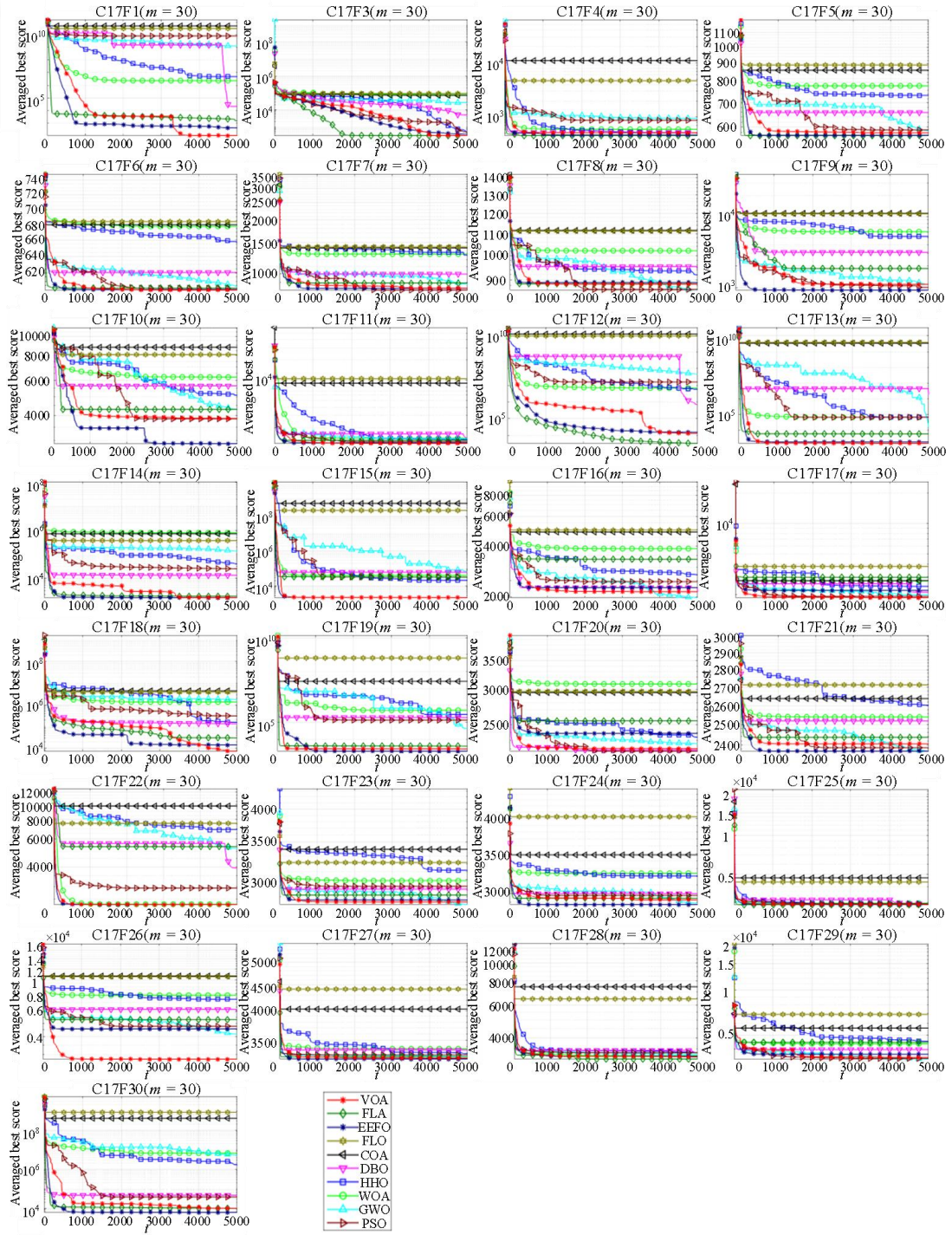


Fig. 3 Convergence analysis for CEC2017 with $m = 30$.

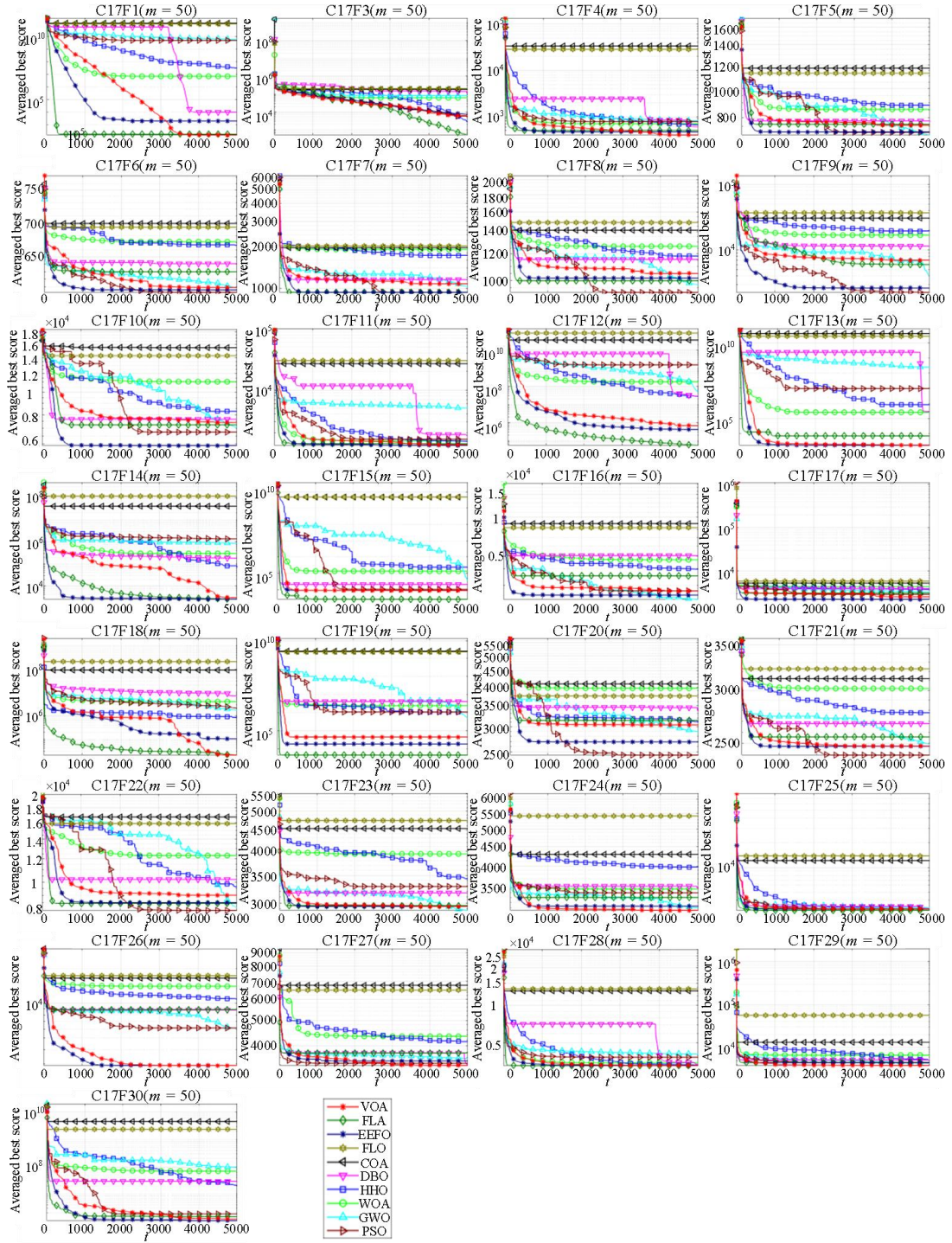


Fig. 4 Convergence analysis for CEC2017 with $m = 50$.

1.1.3 Evaluation on CEC2019, CEC2020 and CEC2022 test suites

The CEC2019, CEC2020, and CEC2022 test suites include 10, 10 and 12 multidimensional complex test functions, respectively. The functions of CEC2019 test suite have fixed m . In CEC2020, m can be set to 5, 10, 15, or 20, while m can be set to either 10 or 20 in CEC2022. In this evaluation, quantitative and convergence analyses are performed on the CEC2019 test suite, the CEC2020 test suite with $m = 10$, and the CEC2022 test suite with $m = 10$. Additionally, convergence analysis is conducted on CEC2020 test suite with $m = 20$ and CEC2022 test suite with $m = 20$, which can supplement the high-dimensional optimization results. N is 100 and T is 5000.

In quantitative analysis, the comparison results of CEC2019 test suite, CEC2020 test suite with $m = 10$ and CEC2022 test suite with $m = 10$ are listed in Table 3, Table 4 and Table 5, respectively. VOA has been ranked as the first best optimizer compared to the other nine optimizers on C19F1-C19F7, C19F10, C20F1, C20F2, C20F6, C20F8-C20F10, C22F1-C22F3, C22F5 and C22F7-C22F12. On other functions (C19F8, C19F9, C20F3-C20F5, C20F7,

C22F6), AVE, STD and BEST indexes of VOA are also very close to the best optimizer, with RANK 2 or 3. In a word, VOA is the best optimizer on CEC2019, CEC2020 and CEC2022 test suites with average rankings of 1.3, 1.4 and 1.167, respectively. And VOA perform the best on 80% (8/10) of CEC2019 test suite, perform the best on 60% (6/10) and second on 40% (4/10) of CEC2020 test suite and perform the best on 83.33% (10/12) and second on 16.67% (2/12) of CEC2022 test suite, which validates the good exploitation and exploration ability of VOA.

Table 3
Quantitative analysis for 10 CEC2019 test suite.

Fun.	Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
C19F1	AVE	1.00E+00	2.55E+00	1.00E+00	1.00E+00	1.00E+00	3.92E+04	1.00E+00	2.36E+05	2.50E+02	2.05E+05
	STD	0.00E+00	3.70E+00	0.00E+00	0.00E+00	0.00E+00	9.31E+04	0.00E+00	3.77E+05	1.32E+03	3.88E+05
	BEST	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	6.44E+00	1.00E+00	1.56E+03
	RANK	1	6	1	1	1	8	1	10	7	9
C19F2	AVE	4.16E+00	4.66E+02	4.19E+00	5.00E+00	5.00E+00	3.48E+02	4.66E+00	6.97E+03	1.90E+02	3.27E+02
	STD	9.81E-02	1.45E+02	1.47E-01	1.13E-05	3.45E-08	2.29E+02	2.07E-01	3.23E+03	1.43E+02	1.56E+02
	BEST	4.04E+00	1.74E+02	3.64E+00	5.00E+00	5.00E+00	4.25E+00	4.43E+00	1.08E+03	8.31E+00	1.05E+02
	RANK	1	9	2	5	4	8	3	10	6	7
C19F3	AVE	1.41E+00	2.22E+00	1.42E+00	3.57E+00	4.92E+00	3.28E+00	1.94E+00	2.20E+00	1.99E+00	2.04E+00
	STD	3.73E-10	2.10E+00	1.44E-01	1.16E+00	1.05E+00	1.99E+00	7.32E-01	1.32E+00	1.71E+00	1.58E+00
	BEST	1.41E+00	1.41E+00	1.00E+00	1.44E+00	2.94E+00	1.00E+00	1.41E+00	1.00E+00	1.00E+00	1.00E+00
	RANK	1	7	2	9	10	8	3	6	4	5
C19F4	AVE	6.17E+00	1.73E+01	9.39E+00	6.59E+01	7.09E+01	1.98E+01	3.19E+01	4.20E+01	1.01E+01	7.27E+00
	STD	2.28E+00	7.35E+00	2.54E+00	1.07E+01	1.41E+01	7.77E+00	1.21E+01	1.58E+01	4.23E+00	2.84E+00
	BEST	2.99E+00	5.97E+00	3.98E+00	4.42E+01	3.57E+01	8.96E+00	1.10E+01	1.49E+01	2.99E+00	2.99E+00
	RANK	1	5	2	9	10	6	7	8	4	3
C19F5	AVE	1.06E+00	1.11E+00	1.06E+00	4.93E+01	4.92E+01	1.13E+00	1.73E+00	1.77E+00	1.34E+00	1.11E+00
	STD	2.95E-02	3.86E-02	3.00E-02	2.13E+01	1.91E+01	8.49E-02	3.25E-01	5.08E-01	2.23E-01	5.92E-02
	BEST	1.00E+00	1.04E+00	1.02E+00	1.88E+01	1.92E+01	1.02E+00	1.27E+00	1.09E+00	1.11E+00	1.01E+00
	RANK	1	3	2	10	9	5	7	8	6	4
C19F6	AVE	1.00E+00	3.56E+00	1.18E+00	9.54E+00	8.86E+00	4.47E+00	5.63E+00	7.28E+00	2.05E+00	1.62E+00
	STD	0.00E+00	2.18E+00	4.20E-01	1.16E+00	1.12E+00	1.39E+00	1.64E+00	1.85E+00	1.05E+00	1.05E+00
	BEST	1.00E+00	1.00E+00	1.00E+00	6.71E+00	5.78E+00	1.85E+00	2.12E+00	3.65E+00	1.05E+00	1.00E+00
	RANK	1	5	2	10	9	6	7	8	4	3
C19F7	AVE	2.70E+02	6.91E+02	2.76E+02	1.61E+03	1.66E+03	8.04E+02	8.43E+02	1.14E+03	5.29E+02	4.17E+02
	STD	1.48E+02	2.25E+02	1.86E+02	2.70E+02	1.84E+02	2.69E+02	2.77E+02	2.62E+02	2.37E+02	1.77E+02
	BEST	4.66E+00	2.43E+02	4.73E+00	1.04E+03	1.28E+03	3.03E+02	2.24E+02	4.76E+02	2.43E+02	1.20E+02
	RANK	1	5	2	9	10	6	7	8	4	3
C19F8	AVE	2.76E+00	3.48E+00	2.55E+00	4.78E+00	4.55E+00	3.64E+00	4.23E+00	4.18E+00	2.97E+00	2.83E+00
	STD	5.00E-01	6.81E-01	3.46E-01	2.21E-01	1.75E-01	5.87E-01	3.59E-01	3.33E-01	5.99E-01	8.18E-01
	BEST	1.77E+00	1.61E+00	1.90E+00	4.28E+00	4.24E+00	2.36E+00	3.33E+00	3.53E+00	1.45E+00	1.15E+00
	RANK	2	5	1	10	9	6	8	7	4	3
C19F9	AVE	1.09E+00	1.05E+00	1.08E+00	2.66E+00	2.84E+00	1.23E+00	1.41E+00	1.38E+00	1.09E+00	1.10E+00
	STD	4.64E-02	3.25E-02	2.09E-02	7.74E-01	7.66E-01	9.67E-02	1.85E-01	1.49E-01	5.08E-02	3.54E-02
	BEST	1.04E+00	1.01E+00	1.04E+00	1.12E+00	1.17E+00	1.05E+00	1.11E+00	1.12E+00	1.05E+00	1.05E+00
	RANK	3	1	2	9	10	6	8	7	4	5
C19F10	AVE	1.19E+00	2.10E+01	1.18E+01	2.12E+01	2.12E+01	2.11E+01	2.10E+01	2.10E+01	2.06E+01	1.85E+01
	STD	1.07E+00	4.62E-02	1.01E+01	1.07E-01	7.16E-02	7.62E-02	3.98E-02	5.36E-01	3.37E+00	6.97E+00
	BEST	1.00E+00	2.10E+01	1.00E+00	2.10E+01	2.11E+01	2.10E+01	2.09E+01	1.82E+01	2.78E+00	1.00E+00
	RANK	1	6	2	10	9	8	5	7	4	3
Mean RANK		1.3	5.2	1.8	8.2	8.1	6.7	5.6	7.9	4.7	4.5
Total RANK		1	5	2	10	9	7	6	8	4	6

Table 4
Quantitative analysis for 10 CEC2020 test suite with $m = 10$.

Fun.	Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
C20F1	AVE	1.00E+02	7.39E+02	1.00E+02	5.74E+09	6.17E+09	4.37E+03	6.56E+04	7.00E+03	9.42E+05	4.02E+07
	STD	0.00E+00	6.90E+02	0.00E+00	2.50E+09	2.67E+09	4.16E+03	5.01E+04	1.22E+04	4.98E+06	2.07E+08
	BEST	1.00E+02	1.10E+02	1.00E+02	4.71E+08	1.10E+09	1.34E+02	5.63E+03	2.91E+02	3.32E+02	1.76E+02
	RANK	1	3	1	9	10	4	6	5	7	8
C20F2	AVE	1.20E+03	1.69E+03	1.21E+03	2.54E+03	2.36E+03	1.63E+03	1.81E+03	1.96E+03	1.49E+03	1.34E+03
	STD	8.63E+01	2.41E+02	7.35E+01	1.87E+02	1.99E+02	2.53E+02	2.44E+02	3.29E+02	2.17E+02	1.55E+02
	BEST	1.10E+03	1.33E+03	1.11E+03	2.09E+03	1.86E+03	1.25E+03	1.29E+03	1.37E+03	1.14E+03	1.12E+03
	RANK	1	6	2	10	9	5	7	8	4	3
C20F3	AVE	7.18E+02	7.19E+02	7.19E+02	7.89E+02	7.85E+02	7.25E+02	7.58E+02	7.71E+02	7.24E+02	7.16E+02
	STD	3.83E+00	5.34E+00	4.13E+00	2.08E+01	1.56E+01	8.12E+00	1.37E+01	2.37E+01	8.07E+00	2.74E+00
	BEST	7.13E+02	7.03E+02	7.12E+02	7.48E+02	7.53E+02	7.06E+02	7.31E+02	7.30E+02	7.15E+02	7.06E+02
	RANK	2	4	3	10	9	6	7	8	5	1
C20F4	AVE	1.90E+03	1.90E+03	1.90E+03	2.22E+04	1.03E+04	1.90E+03	1.91E+03	1.90E+03	1.90E+03	1.90E+03
	STD	2.60E-01	6.59E-01	2.54E-01	2.62E+04	9.30E+03	2.13E+00	2.68E+00	1.73E+00	6.75E-01	8.86E-01
	BEST	1.90E+03	1.90E+03	1.90E+03	2.18E+03	1.98E+03	1.90E+03	1.90E+03	1.90E+03	1.90E+03	1.90E+03
	RANK	2	3	1	10	9	7	8	6	4	5
C20F5	AVE	1.88E+03	2.22E+03	1.72E+03	3.63E+05	1.19E+05	9.20E+03	6.40E+03	6.08E+04	5.10E+03	4.16E+03

	STD	1.43E+02	2.50E+02	2.60E+01	1.77E+05	1.25E+05	1.09E+04	2.94E+03	9.56E+04	2.64E+03	3.63E+03
	BEST	1.71E+03	1.78E+03	1.70E+03	8.07E+03	4.43E+03	1.90E+03	2.18E+03	2.56E+03	2.29E+03	1.92E+03
	RANK	2	3	1	10	9	7	6	8	5	4
C20F6	AVE	1.60E+03	1.73E+03	1.60E+03	2.13E+03	1.93E+03	1.73E+03	1.72E+03	1.72E+03	1.69E+03	1.71E+03
	STD	4.50E-01	1.14E+02	3.83E+00	2.04E+02	9.50E+01	1.02E+02	8.93E+01	8.84E+01	7.54E+01	9.85E+01
	BEST	1.60E+03	1.60E+03	1.60E+03	1.79E+03	1.72E+03	1.60E+03	1.60E+03	1.60E+03	1.60E+03	1.60E+03
	RANK	1	8	2	10	9	7	6	5	3	4
C20F7	AVE	2.13E+03	2.32E+03	2.10E+03	1.67E+05	3.70E+03	4.66E+03	6.32E+03	1.42E+04	6.46E+03	2.25E+03
	STD	4.17E+01	1.78E+02	1.24E+01	3.10E+05	8.54E+02	7.20E+03	3.25E+03	1.05E+04	4.38E+03	1.60E+02
	BEST	2.10E+03	2.10E+03	2.10E+03	2.52E+03	2.48E+03	2.12E+03	2.60E+03	4.17E+03	2.25E+03	2.10E+03
	RANK	2	4	1	10	5	6	7	9	8	3
C20F8	AVE	2.25E+03	2.30E+03	2.29E+03	2.67E+03	2.68E+03	2.30E+03	2.31E+03	2.31E+03	2.30E+03	2.30E+03
	STD	4.57E+01	1.96E+01	2.98E+01	1.63E+02	1.67E+02	2.39E+01	4.97E+00	1.73E+01	3.60E+00	2.71E+01
	BEST	2.20E+03	2.23E+03	2.20E+03	2.40E+03	2.33E+03	2.23E+03	2.30E+03	2.23E+03	2.30E+03	2.20E+03
	RANK	1	4	2	9	10	5	7	8	3	6
C20F9	AVE	2.50E+03	2.74E+03	2.54E+03	2.81E+03	2.77E+03	2.59E+03	2.79E+03	2.73E+03	2.73E+03	2.70E+03
	STD	0.00E+00	6.56E+01	1.03E+02	9.81E+01	9.96E+01	1.24E+02	6.24E+01	9.40E+01	4.53E+01	1.10E+02
	BEST	2.50E+03	2.50E+03	2.40E+03	2.54E+03	2.56E+03	2.50E+03	2.50E+03	2.50E+03	2.50E+03	2.40E+03
	RANK	1	7	2	10	8	3	9	6	5	4
C20F10	AVE	2.90E+03	2.92E+03	2.92E+03	3.21E+03	3.20E+03	2.93E+03	2.93E+03	2.92E+03	2.93E+03	2.92E+03
	STD	1.37E+01	3.19E+01	2.26E+01	1.28E+02	1.30E+02	2.52E+01	2.51E+01	6.80E+01	1.49E+01	2.22E+01
	BEST	2.90E+03	2.90E+03	2.90E+03	3.00E+03	3.01E+03	2.90E+03	2.90E+03	2.60E+03	2.90E+03	2.90E+03
	RANK	1	4	3	10	9	8	7	5	6	2
Mean RANK		1.4	4.6	1.8	9.8	8.7	5.8	7.0	6.8	5.0	4.0
Total RANK		1	4	2	10	9	6	8	7	5	3

Table 5
Quantitative analysis for 12 CEC2022 test suite with $m = 10$.

Fun.	Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
C22F1	AVE	3.00E+02	3.00E+02	3.00E+02	6.93E+03	5.42E+03	3.00E+02	3.00E+02	2.13E+03	5.79E+02	3.00E+02
	STD	0.00E+00	0.00E+00	0.00E+00	1.47E+03	1.54E+03	0.00E+00	1.13E-01	1.30E+03	6.77E+02	0.00E+00
	BEST	3.00E+02	3.00E+02	3.00E+02	4.52E+03	2.69E+03	3.00E+02	3.00E+02	4.23E+02	3.00E+02	3.00E+02
	RANK	1	1	1	10	9	1	6	8	7	1
C22F2	AVE	4.00E+02	4.11E+02	4.02E+02	1.06E+03	9.38E+02	4.08E+02	4.08E+02	4.17E+02	4.13E+02	4.16E+02
	STD	1.01E+00	1.64E+01	2.92E+00	4.27E+02	2.94E+02	1.87E+00	1.25E+01	2.58E+01	1.22E+01	1.73E+01
	BEST	4.00E+02	4.00E+02	4.00E+02	5.13E+02	5.79E+02	4.00E+02	4.00E+02	4.00E+02	4.02E+02	4.00E+02
	RANK	1	5	2	10	9	3	4	8	6	7
C22F3	AVE	6.00E+02	6.01E+02	6.00E+02	6.38E+02	6.41E+02	6.02E+02	6.12E+02	6.21E+02	6.00E+02	6.00E+02
	STD	0.00E+00	1.04E+00	5.35E-05	1.03E+01	6.99E+00	2.94E+00	1.09E+01	1.02E+01	6.62E-01	0.00E+00
	BEST	6.00E+02	6.00E+02	6.00E+02	6.19E+02	6.20E+02	6.00E+02	6.00E+02	6.05E+02	6.00E+02	6.00E+02
	RANK	1	5	3	9	10	6	7	8	4	1
C22F4	AVE	8.11E+02	8.12E+02	8.12E+02	8.43E+02	8.42E+02	8.26E+02	8.24E+02	8.31E+02	8.10E+02	8.11E+02
	STD	4.35E+00	5.79E+00	5.50E+00	7.65E+00	9.08E+00	8.84E+00	9.01E+00	1.14E+01	3.77E+00	4.63E+00
	BEST	8.04E+02	8.04E+02	8.04E+02	8.21E+02	8.20E+02	8.08E+02	8.09E+02	8.10E+02	8.03E+02	8.03E+02
	RANK	2	5	4	10	9	7	6	8	1	3
C22F5	AVE	9.00E+02	9.01E+02	9.00E+02	1.27E+03	1.27E+03	9.07E+02	1.27E+03	1.20E+03	9.01E+02	9.00E+02
	STD	0.00E+00	1.27E+00	1.16E-01	1.72E+02	1.44E+02	1.48E+01	1.80E+02	2.29E+02	6.67E-01	1.28E-01
	BEST	9.00E+02	9.00E+02	9.00E+02	9.84E+02	1.01E+03	9.00E+02	9.03E+02	9.12E+02	9.00E+02	9.00E+02
	RANK	1	5	2	9	8	6	10	7	4	3
C22F6	AVE	1.80E+03	1.96E+03	1.80E+03	6.02E+06	2.28E+06	4.12E+03	3.77E+03	3.56E+03	5.67E+03	5.04E+03
	STD	6.09E+00	5.32E+02	5.37E-01	1.03E+07	2.55E+06	2.29E+03	1.94E+03	2.05E+03	2.05E+03	2.69E+03
	BEST	1.80E+03	1.81E+03	1.80E+03	5.88E+03	5.16E+03	1.92E+03	1.87E+03	1.90E+03	2.00E+03	1.84E+03
	RANK	2	3	1	10	9	6	5	4	8	7
C22F7	AVE	2.00E+03	2.02E+03	2.00E+03	2.09E+03	2.07E+03	2.02E+03	2.03E+03	2.05E+03	2.02E+03	2.01E+03
	STD	1.58E-01	6.97E+00	7.42E+00	2.18E+01	1.17E+01	7.16E+00	5.85E+00	1.69E+01	9.60E+00	1.01E+01
	BEST	2.00E+03	2.00E+03	2.00E+03	2.05E+03	2.05E+03	2.00E+03	2.00E+03	2.02E+03	2.00E+03	2.00E+03
	RANK	1	4	2	10	9	5	7	8	6	3
C22F8	AVE	2.20E+03	2.22E+03	2.20E+03	2.24E+03	2.23E+03	2.22E+03	2.23E+03	2.23E+03	2.22E+03	2.22E+03
	STD	3.75E+00	7.94E+00	7.53E+00	1.87E+01	5.51E+00	5.18E+00	4.68E+00	5.26E+00	7.38E+00	8.16E+00
	BEST	2.20E+03	2.20E+03	2.20E+03	2.23E+03	2.22E+03	2.20E+03	2.20E+03	2.21E+03	2.20E+03	2.20E+03
	RANK	1	5	2	10	9	3	7	8	4	6
C22F9	AVE	2.53E+03	2.53E+03	2.53E+03	2.72E+03	2.71E+03	2.53E+03	2.53E+03	2.54E+03	2.53E+03	2.55E+03
	STD	0.00E+00	2.68E+01	0.00E+00	2.72E+01	3.28E+01	2.68E+01	9.13E-04	3.73E+01	9.59E+00	4.45E+01
	BEST	2.53E+03	2.53E+03	2.53E+03	2.66E+03	2.65E+03	2.53E+03	2.53E+03	2.53E+03	2.53E+03	2.53E+03
	RANK	1	5	1	10	9	5	3	7	4	8
C22F10	AVE	2.50E+03	2.55E+03	2.50E+03	2.60E+03	2.57E+03	2.52E+03	2.56E+03	2.53E+03	2.53E+03	2.54E+03
	STD	4.70E-02	5.94E+01	5.52E-02	7.38E+01	6.99E+01	4.25E+01	6.77E+01	8.46E+01	5.00E+01	5.30E+01
	BEST	2.50E+03	2.50E+03	2.50E+03	2.51E+03	2.51E+03	2.50E+03	2.50E+03	2.50E+03	2.50E+03	2.50E+03
	RANK	1	7	2	10	9	3	8	5	4	6
C22F11	AVE	2.60E+03	2.85E+03	2.62E+03	3.43E+03	3.38E+03	2.78E+03	2.75E+03	2.86E+03	2.89E+03	2.61E+03
	STD	0.00E+00	2.17E+02	6.04E+01	3.40E+02	3.47E+02	1.91E+02	1.41E+02	1.06E+02	1.26E+02	2.75E+01
	BEST	2.60E+03	2.60E+03	2.60E+03	2.96E+03	2.80E+03	2.60E+03	2.60E+03	2.60E+03	2.60E+03	2.60E+03

	RANK	1	6	3	10	9	5	4	7	8	2
C22F12	AVE	2.86E+03	2.86E+03	2.86E+03	2.98E+03	2.92E+03	2.86E+03	2.88E+03	2.87E+03	2.86E+03	2.87E+03
	STD	1.55E+00	2.04E+00	1.92E+00	6.14E+01	2.86E+01	2.53E+00	1.68E+01	1.19E+01	1.95E+00	7.13E+00
	BEST	2.86E+03	2.86E+03	2.86E+03	2.89E+03	2.87E+03	2.86E+03	2.86E+03	2.86E+03	2.86E+03	2.86E+03
	RANK	1	4	2	10	9	5	8	7	3	6
	Mean RANK	1.167	4.583	2.083	9.917	9.0	4.583	5.667	7.083	4.917	4.417
	Total RANK	1	4	2	10	9	4	7	8	6	3

In convergence analysis, convergence curves of CEC2019 test suite, CEC2020 test suite with $m = 10$ and 20, CEC2022 test suite with $m = 10$ and 20 are shown in Fig. 5-Fig. 9, respectively. Among these functions, VOA can quickly find the great solution on C19F1, C19F2, C19F5, C19F6, C19F9, C19F10, C20F1, C20F3, C20F4, C20F6, C20F8-C20F10 and C22F1-C22F12. In addition, VOA can continuously explore the global best solution through the local best solution on C19F3, C19F4, C19F7, C19F8, C20F2, C20F5 and C20F7. Overall, VOA demonstrates excellent convergence speed and results, except on Fig. 7.C20F5 and Fig. 9.C22F5.

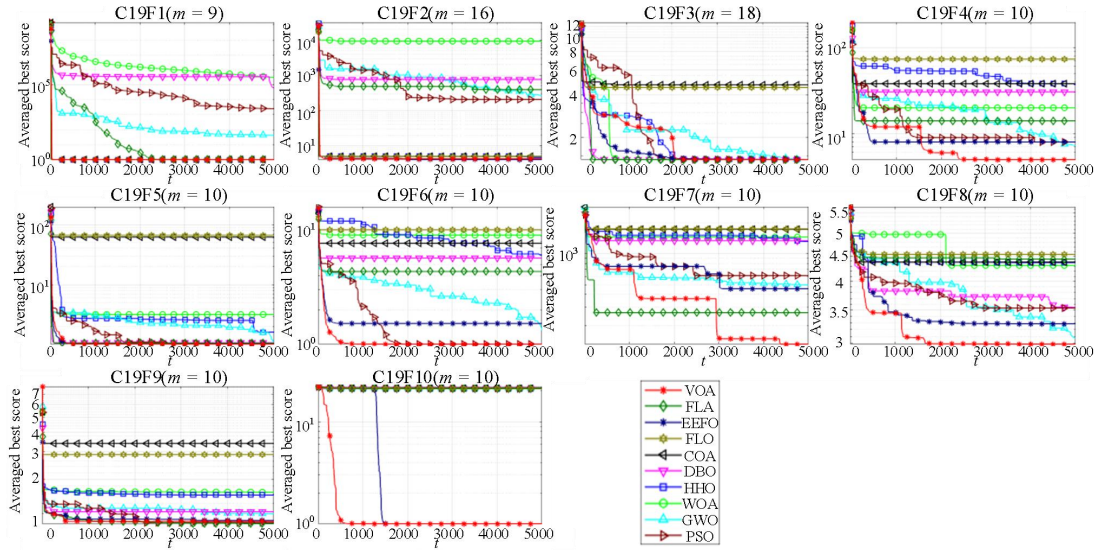


Fig. 5 Convergence analysis for CEC2019.

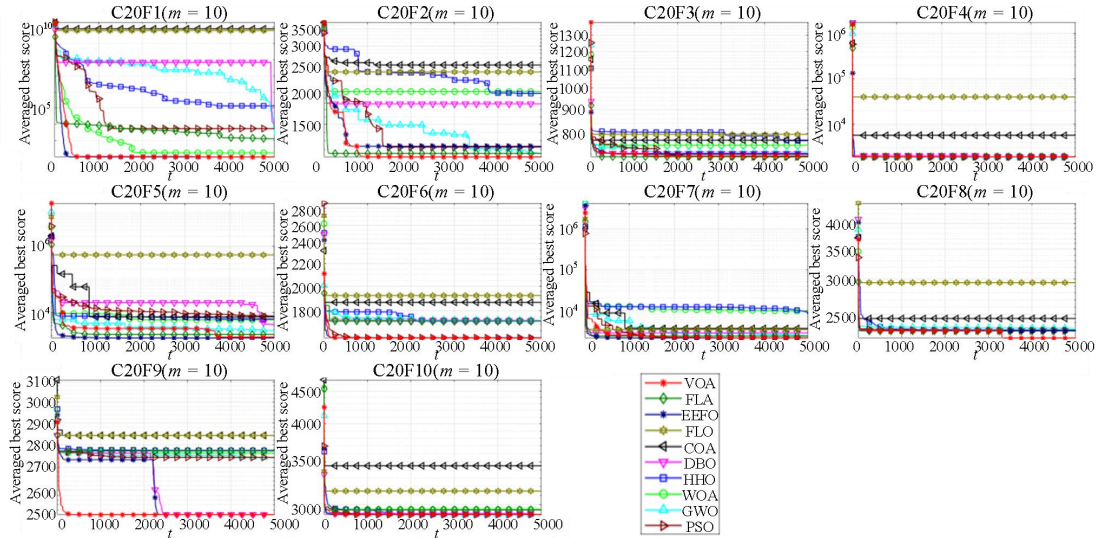


Fig. 6 Convergence analysis for CEC2020 with $m = 10$.

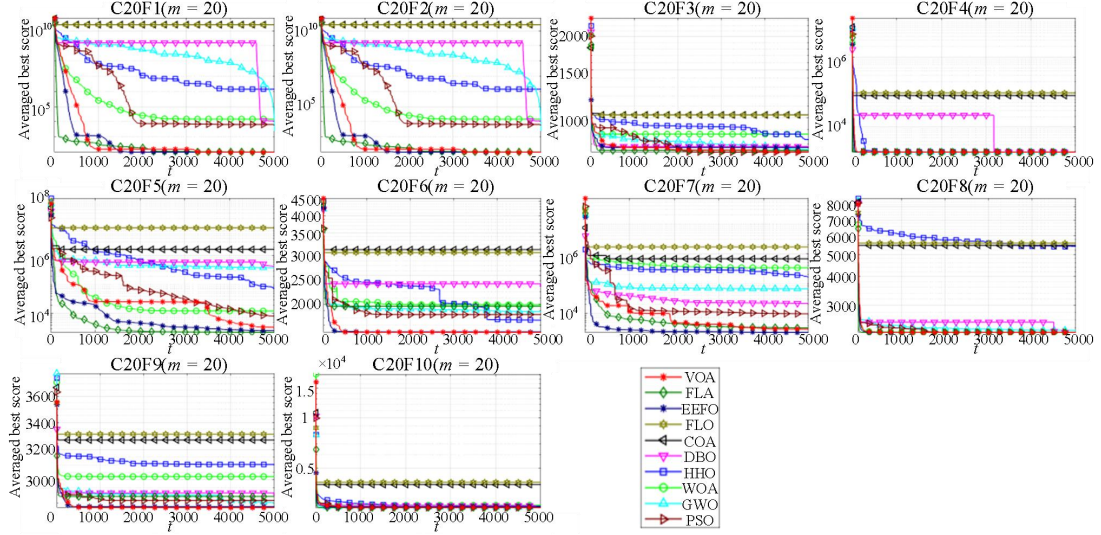


Fig. 7 Convergence analysis for CEC2020 with $m = 20$.

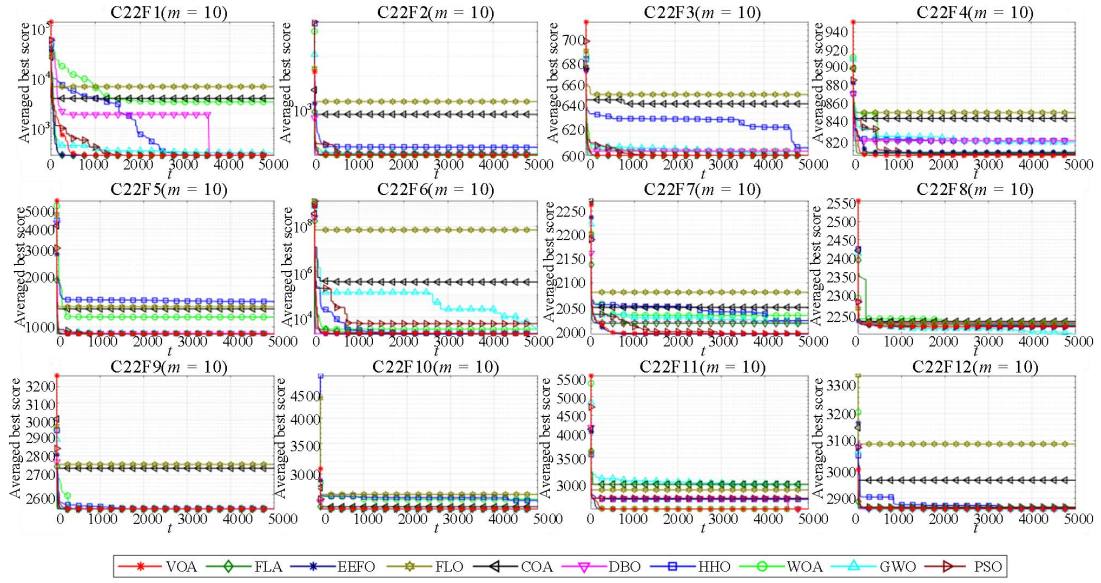


Fig. 8 Convergence analysis for CEC2022 with $m = 10$.

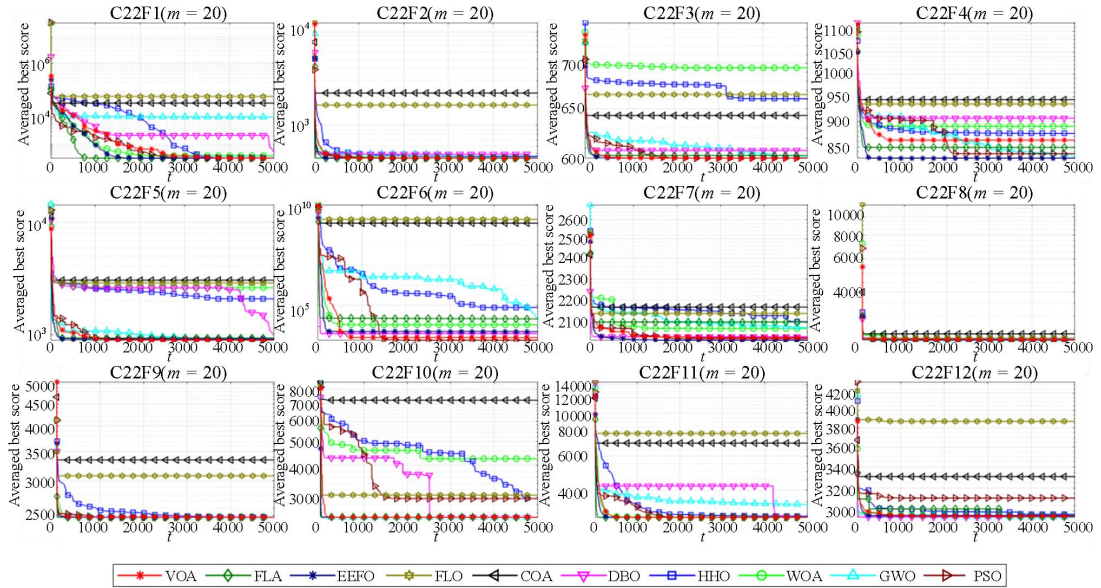


Fig. 9 Convergence analysis for CEC2022 with $m = 20$.

1.2 Statistical analysis

P-value index obtained from Wilcoxon rank sum test is applied to evaluate the differences between VOA and each compared optimizer. The P-value index of 30 repeated runs for each comparison is calculated. When P-value is less than 0.05, VOA has significant difference compared with other optimizers. ‘+’ indicates that VOA has better performance, while ‘-’ indicates that VOA performs worse. ‘+=’ or ‘-=’ signify the P-value is greater than 0.05, which indicates the results of VOA and the compared optimizer are similar, suggesting that they may converge around the same optimum. ‘=’ indicates VOA and compared optimizer have the same result, i.e., they may find the global best solution on this test function at the same time. In statistical analysis, P-values of CEC2017 test suite with $m = 10$, CEC2019 test suite, CEC2020 test suite with $m = 10$ and CEC2022 test suite with $m = 10$ are listed in Table 6-Table 9, respectively.

Table 6
P-values on CEC2017 test suite with $m = 10$ (C17F1-C17F30)

Fun.	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
C17F1	1.21E-12	+ 1.00E+00	= 1.21E-12	+ 1.21E-12	+ 1.18E-12	+ 1.21E-12	+ 1.21E-12	+ 1.21E-12	+ 1.05E-12
C17F3	1.00E+00	= 1.00E+00	= 1.21E-12	+ 1.21E-12	+ 1.00E+00	= 1.21E-12	+ 1.21E-12	+ 1.21E-12	+ 1.00E+00
C17F4	7.73E-08	- 6.12E-10	- 3.02E-11	+ 3.02E-11	+ 7.66E-05	+ 2.37E-10	+ 3.16E-10	+ 3.02E-11	+ 1.77E-11
C17F5	4.37E-10	+ 7.06E-03	+ 2.87E-11	+ 2.87E-11	+ 7.70E-11	+ 2.87E-11	+ 2.87E-11	+ 7.97E-07	+ 2.25E-03
C17F6	1.65E-11	+ 4.71E-08	+ 1.21E-12	+ 1.21E-12	+ 4.57E-12	+ 1.21E-12	+ 1.21E-12	+ 1.21E-12	+ 1.61E-01
C17F7	9.33E-02	+= 4.92E-01	+= 3.02E-11	+ 3.02E-11	+ 5.61E-05	+ 3.02E-11	+ 3.02E-11	+ 3.18E-03	+ 2.89E-03
C17F8	3.01E-07	+ 2.77E-01	+= 2.92E-11	+ 2.92E-11	+ 8.98E-10	+ 3.23E-11	+ 2.92E-11	+ 1.56E-02	+ 5.79E-03
C17F9	6.59E-04	+ 1.00E+00	= 1.21E-12	+ 1.21E-12	+ 2.91E-05	+ 1.21E-12	+ 1.21E-12	+ 1.21E-12	+ 1.00E+00
C17F10	3.46E-10	+ 5.44E-01	+= 3.01E-11	+ 3.01E-11	+ 8.87E-10	+ 1.61E-10	+ 3.46E-10	+ 5.96E-05	+ 2.98E-06
C17F11	9.81E-11	+ 5.50E-04	+ 3.00E-11	+ 3.00E-11	+ 4.47E-11	+ 3.00E-11	+ 3.00E-11	+ 3.00E-11	+ 2.19E-07
C17F12	3.02E-11	+ 1.84E-02	- 3.02E-11	+ 3.02E-11	+ 3.02E-11	+ 3.02E-11	+ 3.02E-11	+ 3.02E-11	+ 5.49E-11
C17F13	4.29E-01	+= 2.37E-10	- 3.02E-11	+ 3.02E-11	+ 1.86E-09	+ 3.02E-11	+ 3.34E-11	+ 3.02E-11	+ 3.82E-10
C17F14	3.16E-10	+ 1.09E-08	- 3.02E-11	+ 3.02E-11	+ 2.39E-08	+ 3.02E-11	+ 3.02E-11	+ 3.02E-11	+ 7.29E-04
C17F15	1.76E-01	+= 4.98E-11	- 3.02E-11	+ 3.02E-11	+ 1.87E-07	+ 1.78E-10	+ 3.02E-11	+ 5.07E-10	+ 2.51E-02
C17F16	7.39E-11	+ 1.30E-01	-= 3.02E-11	+ 3.02E-11	+ 3.02E-11	+ 3.02E-11	+ 3.02E-11	+ 3.02E-11	+ 5.53E-08
C17F17	1.69E-08	+ 4.20E-04	- 3.01E-11	+ 3.01E-11	+ 3.01E-11	+ 3.01E-11	+ 3.01E-11	+ 2.15E-10	+ 5.17E-07
C17F18	2.53E-04	+ 3.02E-11	- 3.02E-11	+ 3.02E-11	+ 4.98E-11	+ 3.02E-11	+ 3.34E-11	+ 3.02E-11	+ 2.61E-10
C17F19	6.36E-05	+ 2.71E-11	- 3.02E-11	+ 3.02E-11	+ 1.49E-06	+ 3.02E-11	+ 3.02E-11	+ 1.86E-09	+ 1.73E-06
C17F20	4.10E-12	+ 7.82E-07	+ 4.10E-12	+ 4.10E-12	+ 4.10E-12	+ 4.10E-12	+ 4.10E-12	+ 4.10E-12	+ 2.99E-11
C17F21	6.05E-10	+ 1.61E-01	+= 1.21E-12	+ 1.21E-12	+ 5.77E-11	+ 1.21E-12	+ 1.21E-12	+ 1.21E-12	+ 1.95E-09
C17F22	3.54E-08	+ 1.61E-03	+ 2.72E-11	+ 2.72E-11	+ 2.16E-10	+ 1.61E-10	+ 1.61E-10	+ 3.17E-10	+ 1.29E-09
C17F23	6.70E-11	+ 9.52E-04	+ 3.02E-11	+ 3.02E-11	+ 3.34E-11	+ 3.02E-11	+ 3.02E-11	+ 1.73E-06	+ 7.39E-11
C17F24	7.74E-12	+ 5.24E-05	+ 2.41E-12	+ 4.21E-12	+ 4.80E-05	+ 3.37E-12	+ 3.02E-12	+ 1.75E-11	+ 7.56E-10
C17F25	6.45E-06	+ 1.53E-02	+ 2.84E-11	+ 2.84E-11	+ 4.45E-07	+ 1.93E-08	+ 4.77E-09	+ 1.34E-09	+ 2.60E-04
C17F26	3.36E-07	+ 1.10E-02	+ 6.43E-12	+ 6.43E-12	+ 2.84E-07	+ 8.46E-10	+ 1.04E-08	+ 6.43E-12	+ 3.58E-05
C17F27	5.84E-06	+ 6.48E-07	+ 3.01E-11	+ 3.01E-11	+ 7.07E-08	+ 3.01E-11	+ 1.20E-10	+ 3.36E-05	+ 2.40E-09
C17F28	7.97E-07	+ 1.90E-01	+= 1.72E-12	+ 1.72E-12	+ 6.12E-12	+ 1.72E-12	+ 1.72E-12	+ 1.72E-12	+ 2.21E-10
C17F29	7.74E-06	+ 1.33E-02	- 3.02E-11	+ 3.02E-11	+ 1.02E-05	+ 4.98E-11	+ 3.02E-11	+ 7.28E-01	-= 4.73E-01
C17F30	2.06E-01	-= 5.75E-02	+= 3.02E-11	+ 2.86E-11	+ 2.86E-10	+ 3.02E-11	+ 3.02E-11	+ 3.02E-11	+ 9.64E-10

Table 7
P-values on CEC2019 test suite (C19F1-C19F10).

Fun.	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
C19F1	1.21E-12	+ 1.00E+00	= 1.00E+00	= 1.00E+00	= 1.93E-10	= 1.00E+00	= 1.21E-12	+ 1.21E-12	+ 1.21E-12
C19F2	3.02E-11	+ 1.54E-01	+= 3.16E-12	+ 1.72E-12	+ 3.16E-10	+ 2.92E-11	+ 3.02E-11	+ 3.02E-11	+ 3.02E-11
C19F3	3.72E-07	- 3.37E-02	+ 3.01E-11	+ 3.01E-11	+ 1.76E-03	+ 3.01E-11	+ 8.46E-09	+ 1.95E-03	+ 5.16E-06
C19F4	1.15E-09	+ 3.48E-05	+ 2.98E-11	+ 2.98E-11	+ 9.78E-11	+ 3.64E-11	+ 2.98E-11	+ 1.33E-05	+ 1.55E-01
C19F5	5.61E-05	+ 8.88E-01	-= 3.02E-11	+ 3.02E-11	+ 6.91E-04	+ 3.02E-11	+ 8.99E-11	+ 4.98E-11	+ 3.88E-04
C19F6	4.57E-12	+ 1.27E-05	+ 1.21E-12	+ 1.21E-12	+ 1.21E-12	+ 1.21E-12	+ 1.21E-12	+ 1.21E-12	+ 3.45E-07
C19F7	3.82E-09	+ 9.35E-01	-= 3.02E-11	+ 3.02E-11	+ 2.61E-10	+ 1.29E-09	+ 4.50E-11	+ 4.12E-06	+ 9.52E-04
C19F8	3.16E-05	+ 8.24E-02	-= 3.02E-11	+ 3.02E-11	+ 6.53E-07	+ 5.49E-11	+ 4.98E-11	+ 1.41E-01	+= 9.82E-01
C19F9	8.88E-06	- 4.29E-01	-= 6.07E-11	+ 4.50E-11	+ 2.39E-08	+ 2.15E-10	+ 8.99E-11	+ 5.69E-01	-= 1.30E-01
C19F10	1.62E-11	+ 5.63E-03	+ 1.62E-11	+ 1.62E-11	+ 1.62E-11	+ 1.62E-11	+ 1.62E-11	+ 1.80E-11	+ 4.53E-08

Table 8
P-values on CEC2020 test suite with $m = 10$ (C20F1-C20F10).

Fun.	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
C20F1	1.21E-12	+ 1.00E+00	- 1.21E-12	+ 1.21E-12	+ 1.15E-12	+ 1.21E-12	+ 1.21E-12	+ 1.21E-12	+ 1.13E-12
C20F2	1.21E-10	+ 3.52E-01	+= 3.02E-11	+ 3.02E-11	+ 1.21E-10	+ 5.49E-11	+ 3.34E-11	+ 5.00E-09	+ 6.36E-05
C20F3	1.58E-01	+= 2.84E-01	+= 3.02E-11	+ 3.02E-11	+ 3.37E-05	+ 3.02E-11	+ 3.02E-11	+ 1.86E-03	+ 1.33E-02
C20F4	1.33E-01	+= 2.97E-01	+= 3.02E-11	+ 3.02E-11	+ 4.57E-09	+ 3.02E-11	+ 4.50E-11	+ 2.78E-07	+ 3.09E-06
C20F5	2.38E-07	+ 1.55E-09	- 3.02E-11	+ 3.02E-11	+ 1.21E-10	+ 3.34E-11	+ 3.02E-11	+ 3.02E-11	+ 6.52E-09
C20F6	6.70E-11	+ 3.79E-01	+= 3.02E-11	+ 3.02E-11	+ 6.70E-11	+ 4.08E-11	+ 4.08E-11	+ 5.49E-11	+ 4.62E-10
C20F7	1.10E-08	+ 2.20E-07	- 3.02E-11	+ 3.02E-11	+ 1.61E-10	+ 3.02E-11	+ 3.02E-11	+ 3.02E-11	+ 4.94E-05
C20F8	7.58E-08	+ 5.79E-04	+ 2.95E-11	+ 2.95E-11	+ 8.23E-08	+ 2.95E-11	+ 2.00E-09	+ 7.00E-09	+ 5.87E-08
C20F9	1.66E-11	+ 7.62E-02	+= 1.21E-12	+ 1.21E-12	+ 1.27E-05	+ 1.21E-12	+ 1.21E-12	+ 1.21E-12	+ 1.28E-08
C20F10	1.17E-03	+ 2.21E-03	+ 2.88E-11	+ 2.88E-11	+ 4.47E-07	+ 6.35E-07	+ 1.97E-07	+ 8.93E-09	+ 1.56E-04

Table 9
P-values on CEC2022 test suite with $m = 10$ (C22F1-C22F12).

Fun.	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
C22F1	1.00E+00	= 1.00E+00	= 1.21E-12	+ 1.21E-12	+ 1.00E+00	= 1.21E-12	+ 1.21E-12	+ 1.21E-12	+ 1.00E+00
C22F2	1.43E-09	+ 9.82E-01	+= 3.02E-11	+ 3.02E-11	+ 3.50E-11	+ 1.46E-10	+ 9.92E-11	+ 5.49E-11	+ 9.65E-11
C22F3	4.56E-12	+ 5.32E-06	+ 1.21E-12	+ 1.21E-12	+ 1.64E-11	+ 1.21E-12	+ 1.21E-12	+ 1.21E-12	+ 1.00E+00
C22F4	7.44E-01	-= 6.36E-01	-= 3.27E-11	+ 3.27E-11	+ 8.71E-09	+ 2.56E-08	+ 4.13E-10	+ 1.26E-01	-= 2.97E-01
C22F5	8.65E-07	+ 2.15E-02	+ 1.21E-12	+ 1.21E-12	+ 5.76E-11	+ 1.21E-12	+ 1.21E-12	+ 1.21E-12	+ 1.10E-02
C22F6	2.61E-10	+ 2.39E-08	- 3.02E-11	+ 3.02E-11	+ 3.02E-11	+ 3.02E-11	+ 3.02E-11	+ 3.02E-11	+ 3.02E-11
C22F7	2.99E-11	+ 1.76E-01	+= 2.99E-11	+ 2.99E-11	+ 2.99E-11	+ 2.99E-11	+ 2.99E-11	+ 3.66E-11	+ 1.60E-10
C22F8	6.05E-07	+ 1.50E-02	- 3.02E-11	+ 3.02E-11	+ 6.70E-11	+ 3.69E-11	+ 3.69E-11	+ 2.15E-10	+ 6.74E-06
C22F9	1.00E+00	= 1.00E+00	= 1.21E-12	+ 1.21E-12	+ 3.06E-04	+ 1.21E-12	+ 1.21E-12	+ 1.21E-12	+ 2.78E-03
C22F10	4.55E-01	+= 2.40E-01	+= 3.02E-11	+ 3.02E-11	+ 3.26E-07	+ 3.02E-11	+ 3.02E-11	+ 1.60E-07	+ 2.39E-04
C22F11	2.42E-09	+ 1.61E-01	+= 1.21E-12	+ 1.21E-12	+ 1.00E-07	+ 1.21E-12	+ 1.21E-12	+ 1.21E-12	+ 3.34E-01
C22F12	1.85E-08	+ 6.38E-09	+ 2.86E-11	+ 2.86E-11	+ 5.24E-09	+ 6.36E-11	+ 6.36E-11	+ 1.31E-07	+ 7.40E-10

Table 10 summarizes the P-value results for 61 functions from Table 6-Table 9. VOA significantly outperforms FLO, COA, DBO, HHO, WOA, and GWO on 57 (93.44%) or more functions. VOA also outperforms FLA and PSO with significant difference on 47 (77.05%) and 44 (72.13%) functions, respectively. VOA has 19 (31.15%) ‘+’ compared with EEFO, while the sum number of ‘+’ and ‘+=’ is 35 (57.38%), which can also be argued that VOA is slightly better than EEFO. Overall, VOA obtains significant performance in statistical analysis.

Table 10
Statistical results of Wilcoxon test based on Table 6-Table 9.

	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
	(+/+/=/-/-)	(+/+/=/-/-)	(+/+/=/-/-)	(+/+/=/-/-)	(+/+/=/-/-)	(+/+/=/-/-)	(+/+/=/-/-)	(+/+/=/-/-)	(+/+/=/-/-)
CEC2017	23/3/1/1/1	10/6/3/1/9	29/0/0/0/0	29/0/0/0/0	28/0/1/0/0	29/0/0/0/0	29/0/0/0/0	28/0/0/1/0	21/2/2/0/4
CEC2019	8/0/0/0/2	4/1/1/4/0	9/0/1/0/0	9/0/1/0/0	9/0/1/0/0	9/0/1/0/0	10/0/0/0/0	8/1/0/1/0	6/3/0/0/1
CEC2020	8/2/0/0/0	2/5/0/0/3	10/0/0/0/0	10/0/0/0/0	10/0/0/0/0	10/0/0/0/0	10/0/0/0/0	10/0/0/0/0	9/0/0/0/1
CEC2022	8/1/2/1/0	3/4/2/1/2	12/0/0/0/0	12/0/0/0/0	11/0/1/0/0	12/0/0/0/0	12/0/0/0/0	11/0/0/1/0	8/1/2/1/0
Total	47/6/3/2/3	19/16/6/6/14	60/0/1/0/0	60/0/1/0/0	58/0/3/0/0	60/0/1/0/0	61/0/0/0/0	57/1/0/3/0	44/6/4/1/6

2. Results on real-world optimization problems

This section evaluates the performance of VOA on 7 real-world optimization problems: the optimal design of industrial refrigeration system, the gas transmission compressor design, the weight minimization of a speed reducer, the multiple disk clutch brake design, the 10-bar truss optimization problem with frequency constraints, the gear train design and the rolling element bearing problem. The maximum number of iterations T are 500 and 5000. The number of population members N is 100. AVE, STD, BEST, RANK and averaged convergence curves of 30 repeated runs are still served as evaluation indexes.

These real-world optimization problems have non-linear constraints. The penalty method could calculate a high value for the infeasible solution and further discard it in the minimization optimization process. Hence, an universal real-world fitness function $F(\vec{X})$ based on some constraints are defined by Eq.(1)-Eq.(3).

$$F(\vec{X}) = F(\vec{X}) + \left(\sum_{i=1}^p k \cdot G_i(\vec{X}) + \sum_{j=1}^q k \cdot H_j(\vec{X}) \right) \quad (1)$$

$$G_i(\vec{X}) = \max(0, g_i(\vec{X}))^2 \quad (2)$$

$$H_j(\vec{X}) = h_j(\vec{X})^2 \quad (3)$$

where $\vec{X}$ stands for the optimization variables. $g_i(\vec{X})$ is the inequality constraint. $h_j(\vec{X})$ is the equality constraint. When the constraint is satisfied, $G_i(\vec{X})$ and $H_j(\vec{X})$ are equal to 0. Otherwise, they are equal to $g_i(\vec{X})^2$ and $h_j(\vec{X})^2$, respectively. p and q represent the number of inequality constraints and equality constraints, respectively. k represents a large positive number. If constraints could not be satisfied, $F(\vec{X})$ will be increased by a large positive number, causing it to become excessively large and be discarded.

2.1 The optimal design of industrial refrigeration system

The optimal design of industrial refrigeration system has 14 design variables and 15 constraints, which can be described as a complex non-linear inequality constrained optimization problem. The mathematical optimization model is defined in Eq.(4).

Minimize:

$$\begin{aligned}
F(\vec{X}) = & 63098.88x_2x_4x_{12} + 5441.5x_2^2x_{12} + 115055.5x_2^{1.664}x_6 + 6172.27x_2^2x_6 \\
& + 63098.88x_1x_3x_{11} + 5441.5x_1^2x_{11} + 115055.5x_1^{1.664}x_5 + 6172.27x_1^2x_5 \\
& + 140.53x_1x_{11} + 281.28x_3x_{11} + 70.26x_1^2 + 281.29x_1x_3 + 281.29x_3^2 \\
& + 14437x_8^{1.8812}x_{12}^{0.3424}x_{10}x_{14}x_1^2x_7x_9^{-1} + 20470.2x_7^{2.893}x_{11}^{0.316}x_1^2
\end{aligned} \tag{4}$$

Subject to:

$$\begin{aligned}
g_1(\vec{X}) &= 1.524x_7^{-1} - 1 \leq 0 \\
g_2(\vec{X}) &= 1.524x_8^{-1} - 1 \leq 0 \\
g_3(\vec{X}) &= 0.07789x_1 - 2x_7^{-1}x_9 - 1 \leq 0 \\
g_4(\vec{X}) &= 7.05305x_9^{-1}x_1^2x_{10}x_8^{-1}x_2^{-1}x_{14}^{-1} - 1 \leq 0 \\
g_5(\vec{X}) &= 0.0833x_{13}^{-1}x_{14} - 1 \leq 0 \\
g_6(\vec{X}) &= 47.136x_2^{0.333}x_{10}^{-1}x_{12} - 1.333x_8x_{13}^{2.1195} + 62.08x_{13}^{2.1195}x_{12}^{-1}x_8^{0.2}x_{10}^{-1} - 1 \leq 0 \\
g_7(\vec{X}) &= 0.04771x_{10}x_8^{1.8812}x_{12}^{0.3424} - 1 \leq 0 \\
g_8(\vec{X}) &= 0.0488x_9x_7^{1.893}x_{11}^{0.316} - 1 \leq 0 \\
g_9(\vec{X}) &= 0.0099x_1x_3^{-1} - 1 \leq 0 \\
g_{10}(\vec{X}) &= 0.0193x_2x_4^{-1} - 1 \leq 0 \\
g_{11}(\vec{X}) &= 0.0298x_1x_5^{-1} - 1 \leq 0 \\
g_{12}(\vec{X}) &= 0.056x_2x_6^{-1} - 1 \leq 0 \\
g_{13}(\vec{X}) &= 2x_9^{-1} - 1 \leq 0 \\
g_{14}(\vec{X}) &= 2x_{10}^{-1} - 1 \leq 0 \\
g_{15}(\vec{X}) &= x_{12}x_{11}^{-1} - 1 \leq 0
\end{aligned}$$

With bounds:

$$0.001 \leq x_i \leq 5, i = 1, \dots, 14$$

Table 11 presents the results of ten optimizers with $T = 500$. VOA obtains the best AVE, STD, BEST and RANK among the compared optimizers. Table 12 extends the evaluation with $T = 5000$. In this case, while DBO and PSO achieve competitive BEST index, VOA still maintains superiority with lower AVE and STD, suggesting consistent optimization stability and reliability over extended trials. These results collectively highlight the effectiveness of VOA in achieving both optimal solutions and minimized result variability.

Table 11
Quantitative analysis on the optimal design of industrial refrigeration system with $T = 500$.

Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	0.037680802	2.59936E+15	0.05083318	3.21459E+16	1.81587E+16	5.42119E+14	1.57811E+14	2.97695E+14	1.8725E+14	1.87237E+14
STD	0.012065063	6.84596E+15	0.013230239	2.02669E+16	1.87483E+16	5.65128E+14	3.5894E+14	4.66756E+14	3.80902E+14	3.80876E+14
BEST	0.032213036	0.032368132	0.036903902	2.22956E+14	11137.01942	0.058448408	0.617742086	0.1564506	0.036750902	0.032213028
RANK	1	8	2	10	9	7	3	6	4	5

Table 12
Quantitative analysis on the optimal design of industrial refrigeration system with $T = 5000$.

Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	0.035258795	2.5862E+15	0.036617624	4.02528E+16	1.84763E+16	2.80856E+14	6.24124E+13	2.18534E+14	3.12062E+13	1.56031E+14
STD	0.003097343	5.62441E+15	0.005065023	2.3585E+16	1.85559E+16	4.36349E+14	2.37518E+14	4.02899E+14	1.70924E+14	3.54861E+14
BEST	0.032213001	0.032213131	0.032259611	3065.517577	2403.214447	0.032213001	0.04978176	0.035014121	0.033529039	0.032213001
RANK	1	8	2	10	9	7	4	6	3	5

Fig. 10 shows the averaged convergence curves of ten optimizers when $T = 500$ and $T = 5000$, respectively. In Fig. 10(a), the averaged convergence curves illustrate that VOA has the fastest convergence speed and effectively finds the optimal solution. In Fig. 10(b), the compared optimizers can not effectively converge except EEFO, while VOA maintains its position as the best optimizer, consistently demonstrating superior convergence behavior even over extended iterations. This solidifies the robustness and effectiveness of VOA in achieving optimal solutions across varying iteration steps.

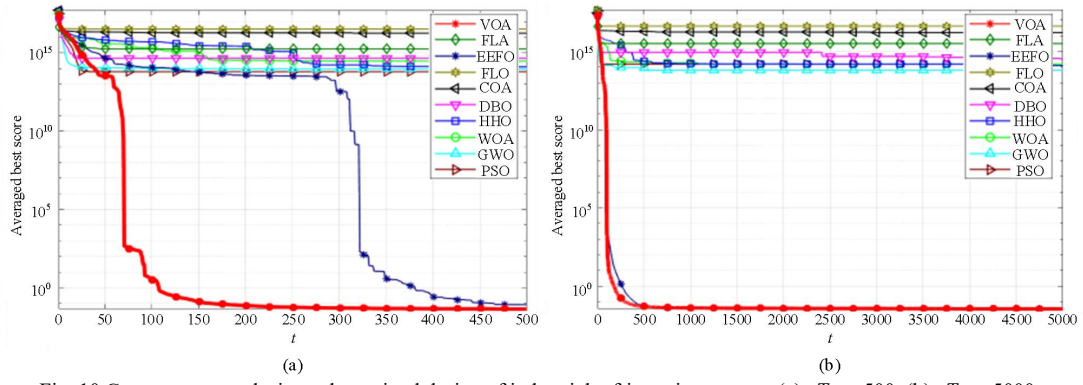


Fig. 10 Convergence analysis on the optimal design of industrial refrigeration system. (a) $T = 500$. (b) $T = 5000$.

2.2 The gas transmission compressor design

The gas transmission compressor design problem is formulated as a minimization problem with 4 design variables and 1 nonlinear constraint, as defined in Eq.(5).

Minimize:

$$F(\vec{X}) = 8.61 \times 10^5 x_1^{1/2} x_2 x_3^{-2/3} x_4^{-1/2} + 3.69 \times 10^4 x_3 + 7.72 \times 10^8 x_1^{-1} x_2^{0.219} - 765.43 \times 10^6 x_1^{-1} \quad (5)$$

Subject to:

$$g_1(\vec{X}) = x_4 x_2^{-2} + x_2^{-2} - 1 \leq 0$$

With bounds:

$$20 \leq x_1 \leq 50, 1 \leq x_2 \leq 10, 20 \leq x_3 \leq 50, 0.1 \leq x_4 \leq 60$$

Table 13 presents the results of ten optimizers with $T = 500$. Despite EEFO also achieving the best AVE and BEST values, VOA ranks first due to its lower STD.

Table 14 shows the results of optimizers with $T = 5000$. VOA, FLA, EEFO, DBO and PSO demonstrate strong performance. However, VOA maintains a lower STD, indicating more reliable outcomes. Overall, quantitative analysis confirms VOA could obtain the superior accuracy, consistency, and robustness.

Table 13
Quantitative analysis on the gas transmission compressor design with $T = 500$.

Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	2964895.416	2978020.804	2964895.416	3098543.976	3098314.538	2983507.497	3005912.102	2989634.487	2965174.375	2964989.88
STD	1.23505E-09	71867.64723	0.000342042	26248.82086	27175.74432	43232.72474	31437.22945	33824.09743	227.4656748	196.9906544
BEST	2964895.416	2964895.43	2964895.416	3017863.891	3029090.226	2964895.481	2966010.709	2965040.451	2964922.715	2964895.788
RANK	1	5	2	10	9	6	8	7	4	3

Table 14
Quantitative analysis on the gas transmission compressor design problem with $T = 5000$.

Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	2964895.416	2964895.416	2964895.416	3093039.693	3096062.369	2981630.435	2980505.516	2965322.153	2964918.853	2964895.464
STD	1.16976E-09	0.000529599	1.71422E-09	28647.05399	33077.28727	43394.79427	20815.27393	446.644167	23.35821268	0.059731752
BEST	2964895.416	2964895.416	2964895.416	3025611.631	2991574.581	2964895.416	2964999.377	2964902.201	2964898.943	2964895.416
RANK	1	3	2	9	10	8	7	6	5	4

Fig. 11 shows the averaged convergence curves. In Fig. 11 (a), for $T = 500$, optimizers like VOA, FLA, EEFO, GWO, and PSO display strong convergence curves, indicating their efficiency in finding optimal solutions within a shorter timeframe. In Fig. 11 (b), for $T = 5000$, these optimizers continue to show excellent stability and consistency. Overall, the convergence analysis demonstrates that VOA, along with FLA, EEFO, GWO, and PSO, perform well in this optimization problem.

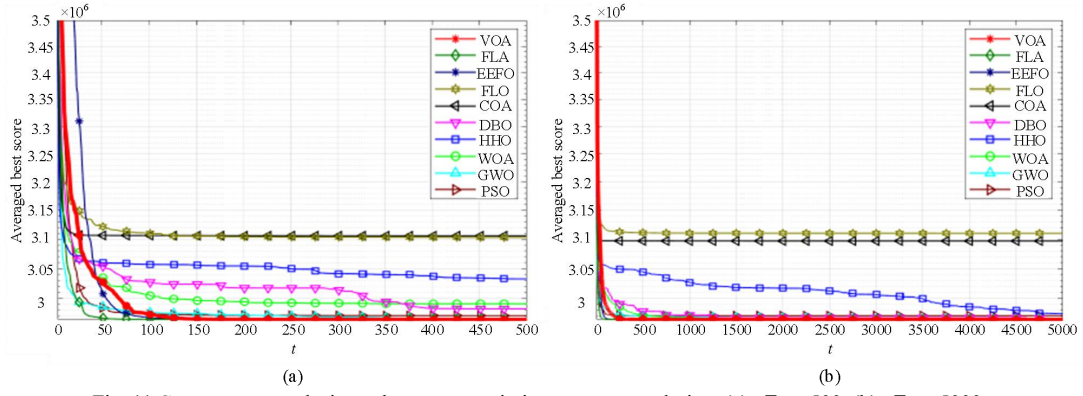


Fig. 11 Convergence analysis on the gas transmission compressor design. (a) $T = 500$. (b) $T = 5000$.

2.3 The weight minimization of a speed reducer

Fig. 12 illustrates an example about the weight minimization of a speed reducer, specifically designed for optimizing a speed reducer used in a small aircraft engine. This optimization involves 7 design variables and 11 constraints, which are selected to achieve the lightest possible configuration while maintaining necessary functional and performance criteria, as defined in Eq.(6).

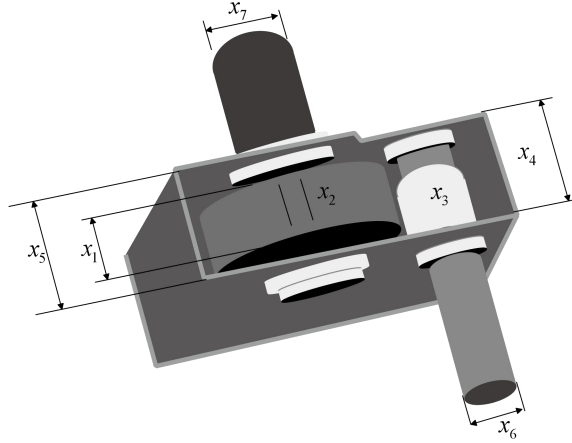


Fig. 12 The weight minimization of a speed reducer.

Minimize:

$$F(\vec{X}) = 0.7854x_2^2x_1(14.9334x_3 - 43.0934 + 3.3333x_3^2) + 0.7854(x_5x_7^2 + x_4x_6^2) - 1.508x_1(x_7^2 + x_6^2) + 7.477(x_7^3 + x_6^3) \quad (6)$$

Subject to:

$$\begin{aligned} g_1(\vec{X}) &= -x_1x_2^2x_3 + 27 \leq 0 \\ g_2(\vec{X}) &= -x_1x_2^2x_3^2 + 397.5 \leq 0 \\ g_3(\vec{X}) &= -x_2x_6^4x_3x_4^{-3} + 1.93 \leq 0 \\ g_4(\vec{X}) &= -x_2x_7^4x_3x_5^{-3} + 1.93 \leq 0 \\ g_5(\vec{X}) &= 10x_6^{-3}\sqrt{16.91 \times 10^6 + 745(x_4x_2^{-1}x_3^{-1})^2} - 1100 \leq 0 \\ g_6(\vec{X}) &= 10x_7^{-3}\sqrt{157.5 \times 10^6 + 745(x_5x_2^{-1}x_3^{-1})^2} - 850 \leq 0 \\ g_7(\vec{X}) &= x_2x_3 - 40 \leq 0 \\ g_8(\vec{X}) &= -x_1x_2^{-1} + 5 \leq 0 \\ g_9(\vec{X}) &= x_1x_2^{-1} - 12 \leq 0 \\ g_{10}(\vec{X}) &= 1.5x_6 - x_4 + 1.9 \leq 0 \\ g_{11}(\vec{X}) &= 1.1x_7 - x_5 + 1.9 \leq 0 \end{aligned}$$

With bounds:

$$2.6 \leq x_1 \leq 3.6, 0.7 \leq x_2 \leq 0.8, 17 \leq x_3 \leq 28, x_4 \leq 8, x_5 \geq 7.3, 2.9 \leq x_6 \leq 3.9, 5 \leq x_7 \leq 5.5$$

Table 15 presents the results of ten optimizers with $T = 500$. In this scenario, VOA outperforms others, achieving the best AVE, STD, BEST and RANK.

Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	2994.42447	2996.902317	2994.425101	9.42534E+12	1.21474E+12	3002.634793	3033.421885	3290.492382	3005.757404	3010.74485
STD	1.80479E-05	10.38813011	0.000991625	1.40838E+13	6.65338E+12	14.2667119	14.86701974	507.8007883	3.588503896	19.58522849
BEST	2994.424466	2994.427085	2994.424549	3457.361278	3027.473854	2994.424466	3006.816081	3000.129839	3000.850361	2994.424466
RANK	1	3	2	10	9	4	7	8	5	6

Table 16 presents the results with $T = 5000$. Both VOA and EEFO obtain the same AVE and BEST performance. However, VOA achieves a lower STD, enhancing its reliability and making it the first top-ranked optimizer. Overall, VOA demonstrates the best performance.

Table 15

Quantitative analysis on the weight minimization of a speed seducer with $T = 500$.

Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	2994.42447	2996.902317	2994.425101	9.42534E+12	1.21474E+12	3002.634793	3033.421885	3290.492382	3005.757404	3010.74485
STD	1.80479E-05	10.38813011	0.000991625	1.40838E+13	6.65338E+12	14.2667119	14.86701974	507.8007883	3.588503896	19.58522849
BEST	2994.424466	2994.427085	2994.424549	3457.361278	3027.473854	2994.424466	3006.816081	3000.129839	3000.850361	2994.424466
RANK	1	3	2	10	9	4	7	8	5	6

Table 16

Quantitative analysis on the weight minimization of a speed seducer with $T = 5000$.

Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	2994.424466	2994.73669	2994.424466	1.23999E+13	3537.919346	3004.388836	3011.781931	3048.04804	2997.379975	3004.630911
STD	7.02E-13	1.704206515	7.96647E-13	1.36578E+13	795.1445533	17.04817889	9.97469375	46.00408689	1.495809908	16.56016614
BEST	2994.424466	2994.424466	2994.424466	4252.734656	3048.156704	2994.424466	2998.838351	3001.305499	2995.233928	2994.424466
RANK	1	3	2	10	9	5	7	8	4	6

Fig. 13 shows the averaged convergence curves of ten optimizers. In Fig. 13(a), when $T = 500$, although FLA, DBO, and PSO have better convergence speed, the convergence speed of VOA is also commendable, which has the best convergence result. In Fig. 13(b), when $T = 5000$, VOA has a perfect convergence curve with great convergence speed and result.

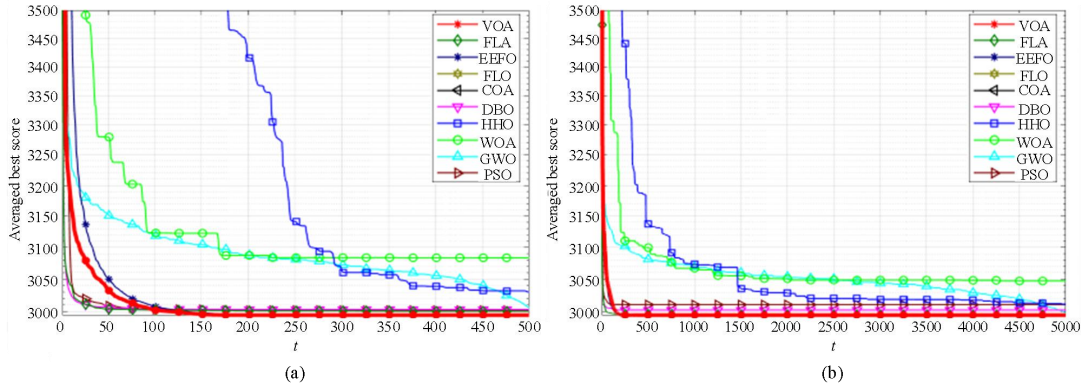


Fig. 13 Convergence analysis on the weight minimization of a speed seducer problem. (a) $T = 500$. (b) $T = 5000$.

2.4 The multiple disk clutch brake design

The multiple disk clutch brake design is to minimize the mass design with 5 integer decision variables and 8 non-linear constraints. The example of this problem is as shown in Fig. 14 and the mathematical model can be defined in Eq.(7).

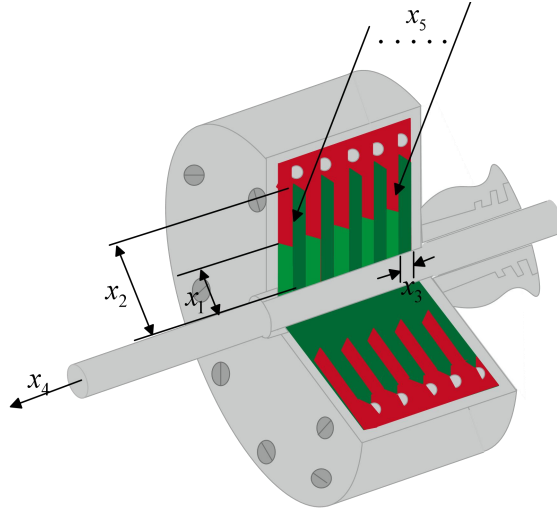


Fig. 14 The multiple disk clutch brake design.

Minimize:

$$F(\vec{X}) = \pi(x_2^2 - x_1^2)x_3(x_5 + 1)\rho \quad (7)$$

Subject to:

$$\begin{aligned} g_1(\vec{X}) &= -p_{\max} + p_{rz} \leq 0 \\ g_2(\vec{X}) &= p_{rz}V_{sr} - V_{sr,\max}p_{\max} \leq 0 \\ g_3(\vec{X}) &= \Delta R + x_1 - x_2 \leq 0 \\ g_4(\vec{X}) &= -L_{\max} + (x_5 + 1)(x_3 + \delta) \leq 0 \\ g_5(\vec{X}) &= sM_s - M_h \leq 0 \\ g_6(\vec{X}) &= T \geq 0 \\ g_7(\vec{X}) &= -V_{sr,\max} + V_{sr} \leq 0 \\ g_8(\vec{X}) &= T - T_{\max} \leq 0 \end{aligned}$$

where

$$M_h = \frac{2}{3} \mu x_4 x_5 \frac{x_2^3 - x_1^3}{x_2^2 - x_1^2} N.mm$$

$$\omega = \frac{\pi n}{30} rad/s$$

$$A = \pi(x_2^2 - x_1^2)mm^2$$

$$p_{rz} = \frac{x_4}{A} N/mm^2$$

$$V_{sr} = \frac{\pi R_{sr} n}{30} mm/s$$

$$R_{sr} = \frac{2}{3} \frac{x_2^3 - x_1^3}{x_2^2 x_1^2} mm$$

$$T = \frac{I_z \omega}{M_h + M_f}$$

$$\Delta R = 20mm, L_{\max} = 30mm, \mu = 0.6, V_{sr,\max} = 10m/s, \delta = 0.5mm, s = 1.5$$

$$T_{\max} = 15s, n = 250rpm, I_z = 55kg.m^2, M_s = 40Nm, M_f = 3Nm, p_{\max} = 1$$

With bounds:

$$60 \leq x_1 \leq 80, 90 \leq x_2 \leq 110, 1 \leq x_3 \leq 3, 0 \leq x_4 \leq 1000, 2 \leq x_5 \leq 9$$

Table 17 presents the results of ten optimizers with $T = 500$. VOA, DBO, HHO, and PSO have the best AVE, STD, BEST, and RANK indexes. Table 18 presents the results of ten optimizers with $T = 5000$. Besides EEFO, DBO, HHO, and PSO, VOA can also find the best solution. From these results, this optimization problem is relatively simple, allowing various optimizers to quickly find the best solution.

Table 17

Quantitative analysis on the multiple disk clutch brake design with $T = 500$.

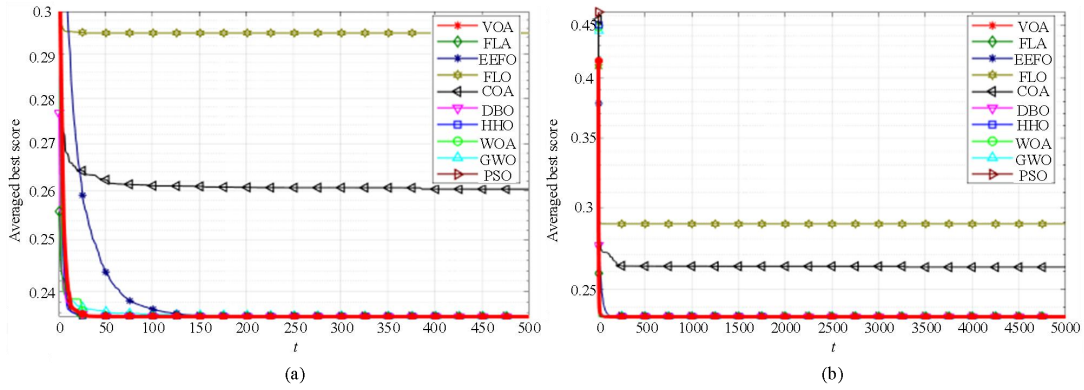
Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	0.235242458	0.235242459	0.235242458	0.291465804	0.289988973	0.235242458	0.235242458	0.235242469	0.235261508	0.235242458
STD	1.4115E-16	3.70137E-09	1.21038E-09	0.027750713	0.046110569	1.4115E-16	1.4115E-16	3.68552E-08	1.67982E-05	1.4115E-16
BEST	0.235242458	0.235242458	0.235242458	0.247025488	0.235242531	0.235242458	0.235242458	0.235242458	0.235244612	0.235242458
RANK	1	6	5	10	9	1	1	7	8	1

Table 18

Quantitative analysis on the multiple disk clutch brake design with $T = 5000$.

Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	0.235242458	0.235242458	0.235242458	0.291063612	0.265215433	0.235242458	0.235242458	0.235242459	0.23524361	0.235242458
STD	1.4115E-16	5.94767E-13	1.4115E-16	0.028170887	0.043955479	1.4115E-16	1.4115E-16	3.55235E-09	1.14984E-06	1.4115E-16
BEST	0.235242458	0.235242458	0.235242458	0.237845302	0.235242531	0.235242458	0.235242458	0.235242458	0.23524257	0.235242458
RANK	1	6	1	10	9	1	1	7	8	1

Fig. 15 presents the averaged convergence curves of ten optimizers. In Fig. 15(a), $T = 500$. The convergence curve of VOA has fast convergence speed. In Fig. 15(a), with $T = 5000$, VOA still has best convergence speed and result. Therefore, VOA has great performance on this problem.

Fig. 15 Convergence analysis on the multiple disk clutch brake design. (a) $T = 500$. (b) $T = 5000$.

2.5 The 10-bar truss optimization problem with frequency constraints

A 10-bar truss structure is shown in Fig. 16, with young modulus $6.98 \times 10^{10} \text{Pa}$, material density 2770kg/m^3 , added mass $M = 454 \text{kg}$ and main bar dimension $l = 9.144 \text{m}$. It has 10 design variables and 3 frequency constraints to minimize the weight of truss structure with satisfying frequency constraints, which can be defined by Eq.(8).

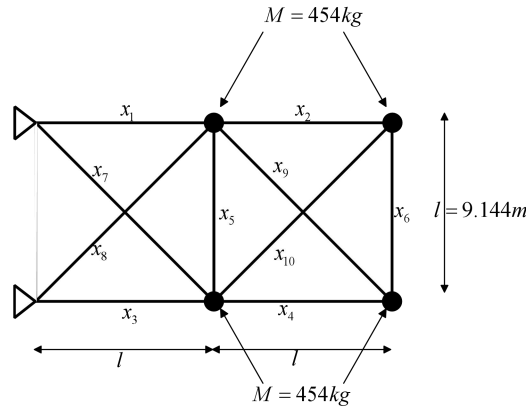


Fig. 16 The 10-bar truss structure.

Minimize:

$$F(\vec{X}) = \sum_{i=1}^{10} L_i(x_i) \rho_i x_i \quad (8)$$

Subject to:

$$g_1(\vec{X}) = \frac{7}{\omega_1(\vec{X})} - 1 \leq 0 \quad (\omega_1(\vec{X}) \geq 7)$$

$$g_2(\vec{X}) = \frac{15}{\omega_2(\vec{X})} - 1 \leq 0 \left(\omega_1(\vec{X}) \geq 15 \right)$$

$$g_3(\vec{X}) = \frac{20}{\omega_3(\vec{X})} - 1 \leq 0 \left(\omega_1(\vec{X}) \geq 20 \right)$$

With bounds:

$$6.45 \times 10^{-5} \leq x_i \leq 5 \times 10^{-3}, i = 1, 2, \dots, 10$$

Table 19 presents the results of ten optimizers with $T = 500$. Although PSO and EEFO obtain the better BEST index than VOA, VOA has the best AVE, STD and RANK. Table 20 shows the results with $T = 5000$. VOA has the best AVE, STD, BEST and RANK. Overall, VOA demonstrates the best performance in both rapid iterations and long-term runs.

Table 19

Quantitative analysis on the 10-bar truss optimization problem with $T = 500$.

Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	525.4877021	528.1684458	527.733554	665.6114135	657.1202582	531.4429152	644.0378277	645.0330625	527.2938283	527.5795137
STD	0.715879939	3.473388335	2.747537899	37.66209296	33.18639608	6.973944421	45.49523875	50.11265855	2.503175581	2.888928914
BEST	524.6348858	524.7087842	524.5817976	598.5542239	585.5309487	524.7564184	568.6648976	560.0997004	524.8941214	524.5041022
RANK	1	5	4	10	9	6	7	8	2	3

Table 20

Quantitative analysis on the 10-bar truss optimization problem with $T = 5000$.

Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	524.6168904	526.2577047	527.0259518	648.7535647	650.0831218	531.1827967	619.2195611	572.8504676	524.6497283	526.7203274
STD	0.12793117	2.758650548	3.029126112	30.75062546	39.47392167	7.063087114	35.6694167	20.74826633	0.134913193	2.835143462
BEST	524.4623486	524.4770957	524.4643232	597.6747441	576.5829156	524.5447536	559.7656493	545.6539983	524.4789113	524.5572912
RANK	1	3	5	9	10	7	8	6	2	4

Fig. 17 presents the averaged convergence curves of ten optimizers. In Fig. 17(a), when $T = 500$, although FLA, DBO, and PSO have faster convergence speed, the convergence speed of VOA remains competitive with the best convergence result. In Fig. 17(b), when $T = 5000$, the performance of VOA are stable and superior. Its ability to maintain effectiveness over extended iterations makes it a reliable choice for optimization problem.

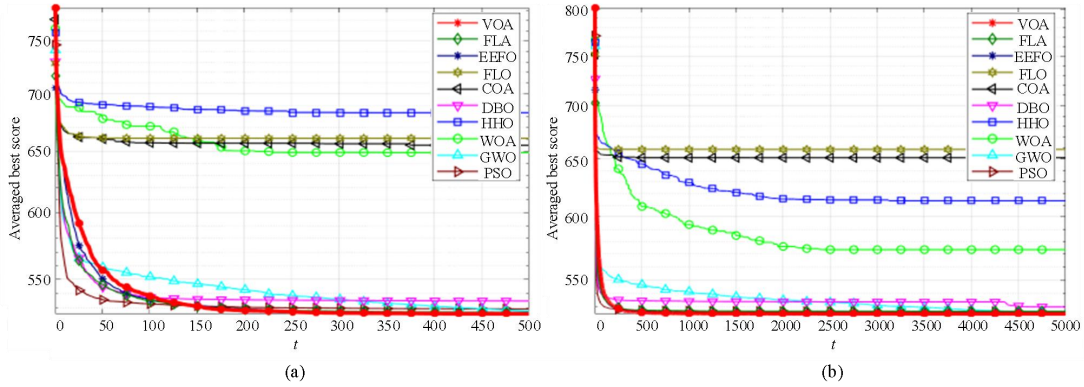


Fig. 17 Convergence analysis on the 10-bar truss optimization problem. (a) $T = 500$. (b) $T = 5000$.

2.6 The gear train design

The gear train design is to minimize the ratio of gears for the arrangement of the compound gear train with 4 integer design variables as shown in Fig. 18. Its mathematical formulation can be simply described by Eq.(9).

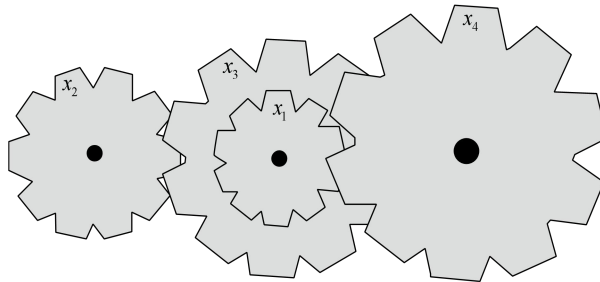


Fig. 18 The gear train design.

Minimize:

$$F(\vec{X}) = \left(\frac{1}{6.931} - \frac{x_1 x_2}{x_3 x_4} \right)^2 \quad (9)$$

With bounds:

$$12 \leq x_i \leq 60, i = 1, 2, 3, 4$$

Table 21 presents the results of ten optimizers with $T = 500$. All optimizers could converge to a solution near the best value of 0, but only VOA and HHO consistently stabilize in finding the best solution. Table 22 shows the results with $T = 5000$. In the long-term iterations, VOA still finds the best solution, with the same results of DBO and HHO.

Table 21

Quantitative analysis on the gear train design with $T = 500$.

Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	0	1.28151E-11	5.7574E-14	4.41615E-05	6.14323E-12	2.31112E-34	0	6.41977E-34	5.00704E-13	7.6332E-16
STD	0	3.33584E-11	8.81501E-14	0.00014816	2.4589E-11	1.26585E-33	0	3.51625E-33	1.22888E-12	3.1476E-15
BEST	0	4.45844E-25	2.55179E-17	0	0	0	0	0	7.52242E-17	3.08149E-33
RANK	1	9	6	10	8	3	1	4	7	5

Table 22

Quantitative analysis on the gear train design with $T = 5000$.

Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	0	6.17213E-16	5.28821E-16	8.10177E-05	2.48999E-19	0	0	5.0331E-33	2.69926E-15	2.56791E-35
STD	0	3.33496E-15	1.16323E-15	0.000230451	9.84989E-19	0	0	2.75674E-32	5.64533E-15	1.4065E-34
BEST	0	0	5.32068E-22	0	0	0	0	0	6.80612E-21	0
RANK	1	8	7	10	6	1	1	5	9	4

Fig. 19 presents the averaged convergence curves of ten optimizers. In Fig. 19(a), when $T = 500$, VOA has the best convergence speed and result. In Fig. 19(b), when $T = 5000$, both VOA and DBO have great convergence curves. This indicates that VOA can maintain high performance while adapting to the complex optimization problem.

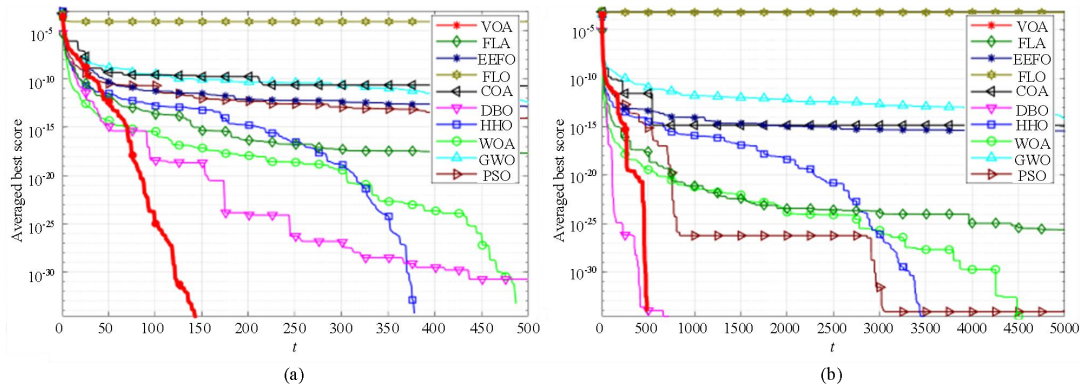


Fig. 19 Convergence analysis on the gear train design. (a) $T = 500$. (b) $T = 5000$.

2.7 The rolling element bearing design problem

The rolling element bearing design is to maximize the dynamic load carrying capacity of a rolling element bearing with 10 design variables, which are demonstrated in Fig. 20. The mathematical model is defined by Eq.(10).

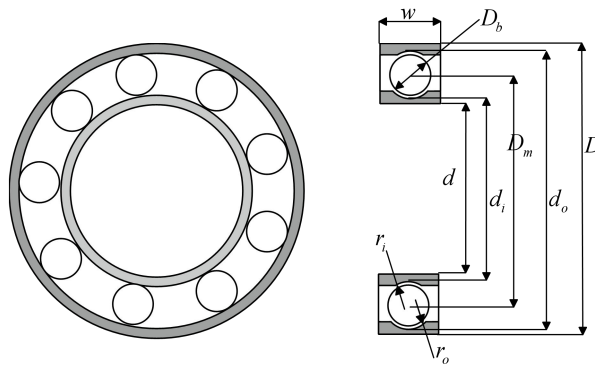


Fig. 20 The rolling element bearing problem.

Maximize:

$$F(\vec{X}) = \begin{cases} f_c Z^{\frac{2}{3}} D_b^{1.8}, & \text{if } D_b \leq 25.4 \text{mm} \\ 3.647 f_c Z^{\frac{2}{3}} D_b^{1.4}, & \text{otherwise} \end{cases} \quad (10)$$

Subject to:

$$\begin{aligned} g_1(\vec{X}) &= Z - \frac{\phi_0}{2 \sin^{-1}(D_b/D_m)} - 1 \leq 0 \\ g_2(\vec{X}) &= K_{D_{\min}}(D-d) - 2D_b \leq 0 \\ g_3(\vec{X}) &= 2D_b - K_{D_{\max}}(D-d) \leq 0 \\ g_4(\vec{X}) &= D_b - w \leq 0 \\ g_5(\vec{X}) &= 0.5(D-d) - D_m \leq 0 \\ g_6(\vec{X}) &= D_m - (0.5+e)(D+d) \leq 0 \\ g_7(\vec{X}) &= \varepsilon D_b - 0.5(D-D_M-D_b) \leq 0 \\ g_8(\vec{X}) &= 0.515 - f_i \leq 0 \\ g_9(\vec{X}) &= 0.515 - f_o \leq 0 \end{aligned}$$

where

$$\begin{aligned} f_c &= 37.91 \left\{ 1 + \left\{ 1.04 \left(\frac{1-\gamma}{1+\gamma} \right)^{1.72} \left(\frac{f_i(2f_o-1)}{f_o(2f_i-1)} \right)^{0.41} \right\}^{10/3} \right\}^{-0.3} \\ \gamma &= \frac{D_b \cos(\alpha)}{D_m}, f_i = \frac{r_i}{D_b}, f_o = \frac{r_o}{D_b} \\ \phi_o &= 2\pi - 2 \times \cos^{-1} \left(\frac{\{(D-d)/2 - 3(T/4)\}^2 + (D/2 - T/4 - D_b)^2 - (d/2 + T/4)^2}{2\{(D-d)/2 - 3(T/4)\}(D/2 - T/4 - D_b)} \right) \\ T &= D - d - 2D_b, D = 160, d = 90, B_w = 30 \end{aligned}$$

With bounds:

$$\begin{aligned} 0.5(D+d) &\leq D_m \leq 0.6(D+d) \\ 0.15(D-d) &\leq D_b \leq 0.45(D-d) \\ 4 &\leq Z \leq 50 \\ 0.515 &\leq f_i \leq 0.6 \\ 0.515 &\leq f_o \leq 0.6 \\ 0.4 &\leq K_{D_{\min}} \leq 0.5 \\ 0.6 &\leq K_{D_{\max}} \leq 0.7 \\ 0.3 &\leq \varepsilon \leq 0.4 \\ 0.02 &\leq e \leq 0.1 \\ 0.6 &\leq \xi \leq 0.85 \end{aligned}$$

Table 23 presents the results of ten optimizers with $T = 500$. VOA, DBO and PSO can find the maximum BEST, but VOA has the best AVE and STD to make it the best optimizer. Table 24 shows the results of optimizers with $T = 5000$. VOA and EEFO obtain the best AVE and BEST indexes. However, VOA has the best RANK due to its lower STD. Therefore, VOA has the best performance with $T = 500$ and 5000 on this problem.

Table 23
Quantitative analysis on the rolling element bearing problem with $T = 500$.

Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	81859.74154	80455.06178	81859.52826	66240.45568	77267.58973	80706.05698	81256.37894	77927.25895	81664.26603	81158.48391
STD	5.2142e-07	3257.636695	0.27625882	5385.72941	2537.399381	2355.036051	628.6720451	3618.357759	116.132706	504.34098
BEST	81859.7416	81859.74158	81859.73786	72509.92382	78896.53451	81859.7416	81859.72191	81325.71388	81831.33902	81859.7416
RANK	1	7	2	10	9	6	4	8	3	5

Table 24
Quantitative analysis on the rolling element bearing problem with $T = 5000$.

Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	81859.7416	81859.73923	81859.7416	65733.15125	78014.98243	81161.11123	81818.24561	80787.25586	81847.90873	81088.35815
STD	2.16E-11	0.010962476	2.75574E-11	5663.712935	1893.404747	568.5517111	77.89004291	1045.771482	7.044792685	473.1137798

BEST	81859.7416	81859.7416	81859.7416	77561.16521	79844.95993	81859.7416	81859.74158	81859.74157	81858.39608	81859.7416
RANK	1	3	2	10	9	6	5	8	4	7

Fig. 21 depicts the averaged convergence curves on this maximum problem. In Fig. 21 (a), when $T = 500$, VOA, FLA and EEFO have good convergence curves with great speed and result. In Fig. 21 (b), VOA keep excellent convergence curve.

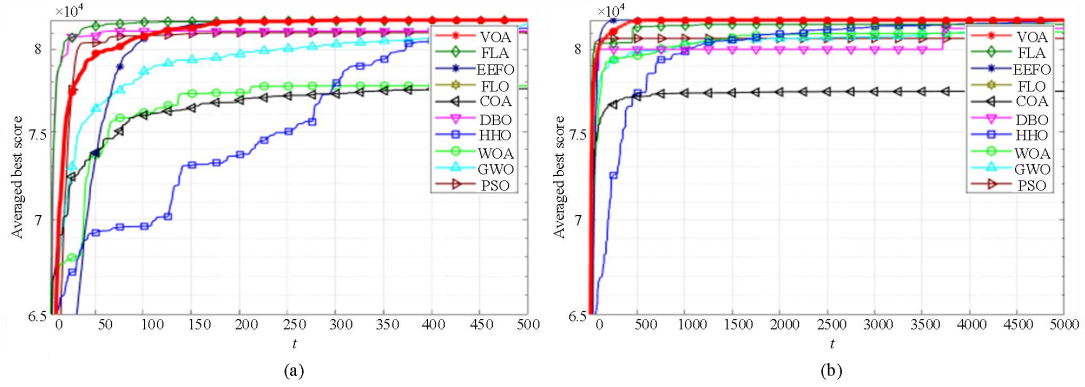


Fig. 21 Convergence analysis on the rolling element bearing problem. (a) $T = 500$. (b) $T = 5000$.

To conclude, this section assesses the performance of VOA across 7 typical real-world optimization problems. When $T = 500$, VOA could find the optimal solution with impressive convergence speed. Even if the iteration step increases to $T = 5000$, VOA can keep the stable performance. These results highlight the effectiveness and generalization ability of VOA across various practical applications, indicating its potential as a valuable optimization algorithm to address complex optimization problems.

3. Discussion on the step response of wind turbine optimization parameters

This section presents the step response curves of the optimal parameters of VOA* under different wind speeds.

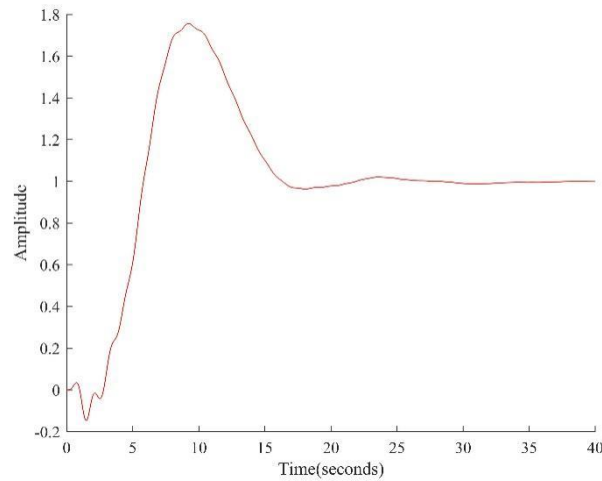


Fig. 22 The step response of controller parameter optimization result at 11m/s wind speed.

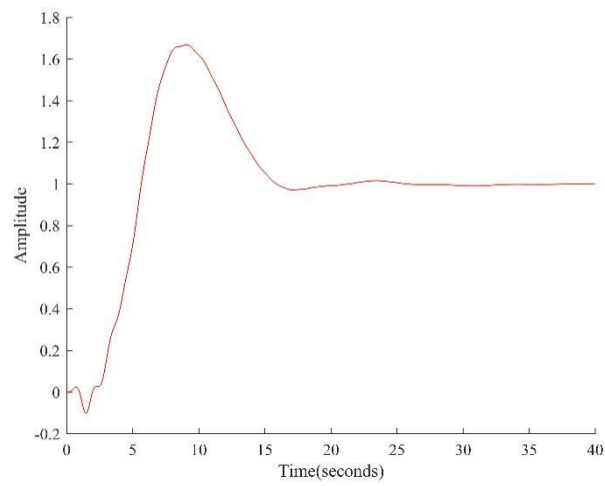
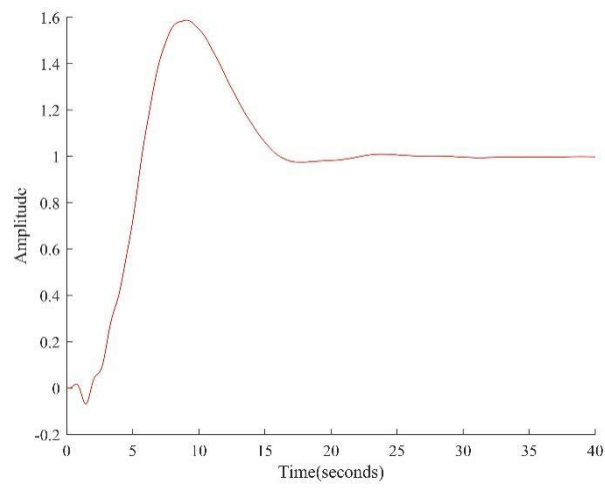


Fig. 23 The step response of controller parameter optimization result at 11.5m/s wind



speed.

Fig. 24 The step response of controller parameter optimization result at 12m/s wind speed.

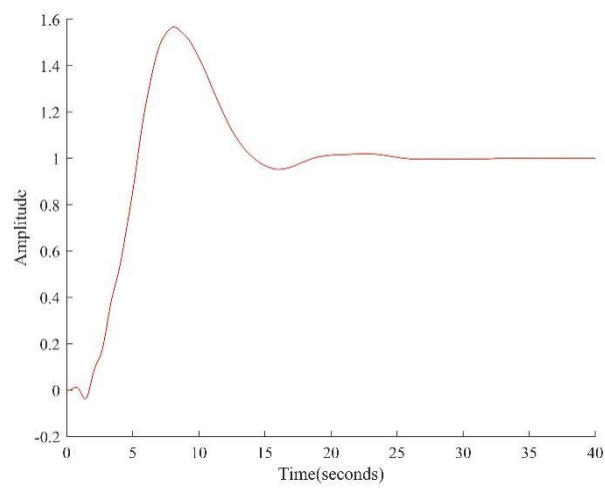


Fig. 25 The step response of controller parameter optimization result at 13m/s wind speed.

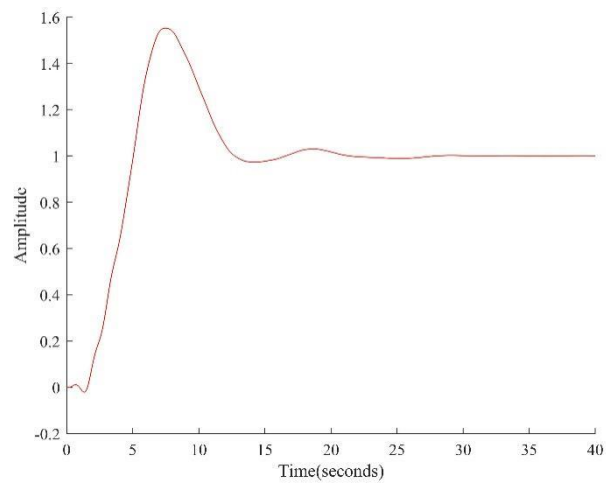


Fig. 26 The step response of controller parameter optimization result at 14m/s wind speed.

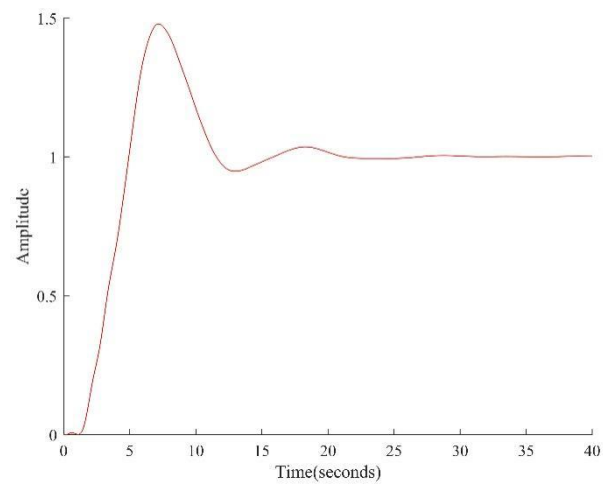


Fig. 27 The step response of controller parameter optimization result at 16m/s wind speed.

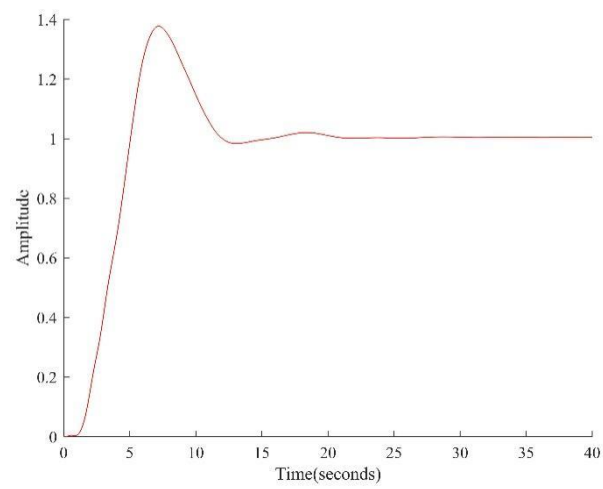


Fig. 28 The step response of controller parameter optimization result at 18m/s wind speed.

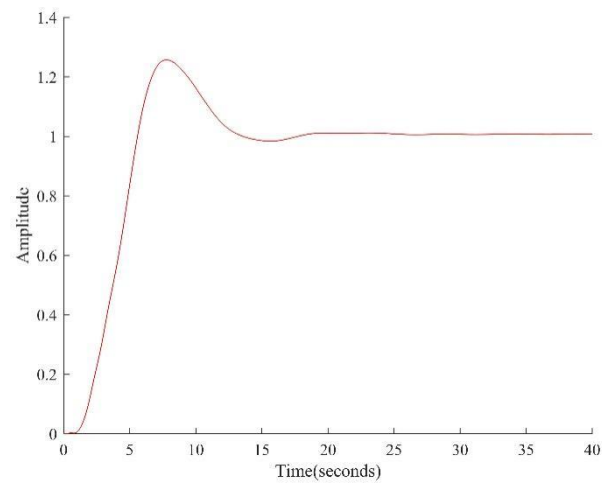


Fig. 29 The step response of controller parameter optimization result at 20m/s wind speed.

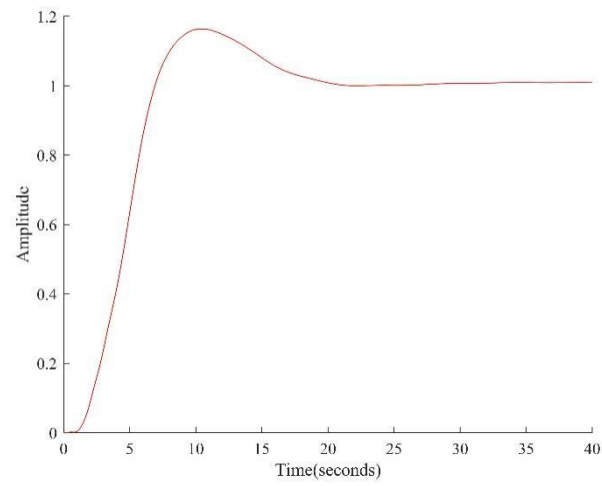


Fig. 30 The step response of controller parameter optimization result at 22m/s wind speed.

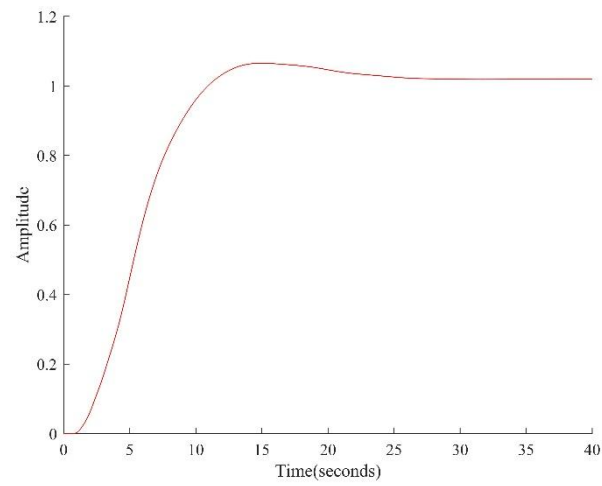


Fig. 31 The step response of controller parameter optimization result at 25m/s wind speed.

4. Discussion on limitations and computational trade-offs

4.1 Analysis of running time complexity

The running time complexity can effectively reflect the efficiency of the algorithm. Therefore, in order to comprehensively understand and evaluate VOA, it is necessary to analyze its running time complexity. Here, the key parameters are set as follows: the number of vines N , the number of vines to do climbing host trees behavior N_c , the number of vines to do leaf growth behavior N_l ($N = N_c + N_l$), the maximum number of iterations T and the number of decision variable m . The running time complexity of VOA is as shown in Eq.(11).

$$\begin{aligned}
O(\text{VOA}) &= O(\text{Problem definition}) + O(\text{Seed germination}) \\
&+ O(\text{Nutrient allocation}) + O(\text{Climbing host trees}) \\
&+ O(\text{Leaf growth}) + O(\text{Drooping}) + O(\text{Swarm behavior}) \\
&= O(1) + O(Nm) + O(T) + O(T \cdot N_c \cdot m) + O(T \cdot N_l \cdot m) + O(TNm) + O(TNm) \\
&= O(1) + O(Nm) + O(T) + O(3TNm) \\
&\cong O(TNm)
\end{aligned} \tag{11}$$

The running time complexity of VOA only grows linearly with the key parameters of the meta-heuristic optimization algorithm, indicating that this algorithm has excellent computational efficiency.

4.2 Quantitative analysis of convergence indicators in the test suites

In the previous convergence analysis, by comparing the convergence curves in the test suites, the convergence ability of VOA was qualitatively analyzed. This subsection introduces the AUC (Area Under the Curve) to conduct quantitative discussion and analysis of the convergence indicators. Due to the different initialization methods of various optimizers, there are significant differences in the initial values, which has a considerable impact on the AUC. Therefore, by incorporating the AUTC (Area Under the Truncated Curve) and the AUNC (Area Under the Normalized Curve), the convergence ability of the optimizers can be comprehensively evaluated. AUTC refers to the data from the first 1% of the truncated convergence curve, which to some extent eliminates the influence of initialization. AUNC refers to the indicator of normalizing the convergence curve and then calculating the area under the curve.

This subsection retains parameters such as number of population members N and maximum number of iterations T mentioned in the previous text. In Table 25 to Table 29, it conducts 30 repetitions of experiments and calculates the average values of AUC, AUTC and AUNC indicators in the test suites. Although, on 23 standard test functions, FLO demonstrated better convergence indicators for some of the functions. Meanwhile, at CEC2017, CEC2019, CEC2020 and CEC2022, the convergence index of EEFO was also quite outstanding. However, in general, VOA has the best or nearly the best convergence indicators in most tests, and as a result, demonstrates an excellent convergence rate overall.

Table 25
Convergence indicators for 23 standard test functions.

Fun.	Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
F1	AUC	1.33E+03	3.58E+05	3.64E+03	3.32E+04	3.40E+04	1.86E+05	4.49E+04	2.03E+05	2.15E+05	9.44E+05
	AUTC	6.33E-07	9.60E+04	6.14E-03	6.06E-06	2.04E-04	2.05E+04	8.96E-02	4.99E+03	1.24E+04	7.73E+05
	AUNC	8.75E-01	6.57E+00	1.59E+00	5.40E-01	5.39E-01	3.50E+00	6.94E-01	3.29E+00	3.44E+00	1.50E+01
F2	AUC	2.36E+01	8.67E+10	7.27E+06	1.75E+12	3.03E+11	2.76E+10	2.82E+11	8.50E+11	6.30E+11	2.13E+11
	AUTC	3.64E-04	2.70E+06	1.10E-01	8.98E-04	6.26E-03	3.75E+01	3.46E-01	7.55E+01	1.79E+02	8.53E+03
	AUNC	1.28E+00	1.01E+00	2.22E+00	5.00E-01	5.00E-01	9.54E-01	5.00E-01	6.19E-01	5.91E-01	5.12E-01
F3	AUC	3.26E+03	1.69E+06	9.51E+04	7.65E+04	7.45E+04	4.92E+06	1.56E+05	3.50E+07	1.01E+06	4.94E+06
	AUTC	2.60E-05	1.12E+06	5.76E+01	7.39E-03	8.05E-04	4.29E+06	1.22E+02	3.41E+07	4.68E+05	4.56E+06
	AUNC	1.35E+00	1.77E+01	1.57E+00	6.93E-01	6.93E-01	5.52E+01	1.40E+00	3.24E+02	9.58E+00	4.54E+01
F4	AUC	1.04E+01	6.31E+03	9.29E+00	6.07E+01	6.16E+01	2.03E+03	1.17E+02	2.61E+04	1.20E+03	1.20E+04
	AUTC	7.31E-04	5.58E+03	5.22E-02	4.26E-04	3.12E-03	1.30E+03	2.36E-01	2.54E+04	5.47E+02	1.15E+04
	AUNC	1.66E+00	7.62E+01	2.10E+00	7.17E-01	7.27E-01	2.43E+01	1.38E+00	3.07E+02	1.43E+01	1.42E+02
F5	AUC	8.13E+04	6.84E+08	3.97E+06	1.11E+08	1.07E+08	9.97E+08	1.13E+08	5.83E+08	4.87E+08	4.68E+08
	AUTC	4.59E+01	7.54E+07	2.55E+03	5.42E+01	5.54E+01	1.81E+08	4.96E+01	1.70E+06	4.58E+06	2.05E+08
	AUNC	6.72E-01	3.69E+00	1.15E+01	5.01E-01	5.03E-01	4.44E+00	5.62E-01	2.70E+00	2.14E+00	2.14E+00
F6	AUC	1.34E+03	3.82E+05	1.95E+03	3.32E+04	3.50E+04	1.87E+05	4.61E+04	2.06E+05	2.10E+05	9.61E+05
	AUTC	3.37E-04	9.95E+04	5.25E+00	4.92E-03	3.33E+00	2.86E+04	1.27E+00	5.50E+03	1.25E+04	7.93E+05
	AUNC	8.73E-01	6.52E+00	2.90E+00	5.39E-01	5.41E-01	3.65E+00	7.19E-01	3.24E+00	3.50E+00	1.58E+01
F7	AUC	7.05E-01	3.43E+02	1.86E+00	5.43E+01	5.46E+01	4.56E+02	6.28E+01	2.88E+02	2.38E+02	4.91E+02
	AUTC	1.52E-01	8.93E+01	5.05E-01	6.59E-02	9.09E-02	8.36E+01	2.70E-01	9.93E+00	5.84E+00	3.14E+02
	AUNC	1.47E+00	4.07E+00	5.92E+00	5.08E-01	5.08E-01	4.62E+00	5.68E-01	2.77E+00	2.34E+00	4.84E+00
F8	AUC	3.02E+04	3.37E+06	3.86E+05	1.67E+06	3.42E+04	3.57E+06	7.39E+04	1.29E+06	8.12E+06	4.57E+06
	AUTC	3.45E+03	3.29E+06	3.03E+05	1.64E+06	3.58E+03	3.49E+06	2.18E+04	1.23E+06	8.02E+06	4.48E+06
	AUNC	3.01E+00	3.48E+02	4.06E+01	1.68E+02	3.44E+00	3.62E+02	7.48E+00	1.29E+02	7.99E+02	4.52E+02
F9	AUC	3.72E+02	8.05E+04	2.26E+02	4.42E+02	6.51E+02	3.52E+04	9.25E+02	1.17E+04	1.08E+04	1.10E+05
	AUTC	3.33E-07	7.72E+04	2.95E-02	3.25E-05	3.27E-01	3.26E+04	1.35E-01	8.50E+03	7.61E+03	1.07E+05
	AUNC	1.04E+00	2.03E+02	1.77E+00	1.04E+00	1.53E+00	8.95E+01	2.17E+00	2.75E+01	2.59E+01	2.63E+02
F10	AUC	7.10E+00	1.88E+04	6.37E+00	2.47E+01	3.03E+01	3.39E+02	4.40E+01	2.40E+02	2.87E+02	4.06E+03
	AUTC	2.41E-04	1.86E+04	3.02E-02	1.72E-04	2.31E-03	2.05E+02	2.63E-01	8.60E+01	1.21E+02	3.89E+03

	AUNC	1.80E+00	9.40E+02	2.32E+00	1.20E+00	1.47E+00	1.67E+01	2.15E+00	1.17E+01	1.40E+01	1.97E+02
	AUC	1.58E+01	3.14E+03	1.42E+01	3.00E+02	3.21E+02	2.05E+03	4.25E+02	1.86E+03	1.93E+03	8.92E+03
F11	AUTC	1.17E-06	8.54E+02	2.90E-02	3.74E-05	2.09E-04	2.81E+02	1.76E-01	8.11E+01	1.48E+02	7.39E+03
	AUNC	1.11E+00	6.49E+00	3.06E+00	5.37E-01	5.43E-01	4.03E+00	7.62E-01	3.38E+00	3.67E+00	1.57E+01
	AUC	1.47E+01	1.04E+09	6.10E+06	2.32E+08	2.42E+08	2.53E+09	2.56E+08	1.20E+09	8.71E+08	3.97E+08
F12	AUTC	6.85E-06	4.01E+07	2.64E-01	2.71E-05	2.70E-02	4.12E+08	2.65E-02	2.17E+05	6.24E+05	1.81E+07
	AUNC	9.26E-01	3.15E+00	1.84E+00	5.00E-01	5.00E-01	5.99E+00	5.20E-01	2.36E+00	1.91E+00	8.29E-01
	AUC	2.17E+02	2.78E+09	2.69E+07	4.78E+08	4.78E+08	4.64E+09	5.08E+08	2.19E+09	1.77E+09	1.12E+09
F13	AUTC	7.70E-05	2.18E+08	2.01E+00	1.78E-04	3.15E-01	7.16E+08	3.04E-01	9.36E+05	6.35E+06	2.24E+08
	AUNC	6.31E-01	3.36E+00	1.78E+00	5.00E-01	5.00E-01	5.16E+00	5.26E-01	2.32E+00	2.17E+00	1.20E+00
	AUC	6.44E+01	3.18E+01	7.28E+01	2.85E+01	2.51E+01	8.80E+01	1.77E+02	6.23E+02	8.50E+02	8.17E+01
F14	AUTC	3.22E-05	3.60E+00	2.64E+01	3.86E-09	6.44E-03	5.20E+01	1.08E+02	5.39E+02	7.96E+02	1.52E+01
	AUNC	1.02E+00	3.78E+00	8.89E+00	1.07E+00	1.07E+00	1.84E+01	1.66E+01	2.64E+01	5.41E+01	4.55E+00
	AUC	7.82E-02	2.85E+00	1.32E-01	1.74E-01	1.37E-01	7.50E-01	1.65E-01	4.66E-01	8.88E-01	4.25E-01
F15	AUTC	8.80E-04	2.76E+00	4.17E-02	1.22E-01	5.75E-02	6.74E-01	6.72E-02	3.02E-01	7.76E-01	3.10E-01
	AUNC	1.20E+00	8.71E+01	5.45E+00	8.91E+00	2.45E+00	7.98E+01	3.27E+00	6.51E+00	2.16E+01	9.18E+00
	AUC	5.71E-01	1.84E-01	4.58E-01	6.06E-01	6.65E-01	2.72E-01	8.36E-01	7.51E-01	5.83E-01	8.02E-01
F16	AUTC	5.20E-05	6.98E-04	7.29E-02	8.55E-02	8.30E-03	5.29E-05	3.50E-03	3.34E-03	7.53E-03	1.36E-01
	AUNC	7.03E-01	2.55E+00	5.30E+00	1.84E+00	1.10E+00	1.44E+00	1.18E+00	1.34E+00	1.34E+00	2.35E+00
	AUC	3.80E-01	2.80E-01	1.34E+00	9.17E-01	1.01E+00	2.79E-01	1.48E+00	2.10E+00	2.16E+00	1.17E+00
F17	AUTC	4.43E-05	3.79E-04	4.03E-01	1.87E-01	1.71E-01	1.92E-05	2.20E-01	3.42E-01	8.02E-01	1.75E-01
	AUNC	1.03E+00	1.64E+00	8.25E+00	6.12E+00	2.11E+00	1.31E+00	2.64E+00	2.37E+00	3.96E+01	5.10E+00
	AUC	3.66E+01	3.93E+00	1.25E+01	2.27E+02	3.30E+01	4.71E+00	7.03E+01	2.13E+02	1.38E+01	2.53E+01
F18	AUTC	2.10E-04	1.07E-02	5.05E-01	2.07E+02	1.27E+01	7.14E-04	1.30E+01	1.28E+02	8.27E-02	1.68E+00
	AUNC	8.66E-01	2.44E+00	3.03E+00	1.17E+01	2.29E+00	1.37E+00	1.92E+00	4.05E+00	1.49E+00	2.22E+00
	AUC	4.62E-01	9.43E-02	6.42E-01	5.68E+01	8.46E+00	1.30E-01	4.37E+00	2.93E+00	3.49E+00	7.82E-01
F19	AUTC	7.59E-03	1.47E-03	5.44E-02	5.57E+01	7.66E+00	7.31E-03	3.42E+00	2.24E+00	2.73E+00	8.67E-02
	AUNC	1.31E+00	1.52E+00	3.97E+00	1.97E+02	3.48E+01	1.54E+00	1.66E+01	1.27E+01	1.78E+01	3.30E+00
	AUC	1.91E+01	6.06E+01	2.08E+01	4.70E+02	4.62E+02	7.25E+01	1.47E+02	5.53E+01	8.37E+01	6.07E+01
F20	AUTC	1.54E+01	5.89E+01	1.56E+01	4.64E+02	4.55E+02	6.98E+01	1.40E+02	5.04E+01	7.88E+01	5.56E+01
	AUNC	1.62E+01	1.50E+02	2.14E+01	4.00E+02	3.73E+02	1.29E+02	1.06E+02	4.15E+01	6.63E+01	4.99E+01
	AUC	2.44E+01	3.97E+03	1.47E+02	1.57E+01	2.02E+01	3.65E+03	4.78E+03	9.42E+02	2.34E+03	3.83E+03
F21	AUTC	2.99E-03	3.91E+03	8.72E+01	7.90E-02	1.09E+00	3.59E+03	4.71E+03	8.79E+02	2.25E+03	3.76E+03
	AUNC	3.01E+00	4.44E+02	1.70E+01	1.68E+00	2.14E+00	3.94E+02	5.03E+02	1.02E+02	2.46E+02	4.20E+02
	AUC	2.72E+01	4.19E+03	1.76E+02	1.62E+01	1.91E+01	4.12E+03	5.00E+03	6.02E+02	1.66E+03	2.24E+03
F22	AUTC	6.07E-03	4.14E+03	1.09E+02	1.07E-01	8.18E-01	4.05E+03	4.93E+03	5.43E+02	1.57E+03	2.17E+03
	AUNC	1.89E+00	4.56E+02	2.09E+01	1.70E+00	1.98E+00	4.45E+02	5.27E+02	6.30E+01	1.73E+02	2.37E+02
	AUC	2.90E+01	3.98E+03	1.34E+02	1.71E+01	1.96E+01	3.61E+03	4.90E+03	8.71E+02	1.71E+03	1.63E+03
F23	AUTC	3.37E-02	3.92E+03	7.42E+01	8.21E-02	1.21E+00	3.54E+03	4.84E+03	7.99E+02	1.62E+03	1.56E+03
	AUNC	3.16E+00	4.39E+02	1.66E+01	1.79E+00	2.08E+00	3.90E+02	5.09E+02	9.04E+01	1.80E+02	1.71E+02

Table 26
Convergence indicators for 29 CEC2017 test suite with $m = 10$.

Fun.	Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
	AUC	1.49E+10	1.52E+10	7.96E+10	2.95E+13	2.98E+13	7.54E+11	6.08E+11	1.02E+11	3.79E+11	6.11E+10
C17F1	AUTC	5.81E+06	9.41E+06	6.49E+08	2.92E+13	2.95E+13	7.16E+11	3.77E+11	2.99E+09	3.28E+11	2.32E+10
	AUNC	2.08E+00	2.34E+00	6.32E+00	1.96E+03	2.07E+03	8.82E+01	4.21E+01	6.61E+00	2.45E+01	3.43E+00
	AUC	1.27E+06	6.12E+05	6.01E+05	4.80E+07	3.60E+07	2.40E+06	3.26E+06	1.55E+06	3.68E+06	3.89E+05
C17F3	AUTC	6.76E+04	6.66E+04	1.47E+05	4.73E+07	3.55E+07	1.35E+06	2.59E+06	5.14E+05	3.02E+06	4.28E+04
	AUNC	1.83E+01	1.26E+01	1.99E+01	6.18E+02	6.30E+02	4.84E+01	5.55E+01	2.17E+01	4.89E+01	6.56E+00
	AUC	2.72E+04	1.22E+04	9.32E+03	2.00E+06	2.02E+06	1.16E+05	1.21E+05	7.21E+04	6.37E+04	3.30E+04
C17F4	AUTC	2.63E+03	9.96E+03	2.27E+03	1.98E+06	2.00E+06	1.12E+05	1.02E+05	6.40E+04	6.02E+04	2.94E+04
	AUNC	1.37E+01	2.61E+01	9.74E+00	1.25E+03	1.43E+03	1.24E+02	7.48E+01	6.51E+01	5.06E+01	2.45E+01
	AUC	5.00E+04	9.94E+04	4.14E+04	3.60E+05	3.43E+05	1.39E+05	2.44E+05	2.07E+05	1.14E+05	5.98E+04
C17F5	AUTC	4.65E+04	9.73E+04	3.85E+04	3.56E+05	3.40E+05	1.37E+05	2.40E+05	2.04E+05	1.11E+05	5.74E+04
	AUNC	3.29E+02	9.01E+02	3.48E+02	2.53E+03	2.47E+03	1.22E+03	1.75E+03	1.45E+03	8.71E+02	4.23E+02
	AUC	1.07E+03	3.30E+03	1.52E+03	1.92E+05	1.89E+05	1.00E+04	1.42E+05	1.42E+05	2.26E+04	1.02E+03
C17F6	AUTC	2.11E+02	2.76E+03	2.94E+02	1.90E+05	1.87E+05	9.36E+03	1.40E+05	1.40E+05	2.16E+04	2.19E+02
	AUNC	1.06E+01	5.34E+01	2.09E+01	2.13E+03	2.17E+03	1.24E+02	1.57E+03	1.58E+03	2.65E+02	1.12E+01
	AUC	1.21E+05	1.08E+05	1.04E+05	4.31E+05	4.12E+05	1.32E+05	3.95E+05	3.88E+05	1.79E+05	9.00E+04
C17F7	AUTC	1.15E+05	1.04E+05	9.94E+04	4.26E+05	4.07E+05	1.27E+05	3.89E+05	3.82E+05	1.75E+05	8.59E+04
	AUNC	2.37E+02	3.52E+02	5.51E+02	1.08E+03	1.02E+03	3.93E+02	9.53E+02	8.71E+02	4.26E+02	2.37E+02
	AUC	5.46E+04	8.39E+04	4.78E+04	2.27E+05	2.09E+05	1.34E+05	1.49E+05	1.88E+05	8.41E+04	5.10E+04
C17F8	AUTC	4.17E+04	8.15E+04	4.52E+04	2.25E+05	2.07E+05	1.31E+05	1.46E+05	1.86E+05	8.15E+04	4.87E+04
	AUNC	3.60E+02	7.83E+02	4.70E+02	1.76E+03	1.71E+03	1.18E+03	1.16E+03	1.48E+03	6.42E+02	4.05E+02
	AUC	2.73E+04	1.58E+04	1.45E+04	2.16E+06	2.10E+06	2.13E+05	2.40E+06	2.00E+06	4.38E+04	9.20E+03
C17F9	AUTC	3.28E+02	5.49E+02	5.19E+02	2.13E+06	2.07E+06	1.79E+05	2.35E+06	1.94E+06	3.34E+04	1.08E+02
	AUNC	2.82E+00	6.41E+00	7.35E+00	5.85E+02	5.69E+02	5.90E+01	6.63E+02	5.05E+02	1.48E+01	2.40E+00
	AUC	1.61E+06	3.65E+06	1.38E+06	7.81E+06	6.87E+06	3.94E+06	5.31E+06	4.45E+06	3.19E+06	2.64E+06
C17F10	AUTC	1.53E+06	3.59E+06	1.30E+06	7.72E+06	6.79E+06	3.88E+06	5.22E+06	4.38E+06	3.10E+06	2.57E+06
	AUNC	6.39E+02	1.59E+03	5.92E+02	3.09E+03	2.72E+03	1.70E+03	2.12E+03	1.79E+03	1.25E+03	1.10E+03
	AUC	6.14E+04	1.39E+05	3.77E+04	5.11E+06	2.09E+06	3.29E+05	5.27E+05	5.47E+05	2.14E+05	5.36E+04
C17F11	AUTC	1.64E+04	1.23E+05	1.71E+04	5.04E+06	2.05E+06	3.09E+05	4.65E+05	5.08E+05	1.91E+05	3.56E+04

	AUNC	6.69E+00	5.77E+01	1.34E+01	6.72E+02	3.60E+02	1.14E+02	1.01E+02	8.21E+01	3.76E+01	8.61E+00
	AUC	4.74E+09	9.09E+08	3.38E+09	4.30E+11	5.01E+11	4.86E+09	2.06E+10	1.44E+10	7.86E+09	3.51E+09
C17F12	AUTC	1.71E+08	6.26E+07	4.43E+07	4.24E+11	4.94E+11	3.59E+09	1.56E+10	1.11E+10	5.27E+09	9.36E+08
	AUNC	2.69E+00	2.80E+00	4.03E+00	3.16E+02	3.24E+02	5.68E+00	1.61E+01	1.16E+01	6.02E+00	4.24E+00
	AUC	1.25E+08	2.85E+07	6.50E+07	2.54E+09	3.29E+08	5.37E+07	2.30E+08	2.22E+08	1.47E+08	1.79E+08
C17F13	AUTC	2.08E+07	3.25E+06	1.17E+05	2.40E+09	2.20E+08	2.58E+07	8.63E+07	8.47E+07	4.30E+07	2.55E+06
	AUNC	1.60E+00	2.36E+00	2.84E+00	2.80E+01	1.26E+01	6.48E+00	4.03E+00	2.86E+00	3.25E+00	2.11E+00
	AUC	3.31E+06	7.52E+05	4.20E+05	1.12E+07	1.91E+06	1.22E+06	1.71E+06	1.83E+06	4.43E+06	1.58E+06
C17F14	AUTC	4.18E+04	3.35E+05	1.61E+04	9.84E+06	5.58E+05	7.50E+05	7.27E+05	6.32E+05	2.95E+06	3.37E+05
	AUNC	5.72E+00	2.81E+01	3.88E+00	2.59E+02	1.45E+01	5.11E+01	3.32E+01	1.15E+01	8.59E+01	4.48E+00
	AUC	4.28E+06	1.46E+06	1.08E+06	5.17E+07	2.12E+07	2.51E+06	1.85E+07	6.99E+06	8.56E+06	4.33E+06
C17F15	AUTC	2.49E+05	2.95E+05	1.08E+04	4.83E+07	1.74E+07	1.61E+06	1.51E+07	3.47E+06	4.41E+06	7.39E+05
	AUNC	2.13E+00	1.06E+01	2.90E+00	9.36E+01	1.37E+02	1.19E+01	3.92E+01	2.06E+01	2.58E+01	8.87E+00
	AUC	4.18E+04	6.06E+05	9.90E+04	2.04E+06	1.73E+06	6.00E+05	1.25E+06	9.89E+05	3.58E+05	8.94E+05
C17F16	AUTC	2.37E+04	5.93E+05	8.27E+04	2.02E+06	1.70E+06	5.89E+05	1.22E+06	9.69E+05	3.45E+05	8.76E+05
	AUNC	3.96E+01	7.76E+02	1.14E+02	2.00E+03	1.73E+03	8.14E+02	1.33E+03	1.07E+03	3.57E+02	8.35E+02
	AUC	1.09E+05	2.96E+05	3.98E+04	4.63E+05	3.73E+05	2.87E+05	3.95E+05	4.06E+05	2.56E+05	1.54E+05
C17F17	AUTC	1.02E+05	2.88E+05	3.35E+04	4.57E+05	3.67E+05	2.80E+05	3.86E+05	3.98E+05	2.49E+05	1.48E+05
	AUNC	1.60E+02	7.22E+02	1.09E+02	9.01E+02	6.70E+02	6.37E+02	6.72E+02	7.43E+02	4.45E+02	2.95E+02
	AUC	5.54E+08	5.57E+07	2.37E+08	4.68E+10	2.09E+09	1.20E+08	2.77E+08	3.11E+08	4.59E+08	5.12E+08
C17F18	AUTC	1.56E+06	4.64E+06	1.55E+05	4.60E+10	1.70E+09	5.78E+07	5.04E+07	7.55E+07	1.67E+08	3.91E+07
	AUNC	1.60E+00	2.72E+00	2.92E+00	2.03E+02	3.60E+01	3.28E+00	1.28E+00	1.57E+00	3.46E+00	1.75E+00
	AUC	1.89E+07	2.57E+06	7.34E+06	2.06E+08	2.29E+07	2.71E+07	9.98E+07	8.62E+07	2.39E+07	1.11E+07
C17F19	AUTC	1.11E+05	2.75E+05	4.64E+03	1.92E+08	8.41E+06	2.16E+07	7.04E+07	5.57E+07	9.48E+06	1.29E+06
	AUNC	1.90E+00	4.38E+00	3.27E+00	2.72E+02	2.40E+01	1.24E+02	7.69E+01	8.52E+01	6.82E+00	2.93E+00
	AUC	1.95E+04	3.79E+05	1.94E+04	1.01E+06	8.08E+05	2.66E+05	7.45E+05	5.93E+05	2.76E+05	1.36E+05
C17F20	AUTC	1.16E+04	3.71E+05	1.25E+04	9.96E+05	7.99E+05	2.59E+05	7.33E+05	5.81E+05	2.67E+05	1.28E+05
	AUNC	4.07E+01	1.06E+03	5.44E+01	2.24E+03	1.94E+03	6.42E+02	1.79E+03	1.38E+03	6.34E+02	3.75E+02
	AUC	5.14E+05	5.28E+05	5.11E+05	1.10E+06	1.13E+06	6.20E+05	1.10E+06	8.14E+05	9.81E+05	8.91E+05
C17F21	AUTC	5.03E+05	5.18E+05	5.02E+05	1.09E+06	1.11E+06	6.09E+05	1.09E+06	8.02E+05	9.70E+05	8.81E+05
	AUNC	1.51E+03	1.85E+03	1.73E+03	3.30E+03	3.58E+03	1.97E+03	3.28E+03	2.45E+03	3.06E+03	2.83E+03
	AUC	2.64E+05	4.87E+05	4.43E+05	2.61E+06	2.44E+06	5.62E+05	6.24E+05	5.40E+05	5.65E+05	5.26E+05
C17F22	AUTC	2.48E+05	4.80E+05	4.30E+05	2.58E+06	2.41E+06	5.53E+05	6.00E+05	5.28E+05	5.55E+05	5.17E+05
	AUNC	1.29E+02	7.06E+02	3.74E+02	1.68E+03	1.56E+03	5.72E+02	4.30E+02	3.84E+02	3.56E+02	3.88E+02
	AUC	1.56E+06	1.62E+06	1.56E+06	2.01E+06	1.93E+06	1.64E+06	1.80E+06	1.74E+06	1.63E+06	1.60E+06
C17F23	AUTC	1.54E+06	1.60E+06	1.54E+06	1.99E+06	1.91E+06	1.62E+06	1.78E+06	1.72E+06	1.61E+06	1.58E+06
	AUNC	3.13E+03	4.12E+03	3.26E+03	4.02E+03	3.68E+03	3.76E+03	3.54E+03	3.35E+03	3.19E+03	3.12E+03
	AUC	5.25E+05	1.74E+06	1.01E+06	2.25E+06	2.00E+06	1.39E+06	1.89E+06	1.76E+06	1.76E+06	1.55E+06
C17F24	AUTC	5.07E+05	1.72E+06	9.88E+05	2.22E+06	1.98E+06	1.37E+06	1.86E+06	1.74E+06	1.74E+06	1.53E+06
	AUNC	9.70E+02	4.09E+03	1.96E+03	4.00E+03	3.59E+03	2.98E+03	3.38E+03	3.16E+03	3.26E+03	2.74E+03
	AUC	2.02E+06	2.17E+06	2.11E+06	3.38E+06	3.51E+06	2.14E+06	2.22E+06	2.20E+06	2.19E+06	2.12E+06
C17F25	AUTC	1.99E+06	2.14E+06	2.08E+06	3.35E+06	3.48E+06	2.11E+06	2.18E+06	2.17E+06	2.16E+06	2.10E+06
	AUNC	1.03E+03	2.82E+03	1.86E+03	2.39E+03	2.32E+03	2.00E+03	1.52E+03	1.47E+03	1.47E+03	1.35E+03
	AUC	1.37E+06	2.57E+06	1.53E+06	5.97E+06	6.08E+06	2.20E+06	3.31E+06	2.85E+06	2.06E+06	1.74E+06
C17F26	AUTC	1.33E+06	2.54E+06	1.49E+06	5.91E+06	6.02E+06	2.17E+06	3.26E+06	2.80E+06	2.03E+06	1.72E+06
	AUNC	6.52E+02	1.73E+03	8.65E+02	2.90E+03	2.80E+03	1.62E+03	1.64E+03	1.39E+03	9.85E+02	9.38E+02
	AUC	1.95E+06	1.97E+06	1.97E+06	2.66E+06	2.26E+06	2.01E+06	2.21E+06	2.06E+06	1.97E+06	2.04E+06
C17F27	AUTC	1.93E+06	1.95E+06	1.95E+06	2.63E+06	2.24E+06	1.99E+06	2.18E+06	2.04E+06	1.95E+06	2.02E+06
	AUNC	3.39E+03	4.33E+03	3.35E+03	4.24E+03	3.46E+03	3.98E+03	3.58E+03	3.34E+03	3.05E+03	3.27E+03
	AUC	1.56E+06	2.66E+06	1.60E+06	4.43E+06	4.18E+06	2.56E+06	2.64E+06	2.45E+06	2.74E+06	2.27E+06
C17F28	AUTC	1.54E+06	2.63E+06	1.57E+06	4.39E+06	4.13E+06	2.54E+06	2.61E+06	2.41E+06	2.71E+06	2.24E+06
	AUNC	1.39E+03	3.75E+03	1.46E+03	3.77E+03	3.39E+03	2.88E+03	2.22E+03	2.11E+03	2.26E+03	1.87E+03
	AUC	1.27E+06	1.57E+06	1.31E+06	2.37E+06	2.14E+06	1.53E+06	1.94E+06	1.91E+06	1.45E+06	1.41E+06
C17F29	AUTC	1.24E+06	1.55E+06	1.29E+06	2.34E+06	2.11E+06	1.51E+06	1.91E+06	1.88E+06	1.43E+06	1.39E+06
	AUNC	1.53E+03	2.30E+03	1.52E+03	2.47E+03	2.38E+03	2.18E+03	2.13E+03	2.13E+03	1.66E+03	1.49E+03
	AUC	2.56E+08	3.05E+09	3.32E+08	7.90E+10	1.03E+10	2.08E+09	6.83E+09	1.49E+09	1.82E+09	1.33E+09
C17F30	AUTC	4.88E+07	2.96E+09	2.50E+07	7.82E+10	9.96E+09	1.98E+09	6.11E+09	1.29E+09	1.63E+09	1.06E+09
	AUNC	3.67E+00	1.49E+02	8.27E+00	1.11E+03	1.96E+02	1.09E+02	1.05E+02	2.35E+01	3.36E+01	1.99E+01

Table 27
Convergence indicators for 10 CEC2019 test suite.

Fun.	Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
	AUC	1.50E+09	1.82E+09	1.02E+08	6.42E+08	6.63E+08	7.36E+09	9.28E+08	3.25E+10	2.03E+09	1.17E+10
C19F1	AUTC	1.02E-06	7.62E+07	1.75E-05	0.00E+00	0.00E+00	5.11E+09	1.69E+00	2.03E+10	1.67E+08	8.91E+09
	AUNC	1.23E+00	4.57E+00	1.70E+00	6.19E-01	6.38E-01	1.38E+01	9.30E-01	4.04E+01	2.22E+00	1.92E+01
	AUC	1.01E+05	2.28E+06	2.80E+04	4.88E+04	4.74E+04	2.61E+06	6.35E+04	4.05E+07	2.35E+06	4.45E+06
C19F2	AUTC	1.60E+04	2.10E+06	1.59E+04	1.98E+04	1.98E+04	2.32E+06	1.94E+04	3.99E+07	2.06E+06	4.12E+06
	AUNC	3.41E+00	1.28E+02	2.88E+01	1.78E+00	1.85E+00	1.41E+02	2.60E+00	1.76E+03	9.61E+01	1.78E+02
	AUC	4.37E+03	8.96E+03	4.42E+03	1.46E+04	1.95E+04	1.63E+04	7.56E+03	6.31E+03	9.77E+03	1.21E+04
C19F3	AUTC	4.02E+03	8.52E+03	4.05E+03	1.44E+04	1.93E+04	1.59E+04	7.24E+03	5.99E+03	9.36E+03	1.17E+04
	AUNC	3.87E+02	8.10E+02	4.08E+02	1.28E+03	1.73E+03	1.48E+03	6.67E+02	5.71E+02	8.68E+02	1.07E+03
	AUC	3.83E+04	9.70E+04	4.35E+04	3.41E+05	3.25E+05	1.21E+05	2.26E+05	1.83E+05	1.00E+05	6.24E+04
C19F4	AUTC	3.51E+04	9.49E+04	4.07E+04	3.37E+05	3.21E+05	1.18E+05	2.22E+05	1.80E+05	9.77E+04	5.96E+04

	AUNC	2.54E+02	9.22E+02	3.60E+02	2.41E+03	2.34E+03	1.04E+03	1.66E+03	1.43E+03	7.74E+02	4.59E+02
	AUC	1.60E+03	8.52E+02	1.13E+03	2.46E+05	2.31E+05	6.79E+03	1.15E+04	4.49E+03	7.31E+03	2.80E+03
C19F5	AUTC	4.92E+02	6.22E+02	4.23E+02	2.43E+05	2.29E+05	6.31E+03	8.99E+03	3.74E+03	6.81E+03	2.26E+03
	AUNC	8.82E+00	1.42E+01	1.11E+01	1.52E+03	1.54E+03	8.66E+01	7.89E+01	3.46E+01	5.13E+01	1.89E+01
	AUC	7.44E+02	1.30E+04	8.66E+02	4.17E+04	4.17E+04	1.77E+04	3.17E+04	3.41E+04	1.14E+04	5.07E+03
C19F6	AUTC	4.04E+02	1.27E+04	5.20E+02	4.13E+04	4.13E+04	1.75E+04	3.12E+04	3.36E+04	1.11E+04	4.74E+03
	AUNC	5.41E+01	1.06E+03	6.91E+01	3.18E+03	3.04E+03	1.42E+03	2.42E+03	2.51E+03	8.55E+02	3.80E+02
	AUC	2.22E+06	3.68E+06	1.85E+06	7.97E+06	8.18E+06	4.41E+06	5.52E+06	5.40E+06	3.95E+06	3.07E+06
C19F7	AUTC	2.12E+06	3.60E+06	1.76E+06	7.88E+06	8.09E+06	4.34E+06	5.43E+06	5.32E+06	3.86E+06	2.98E+06
	AUNC	8.59E+02	1.52E+03	7.34E+02	3.05E+03	3.11E+03	1.80E+03	2.11E+03	2.25E+03	1.52E+03	1.16E+03
	AUC	1.04E+04	1.19E+04	9.11E+03	1.85E+04	1.79E+04	1.46E+04	1.78E+04	1.67E+04	1.28E+04	1.09E+04
C19F8	AUTC	1.02E+04	1.17E+04	8.92E+03	1.83E+04	1.77E+04	1.44E+04	1.76E+04	1.65E+04	1.26E+04	1.07E+04
	AUNC	2.32E+03	2.79E+03	2.10E+03	4.16E+03	4.10E+03	3.39E+03	3.99E+03	3.73E+03	2.89E+03	2.44E+03
	AUC	6.58E+02	3.21E+02	5.29E+02	8.09E+03	7.81E+03	1.50E+03	2.23E+03	2.23E+03	8.99E+02	8.12E+02
C19F9	AUTC	3.04E+02	2.89E+02	4.87E+02	8.00E+03	7.73E+03	1.46E+03	2.15E+03	2.18E+03	8.62E+02	7.68E+02
	AUNC	1.22E+02	1.04E+02	1.54E+02	1.83E+03	1.69E+03	3.97E+02	5.10E+02	5.19E+02	2.21E+02	1.82E+02
	AUC	1.75E+04	9.69E+04	5.26E+04	1.01E+05	1.01E+05	1.01E+05	1.00E+05	1.00E+05	1.02E+05	8.91E+04
C19F10	AUTC	1.64E+04	9.59E+04	5.16E+04	1.00E+05	1.00E+05	9.97E+04	9.94E+04	9.93E+04	1.01E+05	8.81E+04
	AUNC	8.34E+02	4.63E+03	2.52E+03	4.82E+03	4.84E+03	4.78E+03	4.78E+03	4.78E+03	4.84E+03	4.25E+03

Table 28

Convergence indicators for 10 CEC2020 test suite with $m = 10$.

Fun.	Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
	AUC	1.23E+10	1.33E+10	7.85E+10	2.63E+13	2.90E+13	7.77E+11	7.84E+11	9.22E+10	3.76E+11	1.20E+11
C20F1	AUTC	4.26E+06	9.42E+06	4.63E+08	2.61E+13	2.87E+13	7.37E+11	4.88E+11	2.89E+09	3.23E+11	7.22E+10
	AUNC	2.55E+00	2.23E+00	6.90E+00	1.63E+03	1.78E+03	8.54E+01	4.70E+01	6.07E+00	2.40E+01	8.47E+00
	AUC	1.02E+06	2.85E+06	8.29E+05	7.09E+06	6.41E+06	3.47E+06	4.66E+06	5.13E+06	2.76E+06	1.85E+06
C20F2	AUTC	9.42E+05	2.78E+06	7.51E+05	7.01E+06	6.33E+06	3.40E+06	4.57E+06	5.05E+06	2.68E+06	1.77E+06
	AUNC	4.24E+02	1.21E+03	3.69E+02	2.94E+03	2.64E+03	1.45E+03	1.94E+03	2.10E+03	1.16E+03	7.59E+02
	AUC	1.16E+05	1.05E+05	9.70E+04	4.51E+05	3.95E+05	1.23E+05	4.07E+05	3.17E+05	1.83E+05	1.42E+05
C20F3	AUTC	1.11E+05	1.01E+05	9.27E+04	4.46E+05	3.91E+05	1.19E+05	4.01E+05	3.11E+05	1.79E+05	1.37E+05
	AUNC	2.67E+02	3.76E+02	5.09E+02	1.15E+03	1.04E+03	3.54E+02	9.90E+02	7.61E+02	4.61E+02	3.53E+02
	AUC	1.05E+06	3.44E+04	4.49E+05	1.29E+08	4.05E+07	2.11E+05	1.25E+06	1.10E+06	7.45E+05	9.59E+05
C20F4	AUTC	4.85E+03	4.32E+03	3.74E+03	1.27E+08	3.92E+07	1.66E+04	8.34E+04	2.41E+04	2.21E+04	1.60E+04
	AUNC	9.54E-01	6.81E+00	1.75E+00	1.26E+02	6.87E+01	1.39E+00	1.29E+00	1.08E+00	7.67E-01	1.00E+00
	AUC	4.03E+07	1.60E+07	1.58E+07	2.14E+09	7.78E+08	7.10E+07	1.76E+08	2.99E+08	5.62E+07	4.69E+07
C20F5	AUTC	1.18E+07	3.72E+06	5.20E+05	2.11E+09	7.52E+08	5.45E+07	1.44E+08	2.41E+08	3.48E+07	2.07E+07
	AUNC	4.02E+00	6.47E+00	5.49E+00	2.92E+02	1.62E+02	2.98E+01	5.85E+01	3.58E+01	1.32E+01	7.79E+00
	AUC	2.55E+04	7.73E+05	4.64E+04	2.23E+06	1.66E+06	7.65E+05	1.07E+06	6.57E+05	5.94E+05	5.31E+05
C20F6	AUTC	1.06E+04	7.62E+05	3.12E+04	2.20E+06	1.64E+06	7.54E+05	1.04E+06	6.44E+05	5.80E+05	5.15E+05
	AUNC	2.65E+01	1.18E+03	5.25E+01	2.06E+03	1.62E+03	1.09E+03	1.03E+03	6.53E+02	6.00E+02	5.79E+02
	AUC	7.77E+06	3.34E+06	3.87E+06	9.32E+08	2.23E+07	3.60E+07	1.50E+08	1.56E+08	3.04E+07	1.11E+07
C20F7	AUTC	9.33E+04	1.22E+06	7.77E+04	9.18E+08	1.64E+07	2.94E+07	1.22E+08	1.12E+08	2.38E+07	3.02E+06
	AUNC	2.11E+00	9.07E+00	3.34E+00	4.19E+02	1.52E+01	3.87E+01	1.29E+02	1.22E+02	1.32E+01	1.05E+01
	AUC	2.28E+05	5.05E+05	4.11E+05	2.29E+06	2.38E+06	5.58E+05	6.01E+05	7.19E+05	5.53E+05	5.14E+05
C20F8	AUTC	2.14E+05	4.98E+05	3.99E+05	2.26E+06	2.35E+06	5.49E+05	5.77E+05	7.04E+05	5.44E+05	5.04E+05
	AUNC	1.11E+02	6.06E+02	3.56E+02	1.45E+03	1.54E+03	5.23E+02	3.66E+02	4.27E+02	3.91E+02	3.50E+02
	AUC	5.00E+05	1.69E+06	9.41E+05	2.15E+06	2.03E+06	1.26E+06	1.77E+06	1.78E+06	1.73E+06	1.73E+06
C20F9	AUTC	4.85E+05	1.67E+06	9.25E+05	2.13E+06	2.01E+06	1.24E+06	1.75E+06	1.76E+06	1.71E+06	1.71E+06
	AUNC	9.52E+02	3.99E+03	1.82E+03	3.92E+03	3.74E+03	2.74E+03	3.32E+03	3.09E+03	3.01E+03	3.09E+03
	AUC	2.05E+06	2.15E+06	2.10E+06	3.48E+06	3.53E+06	2.22E+06	2.22E+06	2.16E+06	2.19E+06	2.14E+06
C20F10	AUTC	2.02E+06	2.13E+06	2.07E+06	3.44E+06	3.50E+06	2.19E+06	2.19E+06	2.13E+06	2.16E+06	2.11E+06
	AUNC	9.44E+02	2.81E+03	1.74E+03	2.17E+03	2.28E+03	2.25E+03	1.58E+03	1.29E+03	1.49E+03	1.35E+03

Table 29

Convergence indicators for 12 CEC2022 test suite with $m = 10$.

Fun.	Index	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
	AUC	1.11E+06	5.05E+05	6.79E+05	3.26E+07	2.53E+07	8.66E+06	6.24E+06	2.43E+07	2.19E+06	1.01E+06
C22F1	AUTC	4.75E+05	6.18E+04	1.81E+05	3.21E+07	2.48E+07	7.87E+06	5.62E+06	2.26E+07	1.71E+06	4.00E+05
	AUNC	1.15E+01	1.45E+01	2.28E+01	6.94E+02	5.48E+02	2.08E+02	1.30E+02	5.84E+02	6.36E+01	1.55E+01
	AUC	1.61E+04	3.21E+04	1.93E+04	3.00E+06	3.20E+06	1.37E+05	1.74E+05	8.87E+04	9.01E+04	1.05E+05
C22F2	AUTC	5.45E+03	3.05E+04	1.05E+04	2.97E+06	3.17E+06	1.33E+05	1.55E+05	7.97E+04	8.54E+04	9.94E+04
	AUNC	7.95E+00	1.06E+02	1.95E+01	1.67E+03	1.67E+03	1.75E+02	1.01E+02	7.52E+01	4.78E+01	5.08E+01
	AUC	1.69E+03	2.45E+03	1.46E+03	2.02E+05	1.82E+05	1.15E+04	1.44E+05	1.29E+05	2.32E+04	7.15E+03
C22F3	AUTC	3.96E+02	1.90E+03	2.64E+02	1.99E+05	1.80E+05	1.07E+04	1.41E+05	1.26E+05	2.22E+04	6.09E+03
	AUNC	1.60E+01	4.03E+01	2.05E+01	2.37E+03	2.25E+03	1.68E+02	1.60E+03	1.60E+03	2.81E+02	8.20E+01
	AUC	7.49E+04	6.12E+04	5.42E+04	2.09E+05	2.01E+05	1.39E+05	1.33E+05	1.77E+05	8.50E+04	6.57E+04
C22F4	AUTC	7.22E+04	5.86E+04	5.17E+04	2.07E+05	1.99E+05	1.36E+05	1.31E+05	1.74E+05	8.26E+04	6.31E+04
	AUNC	6.02E+02	6.24E+02	6.07E+02	1.96E+03	1.93E+03	1.37E+03	1.21E+03	1.55E+03	7.77E+02	5.89E+02
	AUC	2.29E+04	1.78E+04	1.38E+04	1.86E+06	1.81E+06	1.00E+05	2.05E+06	1.55E+06	3.54E+04	3.19E+04
C22F5	AUTC	3.29E+03	4.68E+03	8.29E+02	1.83E+06	1.79E+06	7.33E+04	2.01E+06	1.51E+06	2.66E+04	2.06E+04
	AUNC	5.25E+00	9.81E+00	8.79E+00	6.62E+02	6.84E+02	4.06E+01	6.32E+02	6.31E+02	1.31E+01	1.05E+01
	AUC	1.00E+09	2.23E+08	7.83E+08	8.43E+10	1.60E+10	3.08E+08	9.41E+08	9.04E+08	1.91E+09	1.04E+09
C22F6	AUTC	1.50E+06	2.33E+06	7.08E+05	8.28E+10	1.50E+10	1.79E+07	5.10E+07	1.09E+07	8.11E+08	1.27E+08

	AUNC	1.35E+00	2.01E+00	3.41E+00	1.70E+02	3.21E+01	1.39E+00	1.47E+00	1.59E+00	5.25E+00	2.47E+00
	AUC	1.10E+04	1.08E+05	3.39E+04	4.09E+05	3.55E+05	1.21E+05	2.88E+05	2.59E+05	1.60E+05	8.90E+04
C22F7	AUTC	7.59E+03	1.05E+05	3.04E+04	4.05E+05	3.50E+05	1.19E+05	2.83E+05	2.54E+05	1.56E+05	8.52E+04
	AUNC	4.54E+01	7.08E+02	2.20E+02	1.95E+03	1.84E+03	6.96E+02	1.34E+03	1.21E+03	8.16E+02	4.25E+02
	AUC	2.31E+04	9.41E+04	5.09E+04	1.96E+05	1.62E+05	1.12E+05	1.62E+05	1.54E+05	1.18E+05	9.56E+04
C22F8	AUTC	2.10E+04	9.23E+04	4.90E+04	1.93E+05	1.60E+05	1.10E+05	1.59E+05	1.52E+05	1.16E+05	9.34E+04
	AUNC	1.35E+02	1.12E+03	5.21E+02	1.03E+03	6.70E+02	9.45E+02	9.46E+02	8.36E+02	5.62E+02	4.70E+02
	AUC	1.15E+06	1.15E+06	1.15E+06	2.14E+06	1.99E+06	1.25E+06	1.35E+06	1.19E+06	1.23E+06	1.15E+06
C22F9	AUTC	1.14E+06	1.13E+06	1.14E+06	2.12E+06	1.97E+06	1.24E+06	1.33E+06	1.17E+06	1.21E+06	1.14E+06
	AUNC	1.56E+03	2.40E+03	2.22E+03	3.20E+03	3.14E+03	2.45E+03	2.17E+03	1.86E+03	1.91E+03	1.77E+03
	AUC	5.03E+05	7.98E+05	5.02E+05	8.93E+05	1.01E+06	6.82E+05	9.02E+05	6.71E+05	6.99E+05	6.80E+05
C22F10	AUTC	4.96E+05	7.89E+05	4.96E+05	8.83E+05	9.95E+05	6.73E+05	8.90E+05	6.60E+05	6.91E+05	6.70E+05
	AUNC	9.04E+02	3.19E+03	1.87E+03	1.67E+03	1.49E+03	2.27E+03	1.53E+03	1.38E+03	1.39E+03	1.19E+03
	AUC	2.82E+04	8.02E+05	2.28E+04	2.77E+06	3.30E+06	7.41E+05	7.50E+05	1.11E+06	7.16E+05	6.16E+05
C22F11	AUTC	9.29E+03	7.90E+05	9.25E+03	2.75E+06	3.26E+06	7.29E+05	7.24E+05	1.09E+06	7.02E+05	6.02E+05
	AUNC	1.49E+01	8.47E+02	2.01E+01	1.57E+03	1.95E+03	8.00E+02	4.55E+02	7.06E+02	4.29E+02	3.21E+02
	AUC	8.15E+05	8.20E+05	8.25E+05	1.56E+06	1.06E+06	8.30E+05	9.65E+05	8.81E+05	8.22E+05	8.32E+05
C22F12	AUTC	8.05E+05	8.12E+05	8.15E+05	1.54E+06	1.05E+06	8.21E+05	9.53E+05	8.70E+05	8.14E+05	8.23E+05
	AUNC	2.43E+03	4.13E+03	2.38E+03	3.53E+03	2.57E+03	3.38E+03	2.34E+03	2.20E+03	2.03E+03	1.87E+03

4.3 Analysis of optimization efficiency in the application of wind turbine

In the practical application of wind turbines, it is expected to complete the optimization task within the specified time. The iteration time for different optimizers varies, and it is difficult to comprehensively evaluate the application effect of an optimizer solely based on the number of iterations. Therefore, in this section, based on the actual requirements (30 seconds for each parameter), the optimization duration is set to 90 seconds to evaluate the optimization efficiency of different optimizers in the application of wind turbines. And the number of population members N is set to 30. Furthermore, assign the convergence parameters in the following Eq. (5) and Eq. (11) of VOA a temporal meaning. That is, t represents the current running time; T represents the set optimization duration. At the same time, the convergence parameters of the contrast optimizer will also be corrected in the same way.

$$A = \frac{\exp(1) - \exp(t/T)}{2}$$

$$B = 1 - t/T$$

Table 30 presents the optimization results of the control parameters for wind turbines within an optimization period of 90 seconds. Among them, VOA already has excellent optimization results. After incorporating expert experience, VOA* can consistently outperform all the compared optimizers. Overall, within the specified optimization time frame, VOA and VOA* demonstrated excellent competitiveness, reflecting the optimization efficiency of this algorithm in the application of wind turbines.

Table 30
Analysis of optimization efficiency in the application of wind turbine.

Wind speed = 11 m/s											
Index	VOA*	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	1.55E+03	1.62E+03	1.57E+03	1.56E+03	1.48E+05	1.87E+05	1.45E+05	2.93E+04	7.55E+05	3.62E+04	1.53E+06
STD	1.65E+01	8.94E+01	1.52E+02	1.98E+01	2.20E+05	2.31E+05	4.76E+05	9.52E+04	1.08E+06	5.88E+04	1.14E+06
BEST	1.54E+03	1.54E+03	1.54E+03	1.54E+03	1.70E+03	1.82E+03	1.56E+03	1.79E+03	1.57E+03	1.55E+03	7.69E+04
Wind speed = 11.5 m/s											
Index	VOA*	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	1.44E+03	1.47E+03	1.55E+03	1.74E+03	8.39E+04	7.78E+04	5.84E+04	7.22E+03	9.70E+05	4.41E+04	3.67E+05
STD	4.09E+00	3.00E+01	2.54E+02	2.74E+02	1.08E+05	1.04E+05	3.18E+04	5.51E+03	8.20E+05	3.87E+04	6.57E+05
BEST	1.43E+03	1.43E+03	1.43E+03	1.43E+03	1.58E+03	1.59E+03	1.46E+03	2.03E+03	7.28E+04	1.44E+03	7.28E+04
Wind speed = 12 m/s											
Index	VOA*	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	1.43E+03	1.47E+03	1.43E+03	1.44E+03	4.38E+04	2.37E+04	2.62E+05	3.74E+03	7.30E+05	1.47E+04	7.23E+05
STD	4.05E+00	4.74E+01	4.57E+01	3.79E+01	3.98E+04	4.08E+04	5.09E+05	2.49E+03	6.03E+05	2.97E+04	5.97E+05
BEST	1.39E+03	1.39E+03	1.39E+03	1.44E+03	1.68E+03	1.68E+03	1.42E+03	1.72E+03	6.88E+04	1.42E+03	6.88E+04
Wind speed = 13 m/s											
Index	VOA*	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	1.23E+03	1.23E+03	1.23E+03	1.24E+03	4.10E+03	8.17E+03	6.35E+04	2.96E+03	5.98E+04	1.37E+04	6.31E+05
STD	6.37E+00	8.16E+00	7.45E+00	8.18E+00	4.33E+03	1.06E+04	9.57E-11	2.07E+03	8.47E+03	2.79E+04	5.20E+05
BEST	1.22E+03	1.22E+03	1.22E+03	1.22E+03	1.39E+03	1.73E+03	6.35E+04	1.24E+03	4.46E+04	1.22E+03	6.35E+04
Wind speed = 14 m/s											
Index	VOA*	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO

AVE	1.15E+03	1.16E+03	1.16E+03	1.21E+03	8.01E+03	3.27E+03	3.42E+04	4.79E+03	3.93E+05	2.99E+04	7.34E+05
STD	2.57E+00	1.15E+01	7.94E+00	1.14E+02	8.43E+03	1.53E+03	2.66E+04	7.76E+03	5.10E+05	3.05E+04	3.77E+05
BEST	1.15E+03	1.15E+03	1.15E+03	1.15E+03	1.43E+03	1.34E+03	1.32E+03	1.16E+03	1.32E+03	1.15E+03	6.17E+04
Wind speed = 16 m/s											
Index	VOA*	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	1.04E+03	1.06E+03	1.05E+03	1.04E+03	7.02E+03	5.26E+03	1.48E+04	3.60E+03	1.45E+05	2.36E+04	4.22E+05
STD	2.95E+00	2.24E+01	2.52E+01	7.37E+00	2.48E+03	2.14E+03	2.42E+04	2.76E+03	2.92E+05	3.09E+04	3.33E+05
BEST	1.03E+03	1.05E+03	1.03E+03	1.03E+03	2.57E+03	3.32E+03	1.09E+03	1.12E+03	1.16E+03	1.04E+03	5.76E+04
Wind speed = 18 m/s											
Index	VOA*	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	9.84E+02	1.01E+03	1.12E+03	9.90E+02	4.38E+03	2.31E+03	1.33E+05	2.02E+03	3.85E+05	1.18E+04	5.03E+05
STD	8.01E-01	2.57E+01	3.07E+02	1.88E+01	8.07E+02	9.91E+02	2.66E+05	9.09E+02	3.24E+05	2.41E+04	2.79E+05
BEST	9.81E+02	9.85E+02	9.81E+02	9.83E+02	3.23E+03	1.06E+03	1.04E+03	1.20E+03	5.60E+03	9.86E+02	6.33E+03
Wind speed = 20 m/s											
Index	VOA*	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	1.01E+03	1.01E+03	1.05E+03	1.02E+03	3.18E+03	1.83E+03	2.30E+04	3.10E+03	2.18E+05	1.60E+03	4.41E+05
STD	3.12E+00	3.12E+00	9.14E+01	6.26E+00	8.66E+02	5.16E+02	2.85E+04	4.34E+02	2.92E+05	1.32E+03	2.16E+05
BEST	1.01E+03	1.01E+03	1.01E+03	1.02E+03	1.64E+03	1.33E+03	1.02E+03	2.73E+03	4.52E+03	1.01E+03	5.42E+04
Wind speed = 22 m/s											
Index	VOA*	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	1.23E+03	1.23E+03	1.36E+03	1.43E+03	2.42E+03	2.23E+03	1.22E+05	1.73E+03	3.11E+05	1.16E+04	3.11E+05
STD	5.78E+00	1.57E+01	2.37E+02	2.09E+02	6.90E+02	6.36E+02	2.13E+05	4.75E+02	2.58E+05	2.32E+04	2.58E+05
BEST	1.22E+03	1.22E+03	1.22E+03	1.23E+03	1.57E+03	1.47E+03	1.22E+03	1.32E+03	3.81E+03	1.22E+03	3.86E+03
Wind speed = 25 m/s											
Index	VOA*	VOA	FLA	EEFO	FLO	COA	DBO	HHO	WOA	GWO	PSO
AVE	1.55E+03	1.58E+03	1.64E+03	1.73E+03	1.74E+03	1.85E+03	1.20E+04	1.74E+03	3.96E+05	2.24E+04	1.40E+05
STD	7.66E+01	4.14E+01	6.67E+01	6.32E+01	1.32E+02	1.58E+01	2.33E+04	1.04E+02	2.20E+05	2.84E+04	2.23E+05
BEST	1.50E+03	1.53E+03	1.53E+03	1.67E+03	1.62E+03	1.83E+03	1.50E+03	1.64E+03	1.90E+03	1.67E+03	1.50E+03

4.4 Ablation analysis

This subsection conducts ablation experiments, removing behaviors such as seed germination (VOA_1), climbing (VOA_2), leaf growth (VOA_3), leaf-shaped optimization (VOA_4), heart-shaped optimization (VOA_5), drooping (VOA_6) and swarm function (VOA_7), and performs ablation analysis in the CEC2017 test suite with $m = 10$. In this experiment, N is 100 and T is 5000.

Table 31 presents the results of the ablation experiment. VOA_1 has removed the seed germination behavior, replacing the cubic chaotic initializations with random initializations. In most test functions, the contribution of the initialization method is relatively low. Through other optimization behaviors during long-term iterations, the best value can still be found. However, cubic chaotic initializations can effectively reduce the STD index, improve the optimization accuracy and speed. Moreover, in some complex functions such as C17F10 and C17F12, it exhibits better optimization results. VOA_2 has removed the climbing behavior, causing all individuals of the vine population to engage in the leaf growth behavior and VOA_3 has removed the leaf growth behavior, causing all individuals of the vine population to engage in the climbing behavior. This significantly reduces the final optimization result. VOA_4 and VOA_5 have respectively removed leaf-shaped optimization and heart-shaped optimization, meaning that only one of these optimizations is carried out in the leaf growth behavior, which results in the algorithm lacking the ability for deep exploitation. VOA_6 has removed drooping behavior, which has significantly reduced the optimization capability of VOA. VOA_7 has removed the swarm function (Eq. 20) from the swarm behavior. Like VOA_1, its main impact is on the optimization accuracy and speed of VOA, as well as the optimization results of some complex functions.

In summary, the behaviors of climbing, leaf growth, and drooping are the main optimization behaviors, and they have significant contributions to performance. Seed germination and swarm function also make certain contributions in improving the speed and accuracy of optimization.

Table 31
Ablation analysis for 29 CEC2017 test suite with $m = 10$.

Fun.	Index	VOA	VOA_1	VOA_2	VOA_3	VOA_4	VOA_5	VOA_6	VOA_7
C17F1	AVE	1.00E+02	1.00E+02	7.64E+02	2.26E+03	2.62E+03	1.00E+02	1.00E+02	1.00E+02
	STD	0.00E+00	5.48E-13	1.79E+03	2.41E+03	2.55E+03	2.46E-13	1.07E+00	0.00E+00
	BEST	1.00E+02	1.00E+02	1.00E+02	1.00E+02	1.80E+02	1.00E+02	1.00E+02	1.00E+02
C17F3	AVE	3.00E+02	3.00E+02	3.00E+02	3.00E+02	3.00E+02	3.00E+02	3.00E+02	3.00E+02
	STD	0.00E+00	0.00E+00	5.22E-12	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
	BEST	3.00E+02	3.00E+02	3.00E+02	3.00E+02	3.00E+02	3.00E+02	3.00E+02	3.00E+02
C17F4	AVE	4.00E+02	4.00E+02	4.00E+02	4.03E+02	4.02E+02	4.02E+02	4.00E+02	4.00E+02
	STD	5.16E-02	7.40E-02	9.07E-02	1.02E+00	9.13E-01	9.00E-01	1.02E-01	8.09E-02
	BEST	4.00E+02	4.00E+02	4.00E+02	4.01E+02	4.00E+02	4.00E+02	4.00E+02	4.00E+02
C17F5	AVE	5.05E+02	5.06E+02	5.25E+02	5.07E+02	5.08E+02	5.06E+02	5.14E+02	5.07E+02
	STD	2.30E+00	2.81E+00	9.00E+00	2.65E+00	3.04E+00	2.44E+00	6.47E+00	2.63E+00
	BEST	5.01E+02	5.02E+02	5.08E+02	5.02E+02	5.01E+02	5.01E+02	5.03E+02	5.02E+02
C17F6	AVE	6.00E+02	6.00E+02	6.08E+02	6.00E+02	6.00E+02	6.00E+02	6.01E+02	6.00E+02
	STD	0.00E+00	0.00E+00	5.80E+00	0.00E+00	0.00E+00	0.00E+00	1.14E+00	0.00E+00

	BEST	6.00E+02	6.00E+02	6.00E+02	6.00E+02	6.00E+02	6.00E+02	6.00E+02	6.00E+02
C17F7	AVE	7.19E+02	7.19E+02	7.52E+02	7.19E+02	7.19E+02	7.19E+02	7.29E+02	7.20E+02
	STD	2.70E+00	4.15E+00	1.17E+01	4.32E+00	3.66E+00	3.32E+00	9.24E+00	4.44E+00
	BEST	7.12E+02	7.14E+02	7.33E+02	7.14E+02	7.13E+02	7.12E+02	7.13E+02	7.13E+02
C17F8	AVE	8.07E+02	8.07E+02	8.24E+02	8.08E+02	8.08E+02	8.09E+02	8.13E+02	8.08E+02
	STD	2.56E+00	4.36E+00	7.87E+00	2.63E+00	3.24E+00	3.63E+00	5.71E+00	2.95E+00
	BEST	8.03E+02	8.01E+02	8.06E+02	8.03E+02	8.01E+02	8.04E+02	8.06E+02	8.03E+02
C17F9	AVE	9.00E+02	9.00E+02	9.90E+02	9.00E+02	9.00E+02	9.00E+02	9.01E+02	9.00E+02
	STD	0.00E+00	0.00E+00	1.34E+02	0.00E+00	0.00E+00	0.00E+00	2.30E+00	0.00E+00
	BEST	9.00E+02	9.00E+02	9.00E+02	9.00E+02	9.00E+02	9.00E+02	9.00E+02	9.00E+02
C17F10	AVE	1.14E+03	1.25E+03	1.63E+03	1.27E+03	1.21E+03	1.26E+03	1.35E+03	1.26E+03
	STD	1.20E+02	1.17E+02	2.41E+02	1.23E+02	1.43E+02	1.35E+02	1.93E+02	1.29E+02
	BEST	1.00E+03	1.00E+03	1.12E+03	1.03E+03	1.02E+03	1.02E+03	1.01E+03	1.02E+03
C17F11	AVE	1.10E+03	1.10E+03	1.12E+03	1.10E+03	1.10E+03	1.10E+03	1.11E+03	1.10E+03
	STD	1.31E+00	1.28E+00	1.40E+01	1.24E+00	1.21E+00	1.46E+00	6.95E+00	1.01E+00
	BEST	1.10E+03	1.10E+03	1.10E+03	1.10E+03	1.10E+03	1.10E+03	1.10E+03	1.10E+03
C17F12	AVE	1.37E+03	1.82E+03	1.75E+03	6.61E+03	6.43E+03	1.70E+03	8.69E+03	1.78E+03
	STD	1.19E+02	8.47E+02	2.07E+02	2.30E+03	2.15E+03	2.63E+02	1.07E+04	8.17E+02
	BEST	1.20E+03	1.20E+03	1.33E+03	3.43E+03	2.92E+03	1.32E+03	1.32E+03	1.20E+03
C17F13	AVE	1.38E+03	1.38E+03	1.42E+03	1.82E+03	1.53E+03	1.46E+03	1.37E+03	1.39E+03
	STD	5.46E+01	8.59E+01	1.11E+02	5.32E+02	2.27E+02	1.71E+02	8.67E+01	8.55E+01
	BEST	1.30E+03	1.31E+03	1.32E+03	1.32E+03	1.32E+03	1.31E+03	1.30E+03	1.31E+03
C17F14	AVE	1.42E+03	1.42E+03	1.45E+03	1.42E+03	1.42E+03	1.42E+03	1.42E+03	1.42E+03
	STD	9.07E+00	7.53E+00	1.62E+01	7.66E+00	6.45E+00	8.75E+00	1.16E+01	9.21E+00
	BEST	1.40E+03	1.40E+03	1.41E+03	1.40E+03	1.40E+03	1.40E+03	1.40E+03	1.40E+03
C17F15	AVE	1.51E+03	1.51E+03	1.52E+03	1.52E+03	1.51E+03	1.51E+03	1.51E+03	1.52E+03
	STD	8.07E+00	5.90E+00	1.66E+01	8.47E+00	5.72E+00	5.82E+00	7.11E+00	1.81E+01
	BEST	1.50E+03	1.50E+03	1.50E+03	1.50E+03	1.50E+03	1.50E+03	1.50E+03	1.50E+03
C17F16	AVE	1.60E+03	1.60E+03	1.72E+03	1.60E+03	1.60E+03	1.60E+03	1.62E+03	1.60E+03
	STD	2.47E-01	4.77E-01	9.69E+01	4.87E-01	7.74E-01	2.17E+00	4.58E+01	5.12E-01
	BEST	1.60E+03	1.60E+03	1.60E+03	1.60E+03	1.60E+03	1.60E+03	1.60E+03	1.60E+03
C17F17	AVE	1.71E+03	1.72E+03	1.74E+03	1.71E+03	1.72E+03	1.72E+03	1.73E+03	1.71E+03
	STD	6.95E+00	7.77E+00	2.16E+01	8.18E+00	6.30E+00	9.71E+00	1.62E+01	8.13E+00
	BEST	1.70E+03	1.70E+03	1.71E+03	1.70E+03	1.70E+03	1.70E+03	1.70E+03	1.70E+03
C17F18	AVE	1.84E+03	1.85E+03	1.86E+03	2.00E+03	1.91E+03	1.86E+03	1.85E+03	1.87E+03
	STD	2.74E+01	2.97E+01	4.02E+01	9.54E+01	4.63E+01	3.22E+01	3.52E+01	6.10E+01
	BEST	1.81E+03	1.82E+03	1.82E+03	1.88E+03	1.84E+03	1.82E+03	1.82E+03	1.82E+03
C17F19	AVE	1.91E+03	1.91E+03	1.92E+03	1.91E+03	1.91E+03	1.91E+03	1.91E+03	1.91E+03
	STD	2.46E+00	4.10E+00	1.30E+01	3.60E+00	2.64E+00	4.25E+00	4.10E+00	8.97E+00
	BEST	1.90E+03	1.90E+03	1.90E+03	1.90E+03	1.90E+03	1.90E+03	1.90E+03	1.90E+03
C17F20	AVE	2.00E+03	2.00E+03	2.05E+03	2.00E+03	2.00E+03	2.00E+03	2.01E+03	2.00E+03
	STD	1.27E-01	3.64E+00	4.27E+01	5.70E-02	1.08E-01	3.70E+00	7.93E+00	1.08E-01
	BEST	2.00E+03	2.00E+03	2.00E+03	2.00E+03	2.00E+03	2.00E+03	2.00E+03	2.00E+03
C17F21	AVE	2.20E+03	2.20E+03	2.31E+03	2.20E+03	2.20E+03	2.20E+03	2.22E+03	2.20E+03
	STD	0.00E+00	1.83E-11	5.04E+01	0.00E+00	0.00E+00	0.00E+00	4.63E+01	4.06E-01
	BEST	2.20E+03	2.20E+03	2.20E+03	2.20E+03	2.20E+03	2.20E+03	2.20E+03	2.20E+03
C17F22	AVE	2.23E+03	2.24E+03	2.38E+03	2.23E+03	2.23E+03	2.23E+03	2.30E+03	2.23E+03
	STD	3.29E+01	3.82E+01	2.41E+02	3.77E+01	3.76E+01	4.06E+01	1.99E+01	4.28E+01
	BEST	2.20E+03	2.20E+03	2.24E+03	2.20E+03	2.20E+03	2.20E+03	2.23E+03	2.20E+03
C17F23	AVE	2.61E+03	2.61E+03	2.63E+03	2.61E+03	2.61E+03	2.61E+03	2.62E+03	2.61E+03
	STD	2.40E+00	2.48E+00	1.36E+01	1.63E+00	2.06E+00	2.65E+00	7.93E+00	1.83E+00
	BEST	2.60E+03	2.60E+03	2.61E+03	2.60E+03	2.60E+03	2.60E+03	2.61E+03	2.60E+03
C17F24	AVE	2.50E+03	2.50E+03	2.75E+03	2.50E+03	2.50E+03	2.50E+03	2.70E+03	2.50E+03
	STD	0.00E+00	0.00E+00	7.28E+01	0.00E+00	3.47E-11	0.00E+00	1.01E+02	0.00E+00
	BEST	2.50E+03	2.50E+03	2.50E+03	2.50E+03	2.50E+03	2.50E+03	2.50E+03	2.50E+03
C17F25	AVE	2.90E+03	2.90E+03	2.93E+03	2.90E+03	2.90E+03	2.90E+03	2.93E+03	2.90E+03
	STD	1.38E+01	8.31E+00	2.43E+01	8.22E+00	8.31E+00	8.23E+00	2.34E+01	1.14E+01
	BEST	2.90E+03	2.90E+03	2.90E+03	2.90E+03	2.90E+03	2.90E+03	2.90E+03	2.90E+03
C17F26	AVE	2.81E+03	2.86E+03	2.97E+03	2.86E+03	2.87E+03	2.84E+03	2.94E+03	2.87E+03
	STD	1.13E+02	1.04E+02	2.70E+02	9.78E+01	9.25E+01	1.24E+02	8.25E+01	9.15E+01
	BEST	2.60E+03	2.60E+03	2.60E+03	2.60E+03	2.60E+03	2.60E+03	2.80E+03	2.60E+03
C17F27	AVE	3.09E+03	3.09E+03	3.11E+03	3.09E+03	3.09E+03	3.09E+03	3.10E+03	3.09E+03
	STD	2.83E-00	8.02E-01	3.25E+01	4.11E-01	4.67E-01	7.37E-01	4.35E+00	4.00E-01
	BEST	3.09E+03	3.09E+03	3.09E+03	3.09E+03	3.09E+03	3.09E+03	3.09E+03	3.09E+03
C17F28	AVE	3.09E+03	3.10E+03	3.19E+03	3.10E+03	3.09E+03	3.09E+03	3.17E+03	3.10E+03
	STD	6.19E-08	1.86E-12	7.40E+01	3.58E+00	5.48E+01	5.66E+01	1.69E+02	1.76E+01
	BEST	2.80E+03	3.10E+03	3.10E+03	3.10E+03	2.80E+03	2.80E+03	2.80E+03	3.10E+03
C17F29	AVE	3.15E+03	3.15E+03	3.24E+03	3.15E+03	3.15E+03	3.15E+03	3.15E+03	3.15E+03
	STD	9.07E+00	1.30E+01	5.92E+01	9.59E+00	1.27E+01	1.17E+01	1.34E+01	1.11E+01
	BEST	3.13E+03	3.13E+03	3.16E+03	3.14E+03	3.13E+03	3.13E+03	3.14E+03	3.13E+03
C17F30	AVE	3.66E+03	3.66E+03	1.40E+05	4.66E+03	4.64E+03	3.65E+03	3.18E+04	3.69E+03
	STD	1.44E+02	2.30E+02	3.10E+05	6.82E+02	9.56E+02	3.64E+02	1.49E+05	1.47E+02

	BEST	3.48E+03	3.47E+03	3.40E+03	3.78E+03	3.67E+03	3.44E+03	3.50E+03	3.45E+03
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